# Combined Handouts of Environmental Psychology (PSY511)

Virtual University

## Table of contents

**Lesson 1**
**Introduction to Environmental Psychology**

The origins of environmental psychology can be traced back to the 1960s when researchers from various disciplines such as architecture, sociology, and psychology began to examine how the environment influences human behavior, cognition, and well-being (Steg, van den Berg, & de Groot, 2012). Initially, the focus was on the design and evaluation of built environments, such as homes, offices, schools, and hospitals. This was also known as architectural psychology or human factors (Clayton & Saunders, 2012). Later, the field expanded to include the natural environment, such as parks, forests, and wildlife. This was also called conservation or green psychology (Clayton & Saunders, 2012). Environmental psychology is a diverse and interdisciplinary field that addresses various topics, such as environmental perception, attitudes, and values, environmental behavior and decision-making, environmental stress and coping, environmental restoration and well-being, and environmental education and communication (Steg et al., 2012).
Different environments can arouse different feelings in human beings. Our mental concepts and associations with a particular environmental type i.e., our perceptions about different surroundings also affect our physiology, feelings, and reactions. Any change in the environment also brings about a shift in our thinking styles and adaptive behaviors. Let's consider the following two situations:

1. Suppose you are sitting in your comfortable room with a cup of coffee while working on your laptop. Automatically you will feel pleasant and more motivated to involve yourself in creative tasks.
2. Now imagine that due to some emergency, you have to work on your laptop during a traffic jam while on public transport. You will feel unpleasant, your creative abilities will probably decline and you will think about getting rid of the task as soon as possible.

The first situation will perhaps give rise to feelings such as contentment, peace, joy, and relief. You will be able to think of multiple solutions to a single problem and develop a broadened understanding of the different demands that the environment presents. The second situation will probably give rise to feelings such as aggression, irritability, urgency, and dissatisfaction. It might even make you feel claustrophobic! In conclusion, it is fair to say that our environment and its different aspects can generate different feelings in us and ultimately different behavioral responses. With technology and rapid industrialization, the environment is becoming increasingly complex. The dangers of acid rain, the unreliability of nuclear power plants, and the difficulties in handling and disposing of toxic chemicals are being discussed. Additionally, the pace of suburban sprawl and urban decay is speeding up. Chemicals such as dioxin, formaldehyde, PCB (polychlorinated biphenyl), / and sulfur dioxide, once found only in the laboratories are becoming household words. Human beings are using and abusing the environment openly.

Many people are working hard to fix the environment, but there are still a lot of problems. Scientists, engineers, political and spiritual leaders, and the general public care more about the environment now, but they don't always have the money, skills, or tools to do something about it.

**The Environmental Debate**

There is a debate regarding the environment between two groups of people. The first group has a pessimistic approach and the second group has an optimistic approach. Let's look at both:

- **The Pessimists:**

Some people say that the environment is so damaged that it will not be livable by the year 2100. They are angry about how many animals and plants have died in the last 100 years, how much land has become dry and useless, how many trees have been cut down, and how much water has been polluted by acid rain. They think that the environment is being harmed by the rich and the poor fighting over natural resources. They say that the rich take too much and the poor want to keep what they have, and this makes the environment suffer and lose.

- **The Optimists:**

Some people confidently highlight the remarkable achievements that have been made in improving human conditions and well-being in the fields of agriculture, medicine, and design technology. They believe that human creativity and innovation can overcome any challenge. They argue that there is sufficient energy, but the methods of obtaining it must be improved. There is a balanced population, but the distribution of people needs to be adjusted. There is manageable waste, but the systems of transporting and storing it need to be refined. They have a positive outlook on the future of the environment.

- **Has the debate been resolved?**

The debate on the environmental crisis remains unresolved, as different perspectives and solutions are proposed. Technology may offer some relief to the strain that human activities impose on the earth's resources, or human behavior may undergo significant changes to reverse the environmental damage. The intensity of the debate is increasing, as evidenced by the United Nations Earth Summit held in June 1992 in Rio de Janeiro, which raised global awareness of the situation. Along with this growing concern, there has been a renewed interest in the impact of the environment on human functioning. Various biological, psychological, and social problems have been predicted due to environmental degradation and population growth. Research is accumulating on the physiological, psychological, and behavioral effects of environmental factors such as noise, air pollution, temperature, barometric pressure, and building design. Both short-term and long-term effects are being examined. Moreover, some studies suggest that exposure to certain environmental conditions can have delayed negative consequences that persist even after the exposure has ceased. These are also being investigated with more attention. Likewise, the possibility of changing values, attitudes, and behavior toward the environment is being explored.

**Definition of Environmental Psychology**

To define Environmental Psychology, first, we need to understand the words “Environment” and Psychology separately.

- **Environment:**

Psychologist Kurt Lewin (1951) proposed a simple formula to define behavior (B) as a function of the person (P) and the environment (E): B = f (P, E). He stated that P and E are interdependent variables. However, Wohlwill (1970) observed that the environment was often understood as social or interpersonal influences, or as an unclear concept. Saegert (1987) also criticized that the social sciences defined the environment mainly in terms of social interactions and structures. Thus, the term environment has been vague and ambiguous, referring to a general physical and social context for behavior. The term environment has also been applied to various nonsocial conditions, ranging from climate to bathroom design. All in all, the environment is the natural world that surrounds us and supports our life. It includes all the living and non-living things that affect us, such as animals, plants, water, air, land, and climate. The environment is important for our survival and well-being, as it provides us with resources, services, and beauty.

- **Psychology:**

Psychology is the scientific study of the mind and behavior. Psychology often focuses on understanding a person's emotions, personality, and mind through scientific research, experiments, observation, and trials. Psychology often involves four goals, including the following:

• Describe: This goal involves collecting information and naming, classifying, and summarizing what is observed. For example, a psychologist may describe the symptoms of depression, the stages of memory, or the types of personality.

• Explain: This goal involves understanding the causes and reasons behind the observed behavior. For example, a psychologist may explain why people are more likely to help others in some situations than others, how stress affects the immune system, or what factors influence intelligence.

• Predict: This goal involves forecasting when and how the observed behavior will occur again in the future. For example, a psychologist may predict how a person will perform on a test, how a couple will resolve a conflict, or how a group will make a decision.

• Change: This goal involves modifying, influencing, or controlling the observed behavior to achieve positive outcomes. For example, a psychologist may change a person's habits, attitudes, or emotions, help a client cope with a problem, or design an intervention to prevent or treat a disorder.

Psychology is a science because it uses the scientific method to study and understand human behavior and mental processes. The scientific method is a systematic and objective way of collecting, analyzing, and interpreting data based on empirical evidence and logical reasoning. The scientific method involves four steps: (1) making observations and asking questions, (2) forming hypotheses and making predictions, (3) testing hypotheses and gathering evidence, and (4) drawing conclusions and communicating results. Psychologists use the scientific method to test theories, discover facts, and solve problems in various areas of psychology, such as cognitive, social, developmental, and clinical psychology. By using the scientific method, psychology aims to produce valid, reliable, and generalizable knowledge that can be applied to real-world situations (Cherry, 2020; McLeod, 2020).

- **The relationship between humans and the environment:**

Humans and the environment have a complex and interdependent relationship, as they influence and affect each other in various ways. Humans depend on the environment for their survival and well-being, as it provides them with natural resources, such as water, food, energy, and materials. However, humans also modify and transform the environment for their benefit, often causing harm and degradation to it. For example, humans cause climate change, pollution, deforestation, biodiversity loss, and environmental injustice through their activities and consumption patterns (Clayton & Myers, 2015). These environmental problems have negative impacts on human health, quality of life, and security, as well as on the ecosystems and species that share the planet with humans (Steg, van den Berg, & de Groot, 2012).

Therefore, humans need to find ways to improve their relationship with the environment and make it a win-win situation for both. This means that humans should respect and protect the environment, and use it sustainably and ethically, so that does not compromise its ability to support life and function. This also means that humans should benefit from the environment, and enjoy its beauty, diversity, and services, without harming or exploiting it. A win-win relationship between humans and the environment can be achieved by various means, such as adopting green lifestyles, technologies, and policies, promoting environmental education and awareness, engaging in environmental activism and stewardship, and supporting environmental justice and equity (Clayton & Myers, 2015; Steg et al., 2012).

- **So how can Environmental Psychology be defined?**

Environmental psychology is an interdisciplinary field that studies the complex and dynamic relationships between humans and their environments. It draws from various disciplines, such as biology, geology, psychology, law, geography, economics, sociology, chemistry, physics, history, philosophy, and engineering. It aims to understand how humans and environments affect and interact with each other. It also focuses on solving socially relevant problems and applying knowledge in practical ways. It considers both the influence of the physical environment on human behavior and the influence of human behavior on the physical environment.

***Environmental psychology could, therefore, be defined as a behavioral science that investigates, to enhance, the interrelationships between the physical environment and human behavior.***

*This definition does not show how rich and different environmental psychology is, or its goal to make better ways of understanding. Environmental psychologists want to know how people's bodies, minds, and actions react to their environment.* Researchers in the field, therefore investigate questions that involve physiological content (e.g., changes in heart rate, endocrine functioning, galvanic skin response, mortality)", psychological content (e.g., spatial behavior patterns, mental images, environmental stress, attitude change) behavioral content (e.g. altruism, aggression, performance). They also adopt an interdisciplinary perspective, and consider the meteorological, physical, geographical, architectural, and ecological features of the environment that affect its inhabitants. These aspects are also relevant to other disciplines, such as geology, physics, chemistry, and biology. However, environmental psychology tries to integrate these aspects and understand the interrelated processes that govern organism-environment relationships. Moreover, environmental psychologists balance between solving practical problems and developing broad-based integrative theory. They use theoretical and applied research to inform each other. For instance, they use theory to understand and prevent negative reactions to crowding and use practice to refine theories of crowding. Therefore, environmental psychology is a unique and diverse discipline that draws from various disciplines, methods, concerns, and theories. Considering these factors, and the possibility of change and criticism, we will define environmental psychology as:

***A multi-disciplinary behavioral science, both basic and applied in orientation, whose foci are the systematic interrelationships between the physical and social environments and individual human behavior and experience.***

**Goals of Environmental Psychology:**

The goals of environmental psychology are to:

- Understand how and why our environment impacts us, such as our behavior, emotions, and cognition.
- Apply that knowledge to improve the design and management of environments that promote well-being, productivity, and sustainability, such as green buildings, parks, and public spaces.
- Improve our relationship with the world around us, by fostering environmental awareness, conservation, and social responsibility

**Objectives of the course:**

- To learn about the history, scope, and methods of environmental psychology
- To understand how human behavior and the environment influence and interact with each other
- To explore the interdisciplinary nature of environmental psychology and its connections with other fields
- To develop an environmental perspective in psychology and apply it to various issues and problems
- To examine the theoretical and empirical developments in environmental psychology and their implications
- To anticipate the future trends and challenges in environmental psychology and their solutions
- To analyze the environmental challenges faced by Pakistan and their psychological aspects

**Common Assumptions of Environmental Psychology:**

The term environment is a vague and broad concept that can be interpreted differently by various disciplines. For instance, historians, biologists, architects, sociologists, economists, and psychologists have each adopted their definitions of the environment to suit their research interests and objectives. Historians have explored the environmental Zeitgeist, biologists have examined the ecological niche, architects have analyzed the design features and economists have calculated the ratio of supply to demand to explain how external factors influence human behavior. Despite the ambiguity of the term environment, it is possible to identify some common assumptions that underpin its study in all environmental science, regardless of the specific approach. These assumptions are as follows:

(1) The earth is the only suitable habitat we have.
(2) The earth's resources are limited.
(3) The earth as a planet has been and continues to be profoundly affected by life.
(4) The effects of land use by humans tend to be cumulative.
(5) Sustained life on earth is a characteristic of ecosystems and not of individual organisms or populations.

**We will begin our next lesson by discussing these assumptions in detail.**

## Lesson 2
## Common Assumptions of Environmental Psychology

The term environment is a vague and broad concept that can be interpreted differently by various disciplines. For instance, historians, biologists, architects, sociologists, economists, and psychologists have each adopted their definitions of the environment to suit their research interests and objectives. Historians have explored the environmental Zeitgeist, biologists have examined the ecological niche, architects have analyzed the design features and economists have calculated the ratio of supply to demand to explain how external factors influence human behavior. Despite the ambiguity of the term environment, it is possible to identify some common assumptions that underpin its study in all environmental science, regardless of the specific approach. These assumptions are as follows:

(1) The earth is the only suitable habitat we have.

(2) The earth's resources are limited.

(3) The earth as a planet has been and continues to be profoundly affected by life.

(4) The effects of land use by humans tend to be cumulative.

(5) Sustained life on earth is a characteristic of ecosystems and not of individual organisms or populations.

In the following pages, we will elaborate on each of these assumptions.

**1. The earth is the only suitable habitat we have, and its resources are limited.** It has been proven by research that human beings will also be affected if certain species become extinct. The earth has witnessed various forms of life emerge, evolve, flourish, and go extinct, with humans achieving a dominant position only in the recent past. However, this domination does not disprove two essential realities that must be acknowledged:

***(1) Humans, too, will succumb to extinction either through geologic, meteorological, or interstellar cataclysm, natural biological processes, internecine quarrels, or because the earth's resources will no longer support human life in its present form***

***(2) Although there may be other habitable islands in the cosmic sea, they are spaced at such great distances throughout the universe as to be virtually irrelevant to human survival.***

Despite these realities, the earth is of great significance to us and our future generations. We need to adapt to the opportunities and constraints that the earth offers, as well as the inevitable fact that it is constantly evolving. Moreover, we need to do so in a manner that ensures our survival. Environmental psychology, a new and emerging field, promises to provide insights that will enable us to live on earth in a sustainable way.

**2. The earth as a planet has been profoundly affected by life.** Human contributions to the visual landscape of the planet are everywhere: tall skyscrapers, intricate networks of highways and electrical lines/ engineered lakes/ and the vapor trails of high-flying jet planes are constant reminders of human presence. More subtle indicators are the changing quality of the chemical composition of the atmosphere, geologic changes in the earth’s crust, engineering changes in hydrologic processes and chemical changes in the waters that cover the planet.

Clearing of forests, plowing of land, black-topping of highways and parking lots not only affect the earth visually but change the amount of carbon dioxide in the atmosphere, the light reflecting and absorbing characteristics of the earth's surface and hydrologic drainage patterns. These changes in turn influence the rate of heating and cooling and therefore the temperature of the earth and its atmosphere. Weather and climate are thus influenced by the presence of humans. It is hoped that a more thorough understanding of the ways in which humans influence earth's natural processes will lead to more intelligent and life-preserving behaviors on their part.

**3. The outcomes of land use by humans tend to be cumulative and therefore we have obligations to ourselves as well as future generations to minimize their negative effects.** A number of changes in the environment are brought about by human habitation. For example, fall plowing, the development of sanitation systems and the building of shelters. These practices influence the earth immediately as well as having long-term impact. A dam built today not only supplies electrical energy for today's population but also influences the course of the waterways on which the dams are built, often in irreversible fashion. The conclusion of all this is that while humans have multiplied, their life's resources have shrunk and there is every indication that this trend is continuing. For example, the same conditions that helped to create the Sahara are expanding it southward. Every year two to three more square miles are lost to the drought and sands. The “Thar” Desert of India is advancing at the rate of about one-half mile annually along its entire perimeter. We have an obligation to ourselves and to future generations to see that this trend is halted, if not reversed. An understanding of environmental psychology represents a start toward reversing these trends.

**4. Sustained life on earth is a characteristic of eco system and not of individual organisms and populations.** No single organism, population or species is capable of both producing all its own food and completely recycling all of its own metabolic products. Green plants and light produce carbon dioxide, sugar and water; from sugar and inorganic compounds many organic compounds are manufactured including protein and woody tissue. But no plant can degrade the woody tissue developed in this fashion back to its inorganic compounds. This degradation requires other organisms such as bacteria and fungi. To complete the recycling of chemical elements from inorganic to organic and back to inorganic requires the use of several species. The smallest system capable of complete chemical recycling is known as ECO System. Humans are part of a very complex and delicately balanced ecosystem. Thus, to understand humans fully it is necessary to study them within the context of the ecosystems in which they survive. Environmental psychology is the field that attempts to develop an understanding of this interdependency. It should be clear by now that to comprehend fully the relationship of humans to their environment the student of the environment should be aware of contributions from a large number of disciplines. Environmental studies by their very nature are the domain of a generalist with a strong interdisciplinary interest. All environmental problems must be looked at from numerous perspectives so that a clear and total picture can be put together from the many pieces.

After studying the common assumptions of Environmental Psychology in detail, it is important to state the comprehensive definition of Environmental Psychology once again. Students must focus how these common assumptions are tightly packed in the following definition:

***A multi-disciplinary behavioral science, both basic and applied in orientation, whose foci are the systematic interrelationships between the physical and social environments and individual human behavior and experience.***

This definition mentions that Environmental Psychology is “Basic” and “Applied” in orientation. Let’s briefly look at these two terms:

- **Basic Behavioral Science.**

Basic behavioral science is a term that refers to the scientific study of human behavior and its underlying causes and consequences. Basic behavioral science aims to discover general laws and theories that explain and predict human behavior, as well as to identify the factors that influence and modify it. An example of basic behavioral science is the field of behavioral psychology, which studies how people learn, remember, and behave based on their interactions with the environment.

- **Applied Behavioral Science.**

Applied behavioral science is the use of scientific knowledge and methods from various behavioral science disciplines, such as psychology, sociology, anthropology, and economics, to solve practical problems or create new products or technologies that involve human behavior. An example of applied behavioral science is the field of applied behavior analysis (ABA), which involves applying behavioral principles and techniques in various settings, such as education, health care, criminal justice, and social services, to help individuals with behavioral or developmental challenges.

**Current Events Influencing Environmental Psychology**

If present trends continue: The world will become increasingly crowded, more polluted (Leonard, 1986; Peterson, 1987), less stable ecologically (Manabe & Wetherald, 1986), and more vulnerable to disruption (World Commission on Environment and Development, 1987). Supplies of drinking water will diminish drastically, and despite greater material output, the world's people will become poorer than they are today (Postel, 1985, 1986, 1987). The forests of the world will become increasingly shed as a result of the requirements of wood for building and burning (Bowander, 1987; Myers, 1984) and as a result of the increasing acidity of rainfall worldwide. The need for commercial land will increase and less land will be left behind for farming and cultivation. The following factors are important to be considered in this regard.

- **Population Trends.**
- **Resource Depletion and Environmental Degradation.**
- **Public Policy and the Environment.**
- **Human Behavior and the Environment.**

1. **Population trends**

According to the world population projections, if the current trends continue, the human population will reach 10 billion by 2030 and 30 billion by the end of the twenty-first century. These numbers are close to the estimated maximum number of people that the earth can support. However, some regions of the world have already surpassed the limit of their carrying capacity. The population in Sub-Saharan Africa and the Himalayan regions of Asia has exceeded the ability of their local environment to sustain life. The combination of overgrazing, fuel wood collection, and harmful farming practices has resulted in a series of ecological changes from open woodlands to scrub to fragile semi-arid land to useless weeds and finally to bare soil. The situation is worsened by the burning of animal dung and crop residues for heating and cooking, which deprives the croplands of organic matter, reduces their water retention, and lowers their fertility. In Bangladesh, Pakistan, and large parts of India, the high demand for basic needs by the large population has degraded the quality of the cropland, pasture, forests, and water resources that their livelihood relies on. Let's look at a few population statistics in Pakistan.

| | |
|---|---|
| **Population mid-2007** | 169, 300, 000 |
| **Births per 1000 population** | 31 |
| **Deaths per 1000 population** | 8 |
| **Rate of natural increase (percent)** | 2.3 % |
| **Projected Population, 2025** | 228, 800, 000 |
| **Projected Population, 2050** | 295, 000, 000 |

The dramatic increase in world population has been a driving force for the development of Environmental Psychology.

**In this lecture, only "Population Trends" was discussed in detail. The remaining three current events will be discussed in detail in the next lecture.**

**Lesson 3**
**Theories in Environmental Psychology**

We will begin lesson 3 by discussing the following current events influencing Environmental Psychology.

- Population Trends. (already discussed in detail in the lecture 2)
- Resource Depletion and Environmental Degradation.
- Public Policy and the Environment.
- Human Behavior and the Environment.

**2. Resource Depletion and Environmental Degradation.**
Other examples of serious deterioration in the earth's basic resources can be found throughout the world, including the industrialized nations. Erosion of agricultural soil, salinization of the highly productive irrigated farmland, and Lake Acidification are becoming more prevalent. Extensive loss of tropical forests and more or less permanent soil degradation have occurred throughout much of the Amazon River Basin (McHale, 1979). The world's deserts now make up 800 million hectares increasing in size.

Regional water shortages and deteriorating water quality which is already a problem in many parts of the world, are likely to worsen. The world water withdrawals are expected to increase by 200 to 300 percent in the next two decades, and a large proportion of it will be polluted due to the use of waterways as coolants and as waste carriers. The potential for human conflict over the use of freshwater reservoirs is heightened by the fact that out of the 200 major river basins of the world, 148 are shared by two countries and 52 are shared by three to ten countries. Long-standing difficulties over the shared rivers of Plata (Brazil and Argentina), Euphrates (Syria and Iraq), and Ganges (Bangladesh and India) have the potential to intensify as the need for additional freshwater occurs. Thus, the depletion of natural resources essential for survival has been another factor influencing the development of environmental psychology.

The industrialization, urbanization, and population growth of the planet have resulted in various negative consequences. Chemical and human waste is being produced at rates faster than we can safely dispose of, non-renewable resources are being consumed at increasing rates, plans for reestablishing renewable resources are shortsighted and the fate of our waterways, wildlife, climate, and perhaps the earth itself appears in jeopardy. The deterioration of the environment is thus another reason for the increased interest in environmental psychology.

Let's look at a few facts about population trends in Pakistan.

**Pakistan population trends and environmental indicators**

| | |
|---|---|
| $CO_2$ emissions per capita in 2002 | 0.7 metric tons |
| Natural habitat remaining (%) | 63 % |

*We have consumed 37% of our natural habitat we are left with.

**3. Public Policy and the Environment**

The problems in preserving the carrying capacity of the earth and sustaining the possibility of a decent life for human beings are indeed enormous and about to happen but they are just that—problems. Policy changes coupled with government business and individual actions can do much to alleviate many of them. Reforestation after cutting requires detoxification of chemical by-products before disposal as well as judicious soil management, thereof, the government has started to implement policies in this regard. Interest in energy and material conservation is growing, industrial and household recycling is becoming more prevalent and the need for family planning is becoming better understood. High-yield hybrids are continuously being introduced and methods for farming the seas are being developed. Public policy requires dependable scientific evidence to support its decisions. This has created a pressing need for the advancement of environmental psychology and has fostered its expansion as a field of study. However, inspiring change in people with regard to environmental literacy such as encouraging them to use personal bags for shopping rather than black plastic bags also requires economic resources. This is because we would have to provide an alternative to the black plastic bag business when our efforts will pull down their industry. This calls for a well-rounded environmental policy to be developed by the government and environmentalists combined.

**4. Human Behavior and the Environment**

Although these policy developments are positive steps, they are insufficient to address the increasing difficulties that humanity faces. Needed changes go far beyond the responsibility of one nation, and new initiatives must be taken if worsening poverty, environmental degradation, and international conflict are to be prevented. The solutions if they exist are complex and long-term and tied to the problems of poverty, injustice, and social conflict. The problems of the globe are human problems, many of which have been caused directly or indirectly by human presence. Advanced technology, while potentially a part of the solution to some of these problems, is also the cause of many of them. Because many of these problems are the result of human behavior, the psychologist, whose domain of interest is human behavior, has potentially a great deal to contribute in their resolution.

Environmental psychologists are very much attuned to the health of mother earth. They realize there are no quick fixes and that only through under-standing and changing human behavior can there be any hope of maintaining a habitable planet—that is, even though psychologists recognize that many of the problems are technological they also emphasize that the source of these problems is human behavior. Equally important is their realization that without an understanding of the mechanisms and laws that govern the life-sustaining processes of the earth, there is no hope that any changes in human behaviors or policies will have any medicinal effects.

Psychologists in this regard tend to focus on behavior modification. An example of behavior modification with reference to environmental psychology is the use of token economy to promote pro-environmental behaviors, such as recycling, saving energy, or using public transportation. A token economy is a system in which individuals earn tokens or points that can be exchanged for rewards. For instance, a school may implement a token economy to reward students for bringing reusable water bottles, turning off lights, or taking the bus. The tokens can then be used to buy items from the school store, such as pencils, stickers, or books. This way, the students are motivated to adopt more eco-friendly habits.

**Theories in Environmental Psychology**

Environmental psychology is an interdisciplinary field that benefits from the diverse viewpoints of various disciplines on a common phenomenon, which may result in less biased theories and more universal solutions. However, this diversity also poses challenges for the consistent use of knowledge and the formulation of coherent and comprehensive theories.

Let's begin by discussing four theories that come under the umbrella of Historical Influences.

**Historical Influences**

Thinking among environmental psychologists has been influenced by theories both within and outside of the discipline of psychology. Some of these theories are very:

(i) Broad in scope:

The scope of a theory means the kinds of problems that the theory addresses. Theories that are broad in scope try to understand numerous phenomena all at once. Their natural framework is so vast and so general in its nature that it can accommodate a bulk of information.

(ii) Focused in scope:

Theories with a narrow or focused scope specifically try to understand a particular thing. For example, a specialist in cell biology might conduct research on cells only while a generalist in biology might engage in surface-level research on a wide range of topics.

(iii) Empirical:

These are scientific in nature and are data-driven. For example, the theory of Natural selection by Charles Darwin which is supported by a lot of data driven research.

(iv) Non-Empirical:

These are general theories usually not backed up by data-driven research. They are based on logic and intuition. For example, those proposed by Plato in ancient philosophy.

We will review a number of them to provide the context for a consideration of current theories of environment-behavior relationships. These theories include:

- **Geographical Determinism**
- **Ecological Biology (theories)**
- **Behaviorism (behaviorist perspective or interactionism)**
- **Gestalt psychology**

**Geographical Determinism**

Some historians and some geographers have attempted to account for the rise and fall of entire civilizations on the basis of environmental characteristics. For example, Toynbee (1962) theorized that the environment (specifically, topography, climate, vegetation, availability of water, etc.) presents challenges to its inhabitants. Extreme environmental challenge leads to the destruction of a civilization, whereas too little challenge leads to stagnation of culture. Thus, Toynbee proposed that an intermediate level of environmental challenge enhances the development of civilizations, and extremely diminished or excessive levels are debilitating. The notion of environmental challenge and behavioral response, although rooted in the thinking of such geographical determinists, appears often in one form or another in various theories in environmental psychology. As one example Barry, Child, and Bacon (1959) have suggested that agricultural, non-nomadic cultures seem to emphasize responsibility, obedience, and compliance in child rearing practices, whereas nomadic cultures often emphasize independence and resourcefulness. These differences, they suggest, result from the fact that people who live and work together in organized non-mobile communities require more structured organization and therefore stress the importance of obedience and compliance. On the other hand, independence and resourcefulness are esteemed and taught by nomads in preparation to meet the changing and unpredictable demands of an environment confronted by them. Thus, the environment sets the stage for the development of cultures having the best chance of survival.

This implies that ghetto culture develops a set of skills in its residents that are most appropriate for the ghetto. It can be reasoned that someone who is not proficient in street fighting is unlikely to cope with the conditions of the streets. As we will discuss later in this text, some have extended this argument to other established environments that shape their residents in ways that are seemingly adaptive to those environments but are harmful in the broader context.

**Ecological Biology (theories)**

Environmental Psychology has been greatly influenced by the development of ecological theories, which examine the biological and sociological interrelations between organisms and their environment. Ecological science has shown that organisms are not isolated from their environment, but rather essential to it. This idea of organism-environment mutualism is present in many contemporary environment-behavior theories. The environment and its inhabitants are often treated as distinct elements, but their interdependence is undeniable. These various elements are considered to form a holistic system and any change in one element is expected to affect the whole system.

**We will discuss Behaviorism (behaviorist perspective or interactionism) in detail in the next lecture.**

**Lesson 4**
**Arousal Theories**

**We start this lecture in continuation of the last lecture by discussing Behaviorism (behaviorist perspective or interactionism) that comes under the influences of Historical Influences.**

- **Behaviorism (behaviorist perspective or interactionism)**

It comes from the psychology discipline and involves behaviorists' reaction to the failure of personality theories to account fully for human behavior. It is now generally accepted that considering both the environmental context in which behavior occurs and person variables (i.e., personality, dispositions, attitudes, etc.) leads to more accurate behavior predictions than measurement alone. This is central to most current theories of environmental behavior. Human beings are biological entities that share this environment along with other species. For example, Human beings and elephants have a long and complex history of coexistence, ranging from cultural reverence and mutual benefit to conflict and competition. (King et al., 2016). Human behavioral patterns are characterized by the attitude that they have. For example, if we consider why people use black plastic shopping bags despite knowing their negative consequences, we will conclude that this has to do with their personal preferences and attitudes. Top behaviorist of the 20th century, William James also said:

***"The greatest discovery of my generation is that human beings can alter their lives by altering the way they look at things"***

Let's try to understand the crisscross between environmental and personal factors in predicting human behavior by looking at the biography of Muhammad Ali Jinnah.

*Muhammad Ali Jinnah was born in Karachi on December 25, 1876. He was admitted to bar in England and then he returned to India and became an active supporter of The Indian National Congress. In 1913, he joined the Muslim League. In 1916, he was elected the president of the Muslim League, he was re-elected to this post again in 1920. He played an important role in bringing about the Lucknow Pact of 1916. In the Lucknow Pact, the Congress agreed to ensure a separate electorate for the Muslim community. He resigned from Congress in 1930 and channeled all his efforts towards the ends of the Muslim League. He headed the Muslim League and struggled until the creation of the Independent Pakistan. He was appointed the first Governor general of Pakistan.*

After reading this biography, ask yourself one question: Did Muhammad Ali Jinnah's personal characteristics make him the founder of Pakistan, or was it the result of environmental factors? This question will help you completely understand the role of heredity and environment in shaping human behavior.

- **Gestalt Psychology**

Gestalt psychology was developed primarily in Germany and concurrently with behaviorism. Gestalt psychologists were more concerned with perception and cognition than with overt behavior. Perception is how we make sense of our surroundings, cognition is how the information coming from the environment is processed in our brain, and overt behavior is how we react in response to our information processing. For example, a child might tell his mother that he wants to be a fighter pilot and start watching pilot movies and cartoons because of the kind of information he received and processed about fighter pilots. In the same manner, an employer might reject a very talented candidate for the job just due to his negative perception of him. The most important principle of this body of work was the objects, persons, and settings are perceived as a whole, which is greater than the sum of its parts. From the Gestalt point of view, behavior is rooted in cognitive processes; it is determined not by stimuli, but from the perception of those stimuli. For example, a man might react positively to a high traffic jam because of his positive mood due to his recent promotion, and a man who recently fought with his wife, due to his negative mood, might become angry due to even a minor delay caused by the traffic.

**Mini-Theories**

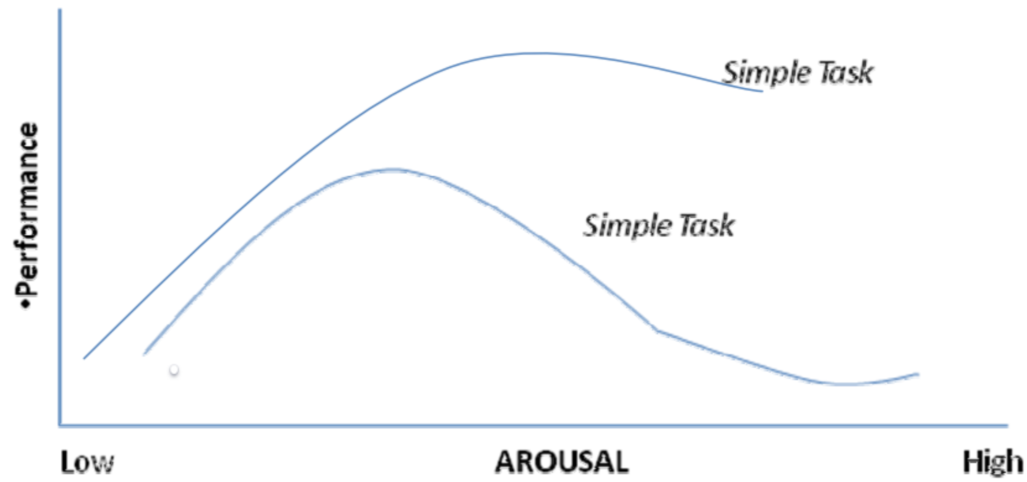
These approaches (the historical influences) tend to be rather broad in scope and lacking in empirical evidence. Each has its focus on convenience and no single perspective is satisfactory in accounting for the complexity of the environment-behavior relationship. No "grand theory" exists that can incorporate the distinctive contributions of each of these influences on environmental psychology. This is so for at least four reasons:
(1) There is not enough data available regarding environment-behavior relationships to lead to the kind of confidence needed for a unifying theory.
(2) The relationships that researchers have looked at are highly varied.
(3) The methods used are inconsistent.
(4) How variables have been measured has not always been compatible from one research setting to the next.

Even though well-articulated, all-encompassing theories are not available at this time, there are several "mini-theories," or mini-approaches, that have been used successfully in conceptualizing specific organism-environment relationships. **Included in these are the arousal approach, the stimulus load approach, the behavioral constraint approach, the adaptation level approach, the environmental stress approach, and the ecological approach.** Each of these can handle some, but not all, of the available data. Some are more useful in dealing with group behavior (the ecological approach), whereas others focus on the individual level of analysis (the stimulus load approach). Some find their greatest utility at the psycho-physiological level (the stress approach); others are useful for accounting for individual differences (the adaptation level approach). Before turning to an orientation that attempts simultaneously to embrace aspects of many of these approaches, a brief description of each of these approaches will be provided.

- **Arousal Theories**

Arousal theories have typically been concerned with the influence of arousal on performance. Generally, performance is maximized at intermediate levels of arousal but falls off as arousal is either increased or decreased. This relationship, sometimes referred to as an inverted-U relationship, has been shown to differ slightly depending on whether performance is measured on simple or complex tasks (see graph Figure) and is often referred to as the Yerkes-Dodson law. These relationships are consistent with other findings that humans tend to seek out intermediate levels of stimulation (Berlyne, 1974), and are reminiscent of Toynbee's assertion, referred to earlier that cultures only develop in environments that provide intermediate environmental challenges.

Performance is predicted to be optimal for both simple and complex tasks at intermediate levels of arousal. Arousal above the level leads to decrements in performance




Performance changes that vary curvilinear with temperature increases have also been shown. One explanation of these findings is that increases in ambient temperature lead to increases in arousal levels. Initially, the higher arousal leads to performance enhancement, but as it increases further, over-arousal occurs, causing performance decrements. Similarly, it has been shown that personal space invasions lead to increases in arousal (Middlemist, Knowles, & Matter, 1976) and performance decrements (Evans & Howard, 1972; McBride, King, & James; 1965). Additionally, increases in noise levels have been associated with changes in arousal and performance (see Evans & Cohen, 1987). Thus, several variables associated with arousal changes are related to performance changes, and performance has consistently been curvilinearly related to arousal.

Other theorists utilizing an arousal perspective have featured physiological responses to environmental stimulation. Changes in heart rate, blood pressure, respiration rate, galvanic skin response, and adrenaline secretion among others have been shown to occur with changes in the environment. Increased ambient temperature leads to blood vessel dilation, perspiration, increased heart rate, and in extreme conditions, lowered blood pressure and insufficient oxygen supplied to the brain. Personal space invasion has been linked to delayed onset and shorter duration of maturation for males. Noise exposure alters blood pressure, heart rhythm, and the flow of gastric juices to the stomach.

Neurologists such as Hebb (1972) have linked arousal with increased activity of the reticular activating system of the brain. Still, other theorists have equated arousal with changes in motor activity or with self-report of arousal. Berlyne (1974) has, for example, characterized arousal as lying on a continuum anchored at one end by sleep and at the other by excitement, and Mehrabian and Russell (1974) have identified arousal as a major component in people's affective responses to their environment.

Independent of the orientation taken concerning arousal, several consistencies are apparent:

1. Changes in arousal are associated with changes in the environment

2. Pleasant as well as unpleasant stimulation increases arousal — that is, room temperatures above 100 degrees Fahrenheit and loud, obnoxious noises influence arousal in ways similar to rollercoaster rides.
3. Changes in arousal lead people to seek information about their internal states (Schacter & Singer, 1962) as well as to seek information from others (Festinger, 1954)
4. People tend to evaluate moderate levels of arousal positively
5. Often great expenditures of energy are utilized by individuals to bring the environment to a level of moderate stimulation.

- **Factors Impacting Arousal Level**

Furthering the point various researchers have tried to identify the impact of different environmental factors on the arousal level:

**I. Temperature:** An increase in ambient temperatures leads to an increase in arousal level. Usually, higher arousal leads to performance enhancement but as it increases further over-arousal occurs causing performance decrease.

**II. Personal Space Invasion:** An optimal personal space invasion can increase performance while too much invasion can cause a decrease in performance. For this reason, cubicles are designed in offices in such a way that colleagues are clustered together yet have a separate working space. This provides an optimal invasion of personal space leading to increased arousal (due to collaboration, relative performance assessment, and comparison) which ultimately leads to increased performance.

**III. Noise Level:** An optimal noise level can increase arousal and performance while too much noise can decrease arousal and performance. For example, you might be able to work better while sitting in a room while occasionally chatting with your colleagues (due to optimal noise level) and you might not be able to work appropriately in a room where there is too much chatter and loud music because noise level is not optimal.

- **Physiological Response to Environmental Stimulation**

Changes in heart rate, blood pressure, respiration rate, galvanic skin response, and adrenaline secretions have been shown to occur with changes in the environment. Every human carry 12 British thermal units of electricity, which can create fluctuations in the body's ecosystem/metabolism or equilibrium when it comes into contact with environmental stimuli.

**I. Temperature**

An increase in ambient temperature leads to the following physiological changes:
• Blood vessel dilation • Pupil dilation • perspiration • Increased heart rate.

**II. Extreme conditions**

Extreme environmental conditions such as food, water, or nutritional scarcity can lead to:
• Lowered blood pressure • Insufficient oxygen reaching the brain due to which a person might collapse.

**III. Invasion in personal space has been linked to**

• Delayed onset and shorter duration of micturition for males.

**IV. Exposure to Noise:**

Extreme noise may:
• Alters blood pressure, heart rhythm, and flow of gastric juices to the stomach which might lead to acidity.

- **Arousal and Nervous System:**

Arousal is linked with increased activity of the reticular activating system of the brain.

- ✓ For arousal theorists, consistency and break inconsistencies are also very important. For example, in a park, we might not notice people engaging in exercise and laughter because that's what everyone is doing but if we see a man crying in the same park, it will be something unusual and inconsistent with what was happening in the environment. It will immediately grab our attention and increase our arousal and ultimately behavior.
- ✓ Changes in arousal are associated with changes in the environment. Pleasant as well as unpleasant stimulation increases arousal, either negative or positive. i.e. room temperatures above 100 degrees might stimulate negative arousal, while a 25-degree temperature may stimulate optimal arousal.
- ✓ Changes in arousal lead people to seek information about their internal states. For example, during times of high arousal and stress, people might experience hold and cold body flashes. This might make them want to inquire about their blood pressure or sugar levels.
- ✓ People tend to associate moderate levels of arousal positively. This is why the interior of many offices is designed to be calm and cool so that people experience positive arousal and ultimately perform better at work. Great expenditure of energy is utilized by individuals to bring such specific environments to a moderate level of stimulation.

## Lesson 5
## Stimulus Load, Behavioral Constraint, and Adaptative Level Theories

- **Stimulus Load Theories**

Central to stimulus load theories is the notion that humans have a limited capacity to process information. When inputs exceed that capacity, people tend to ignore some inputs and devote more attention to others (Cohen, 1978). These theories account for responses to environmental stimulation in terms of the organism's momentary capacity to attend to and deal with salient features of its environment. Generally, stimuli most important to the task at hand are allocated as much attention as needed and less important stimuli are ignored. Our attention is basically like a switch. We can control our attentional capacity according to the demands that we face from the environment. The benefit of this quality of attention is that humans can meet the changing demands of the environment where whereas the downside of this quality of attention is that it becomes limited when environmental demands are too high.

**Example**

For example, while driving during rush-hour traffic a great deal of attention is paid to the cars, trucks, buses, and road signs around us, and less attention is paid to the commentator on the car radio, the kids in the back seat, and the clouds in the sky. If the less important stimuli tend to interfere with the task at hand, then ignoring them will enhance performance, (e.g., ignoring the children's fighting will make you a better and safer rush hour driver. If, however, the less important stimuli are important to the task at hand, then performance will not be optimal; for example, ignoring the road signs because you are attending to the more important trucks, cars, etc., may lead you thirty miles out of your way in getting home (Figure below).

*While driving during rush hour traffic a great deal of attention is paid to the cars, trucks, buses, and road signs and less attention is paid to the commuter or the car radio, the kids in the back seat, and the clouds in the sky.*


Sometimes the organism's capacity to deal with the environment is overtaxed or even depleted. When this occurs only the most important information is attended to, with all other information filtered out. Once attentional capacities have been depleted even small demands for attention can be draining. Thus, behavioral aftereffects including errors in judgment, decreased tolerance for frustration, ignoring others in need of help, and the like, can be accounted for by these theories. For example, the exhausted rush hour driver eventually might reach the point where he or she doesn't notice the traffic light turn from red to green (or worse yet, from green to yellow to red), even though this is a very important stimulus. Additionally, decreased tolerance for frustration may lead to "Laying on the horn" or "lane hopping" and motorists in the break-down lane may be ignored, if not looked upon with dislike.

Read the following explanation given by an athlete about how he tries to limit the focus of his attention on the track and you will understand how attention functions in the real world.

## ATTENTION IN THE REAL WORLD

***"I have learned to cut out all unnecessary thoughts... on the track. I simply concentrate. I concentrate on the tangible - on the track, on the race, on the blocks, on the things I have to do. The crowd fades away and the other athletes disappear and now it's just me and this one lane"***

Stimulus load theories are also able to account for behavioral effects in stimulus-deprived environments (e.g., certain behaviors occurring aboard submarines and in prisons). That is, this approach suggests that understimulation can be just as aversive as overstimulation. So-called cabin fever resulting from monotonous living conditions can also be seen as the result of understimulation. Wohlwill (1966) has argued that environments should be depicted in terms of measurements applied to the dimensions of **intensity, novelty, complexity, temporal variation, surprisingness, and incongruity**, all of which contribute to stimulus load. Subsequent behaviors can then be related to the stimulus properties of environments in systematic and comparable ways. Let's look at these dimensions briefly one by one.

**(i) Intensity:**

It means the range of a stimulus for example if the music is loud it means that it is of high intensity and if the music is of less volume it means that it is of low intensity. We will experience more stimulus load if the music is high and versatile.

**(ii) Novelty:**

It is the uniqueness of a stimulus. For example, if there are 100 channels on the TV, often, the most unique one will grab our attention and contribute to stimulus load.

**(iii) Complexity:**

It is how complicated or multi-faceted a stimulus is. For example, driving through rush hour. Multiple cars, noises, and other traffic-related stimuli come into play and contribute to stimulus load.

**(iv) Temporal Variation:**

This is specifically related to changes in noise level. For example, imagine that you are sitting on a bus, and the AC of that bus is making a repetitive sound. In the beginning, that voice might irritate you but slowly you will get used to it and will no longer be able to notice it as it will get mixed up with other sounds. But suddenly the bus undergoes a jerk and the AC voice gets fixed on its own, then you will automatically feel that something is missing from the environment and you will finally notice the change. These noise variations also contribute to stimulus load.

**(v) Surprisingness:**

This is a stimulus that is quite rare so it invokes the emotion of surprise in a person. For example, imagine that you come to know that a new restaurant has opened in town and that restaurant is made up of ice and chocolate. You will be extremely surprised and feel motivated to visit that restaurant. Such rare stimuli also contribute to stimulus load.

**(vi) Incongruity:**

This is a stimulus that holds together a unique pairing of two different stimuli that normally would not exist together. For example, if you see a three-year-old baby sitting in a cage with a lion, you will feel strange and question yourself how this strange situation came into being. Such unique pairings of stimuli also contribute to stimulus load.

**Stimulus to Behavior-A relationship**

After studying these stimulus characteristics, you might be wondering what is the relationship between stimuli and behavior. This automatically brings us to the implications of stimulus load theories. Since stimuli impact human behavior, we have a lot of control that we can exercise over human behavior by altering the stimuli around us. For example, offices and schools can be designed in such a way that they foster positive and productive behaviors.

- **Behavior Constraint Theories**

Behavior constraint theories focus on the real, or perceived, limitations imposed on the organism by the environment. An example of a real threat would be fear of failing in exams while an example of a perceived threat is a student feels that he will definitely fail, even after all the preparation. According to these theories, the environment can prevent, interfere with, or limit the behaviors of its inhabitants (Rodin & Baum, 1978; Stokols, 1978). Humans and the environment can control each other. For example, a student who is traveling on the train for the first time might feel that he is in danger. However, he might try to negotiate with the environment by making some friends or reading the travel guide for ease and certainty. Friday afternoon rush-hour traffic interferes with rapid commuting; loud, intermittent noises limit effective communication (for example a teacher might not be able to talk to students properly if there is too much noise in the classroom.); over-regimentation (patients being rotated in far-off departments in hospitals for collaborative treatment) in hospitals can interfere with recovery, excessively high ambient temperatures prevent extreme physical exertion, and extremely cold temperatures limit finger handiness. In a sense, these theories deal with situations where persons either lose some degree of control over their environment, or they perceive that they have.

Brehm and Brehm (1981) assert that when we feel that we have lost control over the environment, we first experience discomfort and then attempt to reassert our control. They label this phenomenon as **Psychological Reactance.** (which is the reaction of the person to regain the lost control over the environment. For example, a teacher might first feel discomfort in a noisy class but then she will try to apply different classroom strategies to control the noise level). If the rush-hour traffic interferes with getting home in a timely fashion, we may leave work early, or find alternate, less-congested routes. Loud, intermittent noises may be dealt with by removing their source or by changing environments. Extreme temperatures are handled by adjusting the thermostat. All that is needed is for individuals to perceive that they have lost some degree of control, or for that matter, to anticipate the loss of control, and reactance will occur. If repeated attempts to regain control are unsuccessful, **learned helplessness** may develop (Seligman, 1975). People begin to feel as though their behavior does not affect the environment. They begin to believe they no longer control their destiny, and that what happens to them is out of their control. These feelings can eventually lead to clinical depression, and in the most extreme form can lead people to give up on life, and die.

On the opposite side of the coin, perceived control over one's environment (even when real control does not exist, or is not used) can alleviate the negative outcomes that the environment might otherwise bring about. Perceived control over noise (Glass & Singer, 1972), overcrowding (Langer & Saegert, 1977), and over one's daily affairs (Langer & Rodin, 1976) has been shown to influence positively a variety of behavioral responses. For example, residents of a nursing home who were given greater control over, and responsibility for their well-being displayed enhanced mood and greater activity in comparison with residents who were not given control. Similarly, people who had control over the thermostats in their working and living environments reported fewer health complaints during the winter months than those who did not have control. These results occurred even though they did not manipulate the thermostats and kept their environments at ambient temperatures similar to those without control (Veitch, 1976). Behavior constraint theories thus emphasize those factors (physical as well as psychological; real as well as imaginary) associated with the environment that limits human action.

- **Adaptation-Level Theories**

Adaptation theories are similar to stimulus load theories in that an intermediate level of stimulation is postulated to optimize behavior. Excessive stimulation as well as too little stimulation is hypothesized to have deleterious effects on emotions and behaviors. For example, a dull home interior might make a person feel depressed and a highly stimulating interior might make a person restless. An optimal and calm home environment will make people feel comfortable. Major proponents of this position include Helson (1964) and Wohlwill (1974). While all environmental psychologists emphasize the interrelationship of humans to their environment, adaptation-level theorists speak specifically of two processes that make up this relationship**—the processes of adaptation and adjustment.** Organisms either adapt (i.e., change their response to the environment) or they adjust, (i.e., change the environment with which they are interacting). For example, if political figures are in prison and are deprived of the number of social interactions that they used to have in their daily life, they might adapt and adjust by calming down and focusing their energies on reading and writing about their political journeys. Adaptations to decreases in ambient temperature include piloerection (hair on the body standing up or what is commonly called getting "goose pimples"), muscle rigidity, increased motor activity, and vasoconstriction; adjustments include throwing another log on the fire or turning up the thermostat. Either process brings the organism back to equilibrium with its environment.

This perspective values the recognition of personal adaptation levels, meaning how much excitement or activity a person is used to and looks for in their surroundings. This idea helps us understand why two people might react differently to the same situation. For instance, someone who enjoys lots of excitement might find a noisy party fun, but it could be too much for someone who likes things calmer. In the same way, while some might enjoy the busy feeling of shopping right before Christmas, others might not like it when the store gets too crowded. These personal preferences for excitement can lead to very different choices. A person who likes a lot of excitement might go to lively parties, while someone who doesn't might stay away or find a quiet spot if they do go. We've all noticed people who are the center of attention at a party and those who prefer to keep to themselves. The way they act can often be linked to their comfort with different levels of excitement.

**Lesson 6**
**Ecological Theories and Comparison of Theories**

**We start this lecture by discussing environmental stress theories in detail**

**Environmental Stress Theories**
Stress theories emphasize the mediating role of physiology, emotion, and cognition in the organism-environment interaction. Environmental features are seen as influencing, through the senses, the organism, causing a stress response when environmental features exceed some optimal level. For example, initially, a guitar is at rest. But when you start to play the guitar, your finger movement causes stress in the strings. This can be imagined as a disturbance in the homeostasis level (or in other words, the “rest” level) of the guitar. The organism then responds in such a way as to alleviate the stress. For example, before feeling hungry, your body is steady (homeostasis is balanced). When you feel hungry the homeostasis of your body is disturbed (i.e., release of gastric juices in the stomach, body weakness). To alleviate this stress, you look for food. After eating the food again your homeostasis comes back to normal. Part of the stress response is automatic. Initially, there is an alarm reaction to the stressor, wherein various physiological processes are altered. Resistance then follows as the organism actively attempts to cope with the stressor. Finally, as coping resources are depleted, a state of exhaustion sets in (Selye, 1956).

**Types of stress:**

**There are two main types of stress:**

1. **Eustress:**

It is a positive force that provides a sense of well-being. This stress is up to an ideal degree. For example, a girl is participating in a college debate. She feels very anxious and starts imagining/visualizing again and again that she has to go in front of people. This image disturbs her homeostasis. Her physiology, emotions, and cognitions are disturbed and she enters a state of an unbalanced mind. Suddenly she starts to think of strategies through which she can well prepare for her public speaking assignment. It is Eustress that motivates her to prepare for her task. After good preparation and a successful debate her unbalanced mind again becomes balanced and homeostasis is restored.

2. **Distress:**

It is a negative force that can lead to a decline in well-being. This stress is up to an unsafe degree. For example, the girl in the above example goes to the college debate and faints in front of people, runs away from the hall, or starts experiencing a panic attack in the hall. It is distress that leads to her failure in the college debate.

Let's look at these reactions to stress proposed by Hans Selye (1907-1982) in greater detail:

1. **Alarm Reaction:** This is the first reaction to stress when an organism first acknowledges or perceives stress in the environment. It is the mobilization of resources to confront the stress. For example, a child is happily playing in the park. Suddenly he sees a dog. His first reaction (Alarm reaction) to this stress will be that he will start feeling nervous. This

nervousness will prepare his body to deal with the stress and his physiological/mental resources will come into play.

2. **Resistance:** This is the stage where an individual decides his strategy for dealing with the stress. The threat to the environment will lead to more energy consumption. To acquire more energy the physiological reserves of energy will be used and energy consumption will keep on increasing according to the demands of the stressful situation. For example, a man suddenly hears a loud sound. He comes to know that a loin has escaped his cage and sees it coming after him. His resistance reaction will be that he will start to run while continuously consuming his energy by sweating, breathing heavily, muscle movement, or consumption of body glucose.

3. **Exhaustion:** Now imagine that the lion follows the man for 5 hours. This will lead him towards the exhaustion stage. Exhaustion is when stress continues for long periods. Energy is used but it is not refilled. A perfect example would be war/combat between two countries that has continued for 2 years. All the soldiers have constantly faced life-threatening situations for two years and they are being exposed to extreme stress. There is a shortage of food, manpower, and ammunition. So, their resources are not re-filled. At one point they will experience exhaustion or you can say that they will enter into survival mode.

**How do we know that we are experiencing stress?**

**Stress has two common indicators:**

1. **Physiological:** Physiological indicators of stress are critical markers for assessing the body's response to stressors. According to a review article, common physiological indicators include changes in blood pressure, heart rate, and levels of blood fats such as cholesterol and triglycerides (Iqbal et al., 2021). Additionally, variations in blood sugar levels, particularly in the evening, and alterations in appetite, which can contribute to weight gain, are also noted as physiological responses to stress (The Journal of Clinical Endocrinology & Metabolism, 2021).

2. **Emotional:** Emotional indicators of stress are key to understanding how individuals react to stressors. Research has shown that common emotional responses to stress include feelings of irritability, anxiety, and depression (Katana et al., 2019). These emotional states can significantly impact daily functioning and are associated with various mental health outcomes (Sarrionandia et al., 2018).

## Ecological Theories

Central to the thinking of ecological theorists (Barker, 1963, 1968) is the notion of organism-environment fit. An example of a career would be perfect here. A person with excellent analytical, organizational, and mathematical skills will be fit for the job of a banker or a software engineer. Another example would be the selection of a partner for marriage. Two people courting each other might come to know that they are a perfect fit for each other because they have similar values, outlooks, and understanding in life. Environments are designed or grow to accommodate certain behaviors. Behavior settings, as Barker termed them, are evaluated in terms of the goodness of fit between the interdependent environmental features and the behaviors that take place. For example, a schoolyard, a church, a classroom, an office, or an entire business organization might be considered a behavioral setting; each would then be evaluated in terms of how suitable it is for the play behavior of children, how well it accommodates the religious rituals, or how well it serves the functions of the business.

While any number of behaviors can occur within any physical setting, cultural purpose is defined by the interdependency between standing patterns of behavior and the physical setting. Standing patterns of behavior represent the collective behavior of the group rather than just individual behavior. The standing pattern of behavior in a classroom would include lecturing, listening, observing, sitting, taking notes, asking questions, and taking tests; the physical environment of this behavior setting would include the room and such accessories as a lectern, chairs, chalkboard, microphone, overhead projector, and slide screen. Because this standing pattern of behavior occurs primarily in this behavior setting, social ecologists would suggest that knowing about the setting helps us predict what will occur in it. Once individuals making up this classroom leave this physical setting, most of the physical characteristics of the environment remain unchanged, but the behaviors likely will change dramatically in other words, the students will move to a new physical environment eliciting a different standing pattern of behavior.

**Examples:**

1. The initial concept kept in mind when designing prisons is that this is a place where people are confined as a punishment for crimes. Prisons are also built away from housing and commercial buildings so that criminals are kept away from general people to prevent them from harm. However, a lot can change if we start looking at prisons in a different manner. Prisons must not only be considered as “punishment houses” but also as “correctional facilities”. Criminals are people with psychological problems due to which they commit different crimes. It was us who had a wrong perception of them. They are not to be hated. This viewpoint about prisons and prisoners may lead us to change the total structure of prisons. Physical settings can be designed in such a way that different standing patterns of behaviors can be produced in them. Hospitals can be built inside prisons to provide prisoners with complete care. Prisons can organize regular psycho-education lectures, exercise sessions, career guidance, and counseling as well as on-prison working opportunities for prisoners. All this might help them become better citizens after they are released from prison.

2. An environment conducive to agitation and crime will only lead to agitation and crime. Imagine a crowded traffic square of a South Asian country where there is a lot of noise, smoke, chaos, and traffic. Ask yourself. In this environment, there will be more crime or less crime/ more agitation or less agitation. There will be more agitation which will lead to increased frustration and increased chances of crime.

**We will continue with additional details regarding Ecological Theories in the next lecture.**

**Lesson 7**
**The present framework and future directions in Environmental Psychology (I)**

**We will begin this lesson by continuing the topic of Ecological theories**

**Ecological Theories: undermanning (understaffing)/ overmanning (overstaffing)**
According to Barker and Gump (see Gump, 1987), if the number of individuals in a setting falls below some minimum, some or all of the inhabitants must take on more than their share of roles if the behavior setting is to be maintained. **This condition is termed understaffing**. In an environmental context, understaffing is when there are too many resources/duties in the environment and a lesser number of organisms. For example, the college roommate of one of your authors went to a very small high school, which in many ways exemplifies an understaffed setting. This roommate played football and, like most high schoolers, played before fans on Friday night. Also, like most high schools, there was a high school band that performed at half-time. Your author's roommate, however, was also the best trumpet player in the school; so, at half-time, while the rest of the football players were obtaining instructions as to what they should be doing the second half, he was out marching with the band. In understaffed settings, inhabitants often have to assume a variety of roles.

If the number of participants in a setting exceeds, the capacity for that setting, **then the setting is considered overstaffed.** In an environmental context, overstaffing is when there are fewer resources and too many organisms. More or less the population in a geographical environment disturbs the ecosystem. It causes a positive or a negative impact on the standing pattern of behavior. Too many swimmers waiting to get into the pool on a hot summer afternoon, commuters on the five o'clock train, football fans at the Super Bowl, and shoppers in department stores at Christmas time might all represent overstaffed settings.

**Comparison of Theories**

| Theories | Approach |
|---|---|
| Arousal | General |
| Stimulus Load | Specific |
| Adaptation Level | Specific |
| Stress Approach | General |
| Behavioral Constraint | Specific |
| Ecological Approach | General |

**** General in scope:** The scope of a theory means the kinds of problems that the theory addresses. Theories that are broad in scope try to understand numerous phenomena all at once. Their natural framework is so vast and so general that it can accommodate a bulk of information. For example, generalists in biology might engage in surface-level research on a wide range of topics.
**** Specific in scope:** Theories with a narrow or focused scope specifically try to understand a particular thing. For example, a specialist in cell biology might conduct research on cells only.

In the comparison of theories, we discuss what is the area of prediction of these theories. Whatever these theories address they can predict something about those behavioral patterns. We can use this predictive ability to exert control over our environment. Now let's look at the key statements of these theories and their areas of prediction.

| **Theory** | **Key Statement** | **Area of prediction** |
|---|---|---|
| The Arousal Approach | The arousal approach is the most general in identifying physiological and affective mediators of environment-behavior relationships | Increases or decreases in stimulation produce corresponding changes in physiological and psychological arousal and, in turn, produce predictable variations in behaviors such as task performance and aggression. |
| The Stimulus Load Approach | The stimulus load approach is synaptic, focusing on cognitive information-processing abilities. | Yields predictions regarding the social/behavioral consequences of over/under-stimulation—excessive attentional demands have differential effects on the performance of primary versus secondary tasks and the likelihood of attending to various social stimuli. |
| The adaptation level approaches | An intermediate level of stimulation is postulated to optimize behavior. Excessive stimulation as well as too little stimulation is hypothesized to have harmful effects on emotions and behaviors | Predicts consequences of a particular stimulation level depend on the specific level to which a given individual has adapted. |
| The Stress approach | The stress approach incorporates elements of all of the above. That is, stress can be characterized in terms of objective physical and social environmental conditions that deviate from some optimal level (e.g., a noisy and crowded subway) and are thus potentially disruptive to human functioning. | The approach has the further advantage of predicting behavioral coping and the consequences of adaptive and maladaptive coping mechanisms. |
| The Behavior Constraint Approach | The behavioral constraint approach is the most limited of all in scope, (i.e., it is primarily useful in situations where the | When such conditions do exist, the concepts of reactance and learned helplessness yield useful predictions of behavioral responses to such conditions. |

| | | |
|---|---|---|
| | perception of loss of control or threats to control are present). | |
| The Ecological Approach | The ecological approach has the broadest scope with the concept of behavior setting, and as such is a useful descriptive approach to understanding the behaviors of large numbers of people in different settings. | How behavior settings impact behavior. however, this generality limits the approach's ability to account for individual differences in the behavior setting. |

.

**How these theories help in real life: The case of Amir**

After a comparative analysis of these theories let's look at one real-life example to understand how they may be applied in real life.

***Let's suppose that there is a young man named Amir. He is 20 years old and recently got admission to a city university. He then moves to the city, leaving behind his village. Upon his arrival, he discovers that the university is very big. The city is also very large and commercialized, with heavy traffic jams, large shopping malls, parks, and open markets, and is densely populated. He finds out that there is a lot of distance between his residence and the university. To meet his living expenses, he also has to work a part-time job in the evening. He wakes up early in the morning, does breakfast at a local café, and takes on different buses to reach his university. After a hectic schedule of classes, he then again takes different buses to reach his workplace in the evening. After his work, he again takes different buses to reach back home where he barely finds time to have dinner. He then has to study and complete his assignments. After which he is left with very little time to sleep. The next day this routine repeats over and over again. Amir finds the entire environment very confusing, however, he keeps going on with the routine due to his determination towards his studies. But after six months Amir starts to experience headaches, nausea, and vertigo. He starts to dream of his village where he lived a happy life and felt close and connected to people and nature around him. He then decides to move back to his village.***

After reading the case of Amir, keep the following questions in your mind:

1. How would you assess Amir's life circumstances? Happy or Stressful?
2. What features of his physical and social environment influence the quality of his life? Is it a positive influence or a negative influence?
3. What might you infer about his personality and thought processes that may interact with the environment to influence his experiences? Is it a positive inference or a negative inference?
4. Can you apply the concepts of arousal, stimulus load, behavior constraint, adaptation level, and behavior setting to account for Amir's thoughts, feelings, and behaviors?

Just think of these questions from the perspective of an Environmental Psychologist and try to understand how the theories of environmental psychology form the basis of all research surrounding it.

**The present framework and future directions in Environmental Psychology**

- **The main purpose of the present framework:** The present framework for understanding organism-environment relationships.
- **Origin:** It stems from several theoretical positions and databases established within the domains of general and social psychology, as well as from the newly emerging area of environmental psychology. It is developed from theories emphasizing the affective components of the human experience, where overt behaviors, characterized as approach or avoidance responses, are seen as being mediated by the emotion-arousing properties of the environment.
- **Target:** To bring different theories and databases under the umbrella of a single model.
- **Specific focus:** Specifically, environmental factors (e.g., density, personal space, noise, temperature, etc.) will be postulated to influence individual affective states, which in turn will be asserted to influence overt behavior. For example, how much rise in ambient temperature gives how much arousal.

**Following are seven postulates relevant to the present and future framework of Environmental Psychology.**

**First postulate:**
A. The environment consists of both physical and social variables existing in reciprocal relationships with behavior.
B. The physical and social environments can and should be characterized in terms of measurements applied to their salient characteristics.

**Second postulate:**
A. The environment, physical as well as social, typically exerts a steady-state influence on the behavior of its inhabitants.
B. When the measured values of the environment's most salient characteristics undergo dramatic change, the environment's influence on behavior can no longer be characterized as a steady state.
C. Disruptions of steady state occur when present perceptions do not correspond to the desired or expected level of social and physical stimulation.

**Third postulate:**
The influence exerted by the environment is indirect i.e. Environment acts to influence people's emotional states, which in turn mediate their overt behavior.

**Fourth postulate:**

Environment-evoked emotions are best depicted in terms of three distinct dimensions: pleasure – displeasure, degree of arousal, dominance –submissiveness.

**Fifth postulate:**

The environment–influenced behavior of the individual is dependent upon the extent and the configuration of the dimensions of emotions aroused.

**Sixth postulate:**

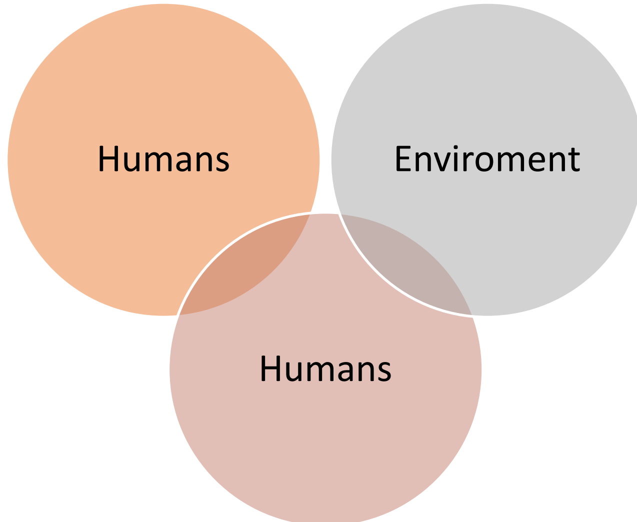
Behavior is goal-directed. That is to say behavior in an environmental setting is performed to increase or decrease the pleasure, degree of arousal, or dominance elicited by changes in the environment in an attempt to return the environmental influence to a steady state.

**Seventh postulate:**

When the goal-directed behavior is ineffective or when great expenditures of energy are required to maintain a steady state, the environment can be considered pathological and a disruptive influence on human functioning. **Let's now look at all of these seven postulates one by one in detail.**

**First postulate:**

**1A: The environment consists of both physical and social variables existing in reciprocal relationships with behavior.** Attempts to understand human/environment relationships that have focused exclusively on either the human components or the environmental aspects have not met with great success. Humans are not only influenced by their environment, but through their behavior, they also alter that environment. The altered environment in turn produces subtle changes in the environmental inhabitant and its behaviors; these behaviors again produce subtle changes in the environment, and so on, indefinitely. For example, the classroom behaviors of teachers and students alike are influenced by such physical properties of the environment as room temperature, chalkboard space, windows (or the lack thereof), the arrangement of desks, available light, and ambient noise level. However, the total classroom environment also includes social and demographic variables (e.g., the age and gender of classmates, the friendship groups, the type of activity being engaged in, and others). Each one of these factors, singly and collectively, influences the behavior that is enacted in this setting. By the same token, the resultant behavior changes the nature of that setting. For example, to enhance discussion, chairs may be moved, thus changing the environment; these changes may lead to changes in friendship groups or to a change in the ambient noise level. Teachers may now have to talk louder, perhaps also changing the tone of their voices, making them appear grumpy; this perception may in turn lead students to avoid the space around the teacher or may keep them from asking questions. Thus, the environment and its inhabitants never stay the same; each is constantly changing as a result of its interactions with the other. Any attempt to understand these relationships, therefore, requires a systematic approach emphasizing the bidirectional, often symbiotic, nature of this interaction.




**1B. The physical environment can and should be characterized in terms of measurement applied to its salient characteristics.** It is an obvious scientific advantage to be able to specify relationships between or among variables in precise mathematical terms. The symbolic representation of the effect x has on y is not as potentially ambiguous if x and y have properties that can be measured by agreed-upon and reliable techniques. Further, the symbolic representation of the effect can assume precise mathematical properties that permit greater precision in prediction. Thus, ambiguity in the discussion of the relationships between variables is reduced by precise measurement.

Statements like "high ambient temperatures may exert a detrimental influence on learning" carry global, but not precise, meaning. How "high" must the temperature be to exert this influence? Do variations in relative humidity combine with temperature differences to produce different effects? Does the clothing the person is wearing moderate these effects? (Or for that matter, determine what temperature is designated as "high"?) Is the length of time of exposure related to these effects? What is meant by the term "learning"? Does it mean something other than "performance”? Arousal levels are measured by self-report inventories as well as physiological arousal readings i.e., heart rate, blood pressure, or blood sugar level can be measured by some instruments. However, the problem here lies in the fact that the cause of these readings remains unknown. We do not know whether the rise in blood pressure is due to the environmental stimuli that we are studying or due to the personal health factors of an individual.

**Second postulate:**

**2A: The environment, physical as well as social, typically exerts a steady-state influence on the behavior of its inhabitants.** This assertion is not unlike the claim of Ittleson, Proshansky, Rivlin, and Winkel (1974) that the environment frequently operates below the level of awareness. The environment is seen as being taken for granted—we operate within an environmental context without paying much attention to it. "Normative" or "normal" behavior occurs in these circumstances. We behave according to the cultural purpose of the setting. However, if the environment changes (or we change environments), we become aware of it because it is at that point that we must consciously begin to adapt or adjust.

Roger Barker's (1968) notion of "behavior settings" is also congruent with this assertion. The arrangement of chairs at a table (Sommer, 1974) or desks in a classroom (Sanders, 1958), for example, have very powerful but non-conscious effects on their users; in a very real sense, we inherit the use of space. Thus, the environment influences our behavior even when its physical and social characteristics are within a normal range, but under these conditions, we are typically unaware of that influence. Thus, you were probably not aware of the fact that on your first day in the room in which this class meets, your behavior of note-taking vs. ballroom dancing was determined, in part, by the setting itself. Nor did you probably give much thought to the fact that you were dancing instead of taking notes the last time you attended a nightclub.

**2B. When the measured values of the environment's most salient characteristics undergo dramatic change, the environmental influence can no longer be characterized as a steady state.** Helson's (1964) adaptation-level theory represents a general framework for the study of diverse responses to any set of stimuli varying along some dimension. Put briefly, Helson maintains that for any specified dimension of stimulus variation, individuals establish an adaptation level (AL; a preferred or expected level of stimulation), which determines their judgmental or evaluative response to a given stimulus located on that dimension. Deviations from AL in either direction (i.e., either increases or decreases in stimulation level) are evaluated positively within a certain range, but beyond these boundaries' changes are experienced as "unpleasant." There are also individual differences in levels of stimulation. For a person living in Jacobabad, a temperature of 35C might be normal because his body has adapted to the temperature of Jacobabad, and for a person living in Murree, a temperature of even 30C might be very high because his body has adapted to the temperature of Murree.

When environmental-stimulus properties change to the extent that they have exceeded the AL boundaries, the individual experiences "unpleasantness," and the influence of the environment is no longer steady-state. It is only at this point that changes in the normative modes of behavior can be expected, that the regularized, routinized behavioral-setting influence begins to break down.

If, as you sit reading these pages, your phone rings instead of the one four doors down the hall, or someone walks into the room wearing an unusual perfume (or no deodorant), or the room temperature increases to 85 degrees Fahrenheit or is reduced to 60 degrees Fahrenheit, or there is a knock on your door, then the environment no longer can be characterized as steady-state. Your response to the environment is now conscious and deliberate. You get up to answer the phone or the door; you look up to see who is wearing the unusual perfume; you open a window or turn up the thermostat (or put on or take off clothes) to return to a condition of thermal comfort. In other words, you become aware of your environment and begin adapting to it or making adjustments to it to achieve a new equilibrium.

**2C. Disruptions from steady-state occur when present perceptions do not correspond to desired or expected levels of physical and social stimulation.** Disruptions in steady-state influence are brought about by the processes of sensation, perception, and cognition. Sensation involves those processes by which the world can be known to the perceiver. These processes typically involve the following modes: touch, smell, taste, vision, and hearing. We know silk by touching it, rotten eggs by smelling it, an orange by tasting it, and so forth. Another example would be two students. For a careless student, 5 hours a day of study can be too much and he might not be able to bear it but for a careful student hour, a day can be unto his satisfaction level. In other words, a careless student needs less study stimulation and a careful student needs more study stimulation.

Environmental cognition involves further processing of information (e.g., storing, organizing, and recalling). It also involves appraisal processes. Is this environment good or bad, cold or hot, strong or weak? It includes the emotional impact of environments, attitudes toward environments, the preferences we have for some environments over others, and the categories we use to organize our knowledge about various settings. Only human beings are capable of adding this element of likeness and dis-likeness, computers cannot do that. Through these various processes, we appraise environments and compare them with mental images of what we desire or expect them to be. Most of the time, because we have planned well, environments are acceptable approximations of what we expect. Under these conditions, the environment exerts a steady-state influence. Disruptions occur when we learn through these processes that present conditions do not correspond to our desired or expected levels of stimulation or are incapable of meeting the objectives of the plan with which we entered the situation.

**We will start our next lecture by discussing the third postulate in detail**

**Lesson 8**
**The present framework of Environmental Psychology (II)**

**We will begin this lecture by discussing the third postulate of Environmental Psychology in detail.**

**Third postulate: The influence exerted by the environment is indirect i.e. Environment acts to influence people's emotional states, which in turn mediate their overt behavior.** The work by Baron and Bell also postulates emotion (affect) as a mediating link between the physical environment and aggressive behavior (e.g., Baron & Bell, 1976; Bell & Baron, 1976). It has been shown that such diverse environmental conditions as temperature (Griffitt, 1970; Griffitt & Veitch, 1971), crowding (Baum & Valins, 1973; Griffitt & Veitch, 1971; Valins & Baum, 1973), noise (Bull, et al., 1972; Geen & O'Neil, 1969; Mathews & Cannon, 1975), air pollution (Rotton, Barry, Frey & Soler, 1976), and radio news broadcasts (Veitch, DeWood, & Bosko, 1977; Veitch & Griffitt, 1976) influence the affective state and interpersonal behaviors of individuals. If you feel annoyed by the phone ringing you might answer it in gruff tones, or not at all; if you are gladdened by the distraction, you will probably answer with greater civility. If you like the smell of an unusual perfume, you are likely to strike up a pleasant conversation; if instead what you are confronted with is the smell of stale cigarettes and body odor, your disposition is likely to be less shining, and your overt behaviors less positive.

**Fourth postulate: Environment-evoked emotions are best depicted in terms of three distinct dimensions: pleasure – displeasure, degree of arousal, dominance –submissiveness.** Russell and Mehrabian (1977) have provided evidence to reconcile these two sets of research findings and have reported that the three dimensions of pleasure-displeasure, degree of arousal, and dominance-submissiveness constitute both the necessary and sufficient dimensions to describe all emotions.

(i) **Pleasure and Displeasure:**

For example, you remember a very sad event in your life. While thinking about that event you might experience pleasure or displeasure based on whether the event was happy i.e. winning a lottery or whether it was sad i.e. losing a mobile phone.

(ii) **Degree of arousal:**

When you remember the event in which you won the lottery, you might experience the same physical symptoms that you experienced when you won the lottery i.e., the same adrenaline rush, rise in blood pressure, or pupil dilation.

(iii) **Dominance-Submissiveness:**

Suppose you are a student who has topped the exam and after 6 months you remember this event. You will suddenly feel that in that particular situation, you might have achieved anything (you will generalize your self-confidence to other situations), and you might start to realize how much control you were able to exert over the environment or how much power you were able to exercise. When you attach this degree of control to the high level of arousal and attach this whole experience with the element of pleasantness, you will feel excited, as if you are dominating the environment. Now imagine the opposite, where you are a student who has failed, you will simply think that everything that happened, is because of external environmental factors. You will feel that you were not able to control the environment and ultimately feel your submissiveness at the hands of the environment.

Results of their studies show that the larger number of dimensions obtained in verbal-report studies can be accounted for by these three dimensions. Thus, self-reports of emotions that employ different words (e.g., joy and happiness) may yield similar underlying configurations of pleasure, arousal, and dominance, which differ primarily in the intensity of affect. Further, differences between global emotional states can be best understood in terms of differences in the underlying configuration of these three dimensions. Thus, the best available evidence to date suggests that the emotional (affective) state of individuals can be adequately described by these three dimensions.

**Fifth postulate: The environment–influenced behavior of the individual is dependent upon the extent and the configuration of the dimensions of emotions aroused.** In most of the studies to date where the emotional state of the individual is seen as a mediating factor for behavior, emotion has been conceptualized in a unidimensional manner. Typically, the one dimension utilized has been pleasure-displeasure, although sometimes a variant of the dominance/submissiveness dimension has been used (e.g., studies of control, constraint, and reactance). However, researchers have seldom considered the interrelationships of these dimensions. The current framework attempts to utilize a multiple-dimension approach to emotion as a way of accounting for more of the reliable variance in the relationship between emotional states and overt behavior.

We would expect different behaviors to ensue as a result of changes in the environment that elicited displeasure, high arousal, and submissiveness ("anxiety") changes resulting in displeasure, high arousal, and dominance ("anger"). A teacher experiencing the former, for example, may attempt to "leave the field," whereas experiencing the latter may lead him or her to combat, or at the very least, to attempt to change the environment. Aggression and excitement have the same physiological arousal but their experience is categorized differently. Re-living the experience is also different from the actual experience.

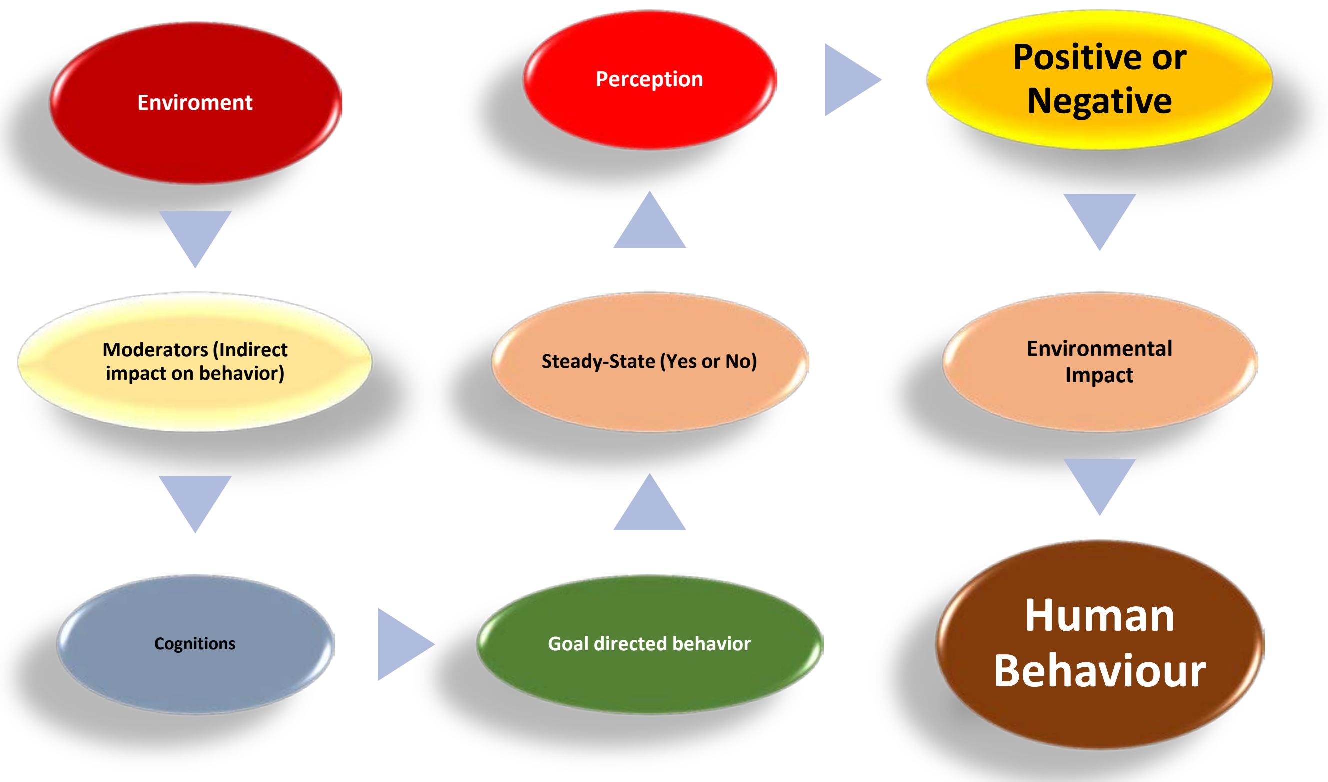
**Sixth postulate: Behavior is goal-directed. That is to say behavior in an environmental setting is performed to increase or decrease the pleasure, degree of arousal, or dominance elicited by changes in the environment in an attempt to return the environmental influence to a steady state.** The underlying assumption is that steady-state environments are preferred and actively sought (i.e., humans don't passively react to environmental stimulation, but rather attempt to alter their experiences to maintain a state of equilibrium). They do this by altering their evaluation, by altering the setting, or by changing settings altogether. For example, a student who completes a very difficult assignment by considering work from an angle of entertainment.
In response to displeasure aroused by stimuli whose preferred values are exceeded (in either direction), the behaving organism attempts to decrease the unpleasantness. For example, the temperature of a particular environmental setting exceeding AL (adaptation level) elicits discomfort. Simultaneously, the degree of arousal and dominance/submissiveness will also be influenced. If the organism experiences great displeasure, high arousal, and high dominance, then aggressive behaviors may occur, which might be directed at removing the social source of that emotion. On the other hand, if displeasure is not felt, if the degree of arousal is low, or if submissiveness is felt, submissive behavior might occur. This could involve withdrawal to a different setting more conducive to the desired emotional configuration.

**Seventh postulate: When goal-directed behavior is ineffective or when great expenditures of energy are required to maintain a steady state, the environment can be considered pathological and a disruptive influence on human functioning.** What are the long-range effects of exposure to a given environment featured by a particular level of intensity, complexity, and inappropriateness of stimulation? According to adaptation level theory, the individual's AL will be shifted to a value corresponding more nearly to that of the environment. This, of course, is what adaptation is. However, for adaptable humans, it is still possible that the range of environmental stimulation could be such that the energies expended in adaptation would have detrimental effects on the individual, or that the behavioral repertoire of the individual is too limited, thereby making adaptation impossible. It is under these conditions that the environment can be considered pathological. Evidence (to be detailed in later chapters) concerning the effects of prolonged noise exposure (e.g., Glass & Singer, 1972; Weybrew, 1967) exemplify this state of affairs, as does the work of Calhoun (1962) who looked at the long-term effects of increased population density on social adaptive behaviors of Norway rats. Another example can be year-long dessert or winter survival courses. The excitement for adventurers is usually high on such training but with time the excitement level dies down and survival becomes difficult.

**Model: How do the seven postulates of the environmental framework interact with each other?**

| **Environment** → | **Moderators** → | **Cognition** → | **Steady State** |
|---|---|---|---|
| • **Noise**<br>• **Pollution**<br>• **Weather**<br>• **Crowing**<br>• **Mountains/Desserts**<br>• **Architecture** | **Social situations:**<br>• **Relationships between different people.**<br>• **Classroom settings.**<br><br>**Personal Variables:**<br>• **Adaptation level. (A person in Murree will feel a temperature of 35C differently than a person in Jacobabad due to their different adaptation levels of temperature)** | • **Expectations.**<br>• **Attaching emotional meanings to environmental stimulus.**<br>• **Goals**<br>• **Moods**<br>• **Sentiments**<br>• **Personal expressions** | **What is the effect of the environment on the human? Is it helping them gain a steady state or is causing disruptions in their behavior?** |

Let's understand this table in the form of a reactive diagram:




**This course has also been divided into chapters. Chapter one consists of (Lesson1-Lesson8). From the next lecture we will start chapter2.**

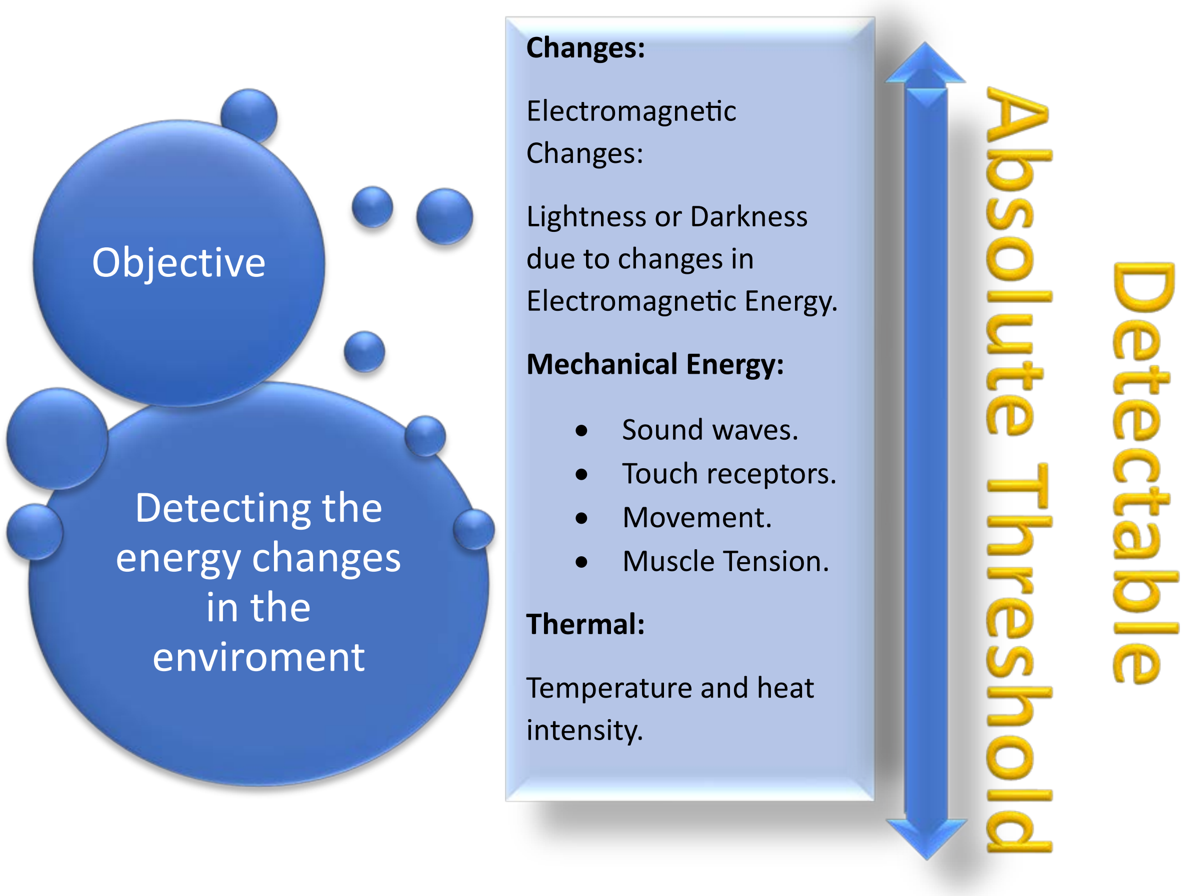
**Lesson 9**
**Chapter 2: Environmental Perception and the use of natural environment.**
**Lesson 9: Elementary Psychophysics/Perception and its Cognitive Bases**

In elementary psychophysics, we try to understand how human beings can psychologically detect physical stimuli (Stimulus Detection). For example, if a student is deeply involved in studies, he might not notice any other sound in the environment such as the sound of the fan in his room or the ticking of the clock. Elementary psychophysics tries to understand at what time and how physical stimuli gather such momentum, that they can be interpreted psychologically by a person.

**Objectives of Elementary Psychophysics:**
There are four objectives of Elementary Psychophysics:

1. **Stimulus Detection:**

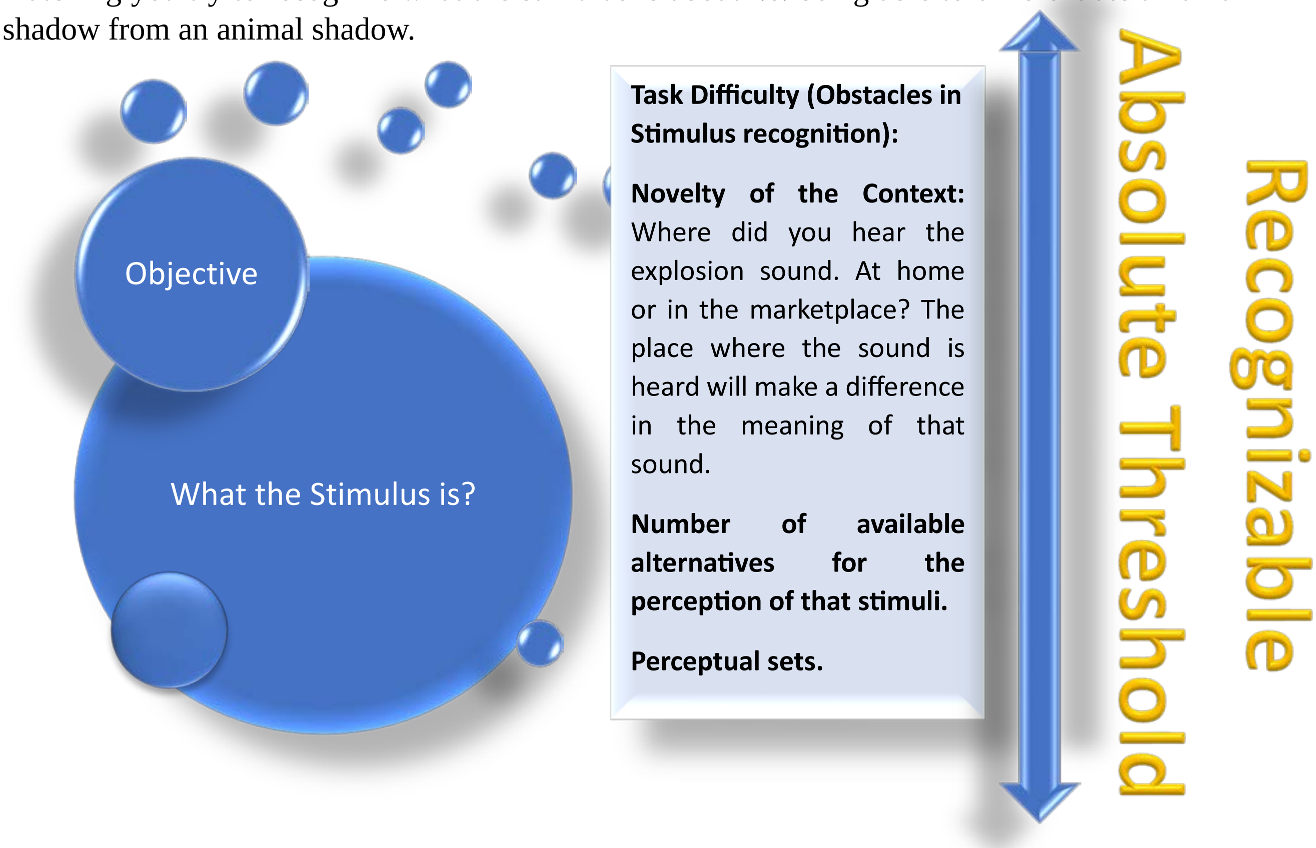
What is it? For example, suppose that you are walking through a dark street at night. At the end of the dark street, you saw something that felt like the figure of an animal. But you are not too sure about it because light is very little. You might dilate your pupils or try to use your depth perception cues in an attempt to detect the stimuli.
The basic objective of stimulus detection is to understand what is happening around us. In Psychophysics this relates to the different types of energy changes that are taking place in the environment. Our receptors can read those energy changes.



Every stimulus is detectable up to a certain point but beyond that range it becomes undetectable. All five senses have their ranges in between which they can be detected. The minimum intensity required by a stimulus to be detected is called the absolute threshold.

2. **Stimulus Recognition:**

Is this stimulation different from that one? For example, when you have accommodated your physiological systems and sensory receptors according to darkness then that information goes inside your brain. That is the time when you perceive those current images and you match those images to the previous images of such things that exist in your mind and then as a result of that matching you try to recognize what the stimulus is about i.e. being able to differentiate a human shadow from an animal shadow.




There are also minimum or maximum ranges for stimuli to become recognizable i.e., is this the picture of an animal or a tree. For example, you are studying and you hear a loud sound outside your home. Now you have to recognize whether it is the sound of a car crash, a cylinder blast or an actual blast. The novelty of the context and available meanings you can give to the explosion sound in that particular situation determines how you can recognize the explosion sound.

3. **Stimulus discrimination:**

How to differentiate between different stimuli? For example, you may need to differentiate between similar stimuli to come to the final decision of what that stimulus actually is. In the explosion sound example, you may think of the sound of a car crash, cylinder blast or bomb blast and do a comparative analysis of these sounds in your mind to make the final decision about what sound it was. This brings if to the concept of Just noticeable difference (JND):

**Just noticeable difference (JND):** An increase in the intensity of a stimulus above its original level required for a person to notice a difference in the level of intensity. The intensity of the sound will increase or decrease concerning the person's distance from the place where the sound is coming from. A person standing near the explosion sound will have more requisite information to perceive that sound accurately/to create such a noticeable difference as compared to the person who was far from the explosion sound.

**Weber's Law:**

High-intensity stimuli require a proportionately larger increase to produce the Just noticeable difference (JND). For example, if you are listening to music in less volume, even a little increase in the volume will be noticed by you but if you are already listening to loud music, a little increase in the volume will not be noticed by you. You need to increase the loud music much more to create the Just noticeable difference between the two loud music sounds.

4. **Stimulus Scaling:**

How much of a certain stimulus is there? For example, when you are passing through a dark street and you look at something that looks like the figure of an animal, scaling would be when you try to find out that is the animal big or small and what type of animal this is. Then you would appropriately design your reaction towards it. It is a process for effective commerce with the environment. The magnitude of the stimulus, as scaled by the perceiver, determines the magnitude of the response deemed necessary to deal with it. For example, if the sound of the explosion is too loud, the person standing near the explosion sound will have a different reaction according to it. The same is the case with low-intensity explosion sound.

**Perception**

**Perception means to look at, to understand, and to know about the world around us. Perception is a sudden realization or an insight experience of our ability to navigate the environment on our own. For example, when a 5-year-old child rides a bicycle for the first time in life without the support of his/her father, he is suddenly surprised and starts to realize how much he/she is capable of.** When we think of perception we usually think first of visual perception. This is probably because we gather so much information from the environment via this sensory system. Except listening to speech, the vast majority of our everyday activity is guided by vision. Vision guides our motor behavior (getting us from here to there while avoiding running into things on the way), providing us with information concerning what is out there and where it is. Additionally, much of our interpersonal behavior is guided by the visual information we receive. Our judgments of the emotional states of others (and ourselves, for that matter), and our intentions, likes, and dislikes, are determined in good part by what we see. No wonder we rely so heavily on our sense of vision to inform us of the world that we inhabit! There is an old adage that "seeing is believing," and when it comes to conflicting cross-modal information, we do indeed tend to believe our eyes over the other senses.

Despite this great reliance on vision, perception is more than just sensory input to the visual system. Often, we must depend on other systems when interacting with the environment. For example, we cannot see heat, so we must depend on thermal receptors on the skin's surface to warn of the dangers of placing the hand on a stove; we cannot see natural gas and must depend on olfactory cues and odorous additives to warn of a gas leak in the home; we cannot see the Civil Defense siren warning of an impending tornado, and therefore must rely on our sense of hearing. But **perception of the environment** is even more than the summation of all these sensory inputs. It involves **labeling, describing, and attaching meaning to the world** around us. Perception, in addition to being sensory, is also highly cognitive.

(i) **Labeling:**

It means to identify what is happening around us. Or what the stimulus is about? For example, after looking at a smiling facial expression, we can identify is it a happy smile or a sarcastic smile.

(ii) **Describing:**

It is the interpretation of details about the context of the stimuli. For example, if we know that a person is smiling because of happiness, our next step would be to look for further details i.e. why is the person happy? Is he happy because he received an increase in his salary or is he happy because he is going on a vacation?

(iii) **Attaching meaning to the world around us:**

It means to make the world around us meaningful. For example, while teaching a child how to cross a road, we also need to teach him the dangerous outcome i.e. fatal accidents that may happen if he does not follow the road rules. The whole experience gives a bulk of information to the child at one time. In other words, we need to go beyond then the present stimuli so that a person can make sense of perceptual experiences.

**Environmental Perception (Cognitive Bases of Perception):**

All environments carry a set of meanings acquired through their specific physical, social, cultural, aesthetic, and economic attributes. Let’s look at them one by one:

**An example of a beautiful picture of a waterfall with one-man swimming in it.**

(i) **Physical attributes:** This means the physical setting of the stimuli. For example, upon looking at a beautiful picture of a waterfall, we might focus on the color of the water, the plants on the waterside, or the mountains behind it.

(ii) **Social Attributes:** This is the social scenario around the perception of the stimuli. For example, if only one person is diving in the waterfall, a happy person might say that this person seems peaceful and happy enjoying himself and a lonely person might think what a lonely person he is! Diving alone in the waterfall.

(iii) **Cultural attributes:** This is the content of the stimuli in relevance to the person perceiving it. For example, while looking at the waterfall picture we might question ourselves that does this waterfall exists in Pakistan or if is it a scene from a western country.

(iv) **Aesthetic attributes:** These are the personal ways we find beauty and meaning in stimuli. For example, a person with a high aesthetic sense might look at the picture of the waterfall and think “I feel full of life when I look at this picture” or “I feel like revisiting my childhood after looking at this picture”.

(v) **Economic Attributes:** This is the assessment of economic resources required to interact with the stimulus. For example, a man might look at the picture of the waterfall and think “How can I visit this waterfall”, “How much expense will it take to dive into this waterfall” “How much travel cost will it take to visit this waterfall. For example, a Pakistani person might immediately think of travel costs if he looks at the picture of Niagara Falls which is located in Canada.

These meanings are extracted from the environment by the perceiver in terms of his or her attitudes, beliefs, values, and physical limitations. We may admire an apple orchard in spring for its floral beauty and its aromatic fragrance, while simultaneously realizing its worth in terms of the honey that will be produced by bees from its nectar and the apples to be harvested in the fall. Additionally, we may see the orchard as symbolic of the economic power of its owner, his or her ability to buy and sell, and ultimately the political influence that person is likely to exert in the community. Finally, we may see it as representing the outcome of years of research in the development of hybrid apples. We perceive all of these meanings and respond to them to some degree as the sight and smell of the spring blossoms reach our eyes and noses.

- **Assessment of Environment:**

Our assessment of the environment is achieved within the context of three broad but not always congruent ways of viewing the world.

(i) **Culture:** First, we develop attitudes as a result of living within a culture. For example, in Pakistan, the culture is different from one city to another city. Our culture defines our norms and values, which, in turn, dictate our general behavioral patterns. As we studied previously nomadic cultures promote independence and how agricultural culture promotes obedience. Now, a person from a nomadic culture, when visiting an agricultural societal environment, might assess the environment as being “suffocating” or” controlling” for its rules like obedience and staying in one place for a long time.

(ii) **Needs and Preferences:** We also assess the environment according to our needs and preferences and behave accordingly. For example, imagine that you are feeling very hungry and the food is not ready. Now, this frustration caused by hunger and non-availability of food will provoke you to think “What a bad day it is today” and you might start watching cooking shows on TV while waiting for your food. Cooking shows also come on TV at specific times before or near dinner time to get more views.

(iii) **Future:** We also assess our environment in terms of how it will affect us in the future. For example, a housewife might think of separating parents’ room from their children’s room because near future children will grow up and require their personal space. Not only is the question of "What's in it for me?" asked, but also, "What effect will my presence and interaction have on the environment being viewed?" If trees are seen only in terms of the apples they will produce and care is not taken to preserve them, soon there will be no apples and eventually no trees. This view of the environment is much like the ecologist's view.

**Contextual and Social Bases of Perception**

It is important to emphasize that perception is contextual. Cultural, social, gender, and individual differences all influence what we do and do not see in our environment. Looking within the context of gender, there are characteristic differences in the ways in which males and females tend to appreciate certain things. This is the reason that cars are designed differently for males and females. For instance, consider how exposure to green spaces (natural environment) versus urban concrete jungles (built environment) affects people's stress levels, attention, and overall well-being. Another example would be how an individual's identification with larger social units (such as community, nation, or world) strengthens in-group solidarity and empathy, influencing their support for environmental protection.

Thus, perception is not simply a matter of the individual responding to sensations created by energy from stimuli impinging on the sensory organs. Rather, this process is embedded in a cultural context, and various social factors have been demonstrated to produce differences in the ways two individuals perceive the same stimulus. Individual differences in backgrounds, experiences, values, and purposes can have a profound influence on the result of the processing of information from the world around us. These differences, however, do not detract from the fact that perception is a fundamental psychological process in which all humans engage.

**Lesson 10**
**Perception and its Cognitive Bases/Theories of Environmental Perception**

## Complex Perceptual Processes

The exercise below illustrates the complexity of processes taken for granted and engaged almost automatically when perceiving individual stimuli in the environment. As indicated, such processes are critical, and psychologists have exerted considerable effort to understand them. However, the primary concern of environmental psychologists is understanding how people perceive molar aspects of the environment. In addition to understanding how discrete stimuli or objects are perceived, environmental psychologists attempt to understand the processes by which whole environments are perceived. In a sense, they are interested in the forest, not just the individual trees. Further, a discussion of these processes will follow since it is important to account for the relationships among the perceptual, cognitive, and evaluative processes that affect experience and determine behavior. Although highly interdependent, they will be discussed separately, making note of their interrelationships where appropriate.

- **Exercise:**

Below is a saying by Masaru Ibuka, a Japanese man who was the founder of an Electronic Company, in 1991.

***We worked furiously [to realize our goals]. Because we didn't fear, we could do something drastic.***

When you initially read this saying it almost seems like an automatic activity. But from a research point of view, the human might go through the following complex perceptual processes when reading the above saying.

1. We Start reading it with a specific purpose.
2. We Screen out other stimuli.
3. Our Eyes translate the light waves reflected off to screen into neural energy.
4. Our sensory pathways transmit a series of electrical impulses to the higher brain centers.
5. The brain organizes and interprets the input.
6. The brain matches the present patterns of sensation with previous images.
7. Finally, Words and phrases became coherent and meaningful.

## Theories of Environmental Perception

- **Gestalt Theory:**

Much research and theory exist regarding the basic processes involved in perception. Among the earliest and most significant contributors to this area was a group of German psychologists working within a framework known as Gestalt theory (e.g., Koffka, 1935; Kohler, 1929). These theorists emphasized the active role of the brain in searching for meaning in stimuli.

**(i) The Principle of Good Form:**

The term Gestalt means "good form" and Gestalt theorists proposed that the brain is organized in such a way as to construct meaning from stimuli, and even to impose meaning where it might not appear to exist objectively to achieve this "good form. For example, read the following sentence:

*THE QUAK BRON FAX JUNS OUER THA LEZY DAG*

Many people can read this sentence correctly as “*The quick brown fox jumps over the lazy dog*”. It is because of the principle of good form that our brain constructs meanings from any stimuli even where they might not appear to exist. This entire sentence was incorrect in spelling but we were still able to read it correctly. Good Form has its benefits as when it helps us to make sense of even a disorganized environment, however, it is negative in a few places as well. For example, a teacher might mark wrong answers as correct due to good form and students might receive unjustified benefits. Good form may be harmful or even dangerous if it interferes while performing detail-orientated tasks like performing heart surgery. Any little mistake to wrong meaning construction by our brain due to good form can lead to heart failure. Due to this good form may also be called a “False Positive” or False Negative”. It is not uncommon for students, when proofreading their term papers, to "read" words that do not exist on the paper. The sentence "The bear climbed up ***a*** tree" might be read, "The bear climbed up ***the*** tree."

**(ii) The Holistic Orientation:**

A second characteristic of the Gestalt approach is its holistic orientation (i.e., the assumption that the perceptual process must be understood in its entirety rather than broken down into discrete elements). This assumption is often expressed in the statement that in perception, "the whole is greater than the sum of the parts." For example, we do not say “I am watching a show on a box which has a screen, buttons, and a black outline” We simply say “I am watching a show on TV”. We perceive things as a whole in one part. Look at the following figure:

We know that this image is that of a star (because of good form) even though there are some blank spaces left in it. Good form does not only imply physical stimuli but applies to our cognitive processes as well. For example, a student who has passed college remembers her school days as the best days of her life. However, she might not remember every detail. She forms a complete image of her school time as a happy image, ignoring the little positives or negatives that come along the way.

**(iii) Size Constancy:**

Another illustration of the tendency to impose meaning on sensory inputs is the principle of size constancy, in which changes in the size of the retinal image of an object are interpreted as changes in distance rather than changes in the size of the object. In this case, the brain is actually "overriding" sensory inputs to maintain a Gestalt. For example, look at the following image:

Now, due to the principle of size constancy, you might perceive that the hand on the left side is far away and the hand on the right side is near to us. In a literal sense, the size of both hands is different. However, our brain knows that they are the hands of the same person so they will be the same.

- **Functionalism:**

A theoretical orientation that differs from the previous approach views perception as a much more direct process. It involves less mediation by higher brain centers to perceive meaning in the environment.

1. It is argued that meaning already exists in the environment and that our sensory mechanisms are "prewired" to respond to meaningful aspects of our environment. For example, due to our past experiences, we might assume the content of the book just by looking at its title. Another example is the generational experience example. Due to the experiences of our ancestors, we naturally perceive our environment. We have pre-formed meaning. So, there is no need to use higher brain centers and perception occurs in sensory areas or organs directly. Due to this, the perception process seems automatic.

2. This approach is related to **ecological biology,** which studies organisms' adaptation to their environment. For example, the concept of an **ecological niche** refers to the instinctive tendency observed in animals to seek out that area of their environment that affords them the greatest chance of survival. For example, animals have the same methods of hunting that they used 100 years ago. Due to this reason, tigers live in environments where there is a widespread landscape and other animals are co-habituating so that they have plenty of area to search and hunt and there is plenty of prey, ultimately, increasing their chances of healthy survival. But this is not true for humans as our behavior constantly keeps on changing. For example, the working ratio in today's world is different than it was in the time of our grandparents. They used to work in the day and the rest of their time was for their families. However, nowadays people do not have much time for their families, and

multitasking is highly appreciated. There was a time when a movement named **“Ethology”** started in Psychology. In this movement, psychologists attempted to study instinctual behavior in humans. For example, the instinct of aggression or the will to power. But with time it became evident that humans cannot be studied in terms of instincts. Because we know that instinctual behavior is constant, and human behavior keeps on changing.

3. Gibson (1979) applied this notion to human perception. She suggests that humans are innately endowed with the ability to perceive those aspects of our environment that have **functional value** for them. Thus, according to this view, an infant should be born with the ability to perceive its mother's face, since this stimulus has obvious survival value. Because as compared to animal babies, human babies depend a lot on their mothers to survive and to learn to function in their environment. In other words, mothers have a high functional value for children and most children recognize their mothers as soon as they are born

**Lesson 11**
**Learning Theory/Probabilistic Functionalism and Environmental Cognition**

**Learning Theory:**
Much research on human perception concerns the role of experience in perceptual development. From this perspective, our perceptions are not innately determined, but rather we must learn to perceive critical aspects of our environment. For example, the **principle of size constancy** referred to above is not seen as an inborn perceptual ability, but one that develops only through the experience of seeing many objects from a variety of distances. Gradually the infant learns that the objects are not growing or shrinking, but remaining a constant size regardless of their distance. This happens even though the retinal images, and thus the neural impulses sent to the brain, vary dramatically.

Gestalt and Learning perspectives have a lot in common. According to the Gestalt theory, Perception is done almost within seconds along with the involvement of higher brain centers because it holds the belief that perception is done directly via sensory organs and receptors as well as with the help of information we hold from past experiences. Learning theory also places a lot of emphasis on direct information processing, past experiences, and specialized interpretation of information. Functionalists on the other hand are different in the sense that they do not focus on the involvement of higher brain centers and place more emphasis on mechanical instinctual behavior.

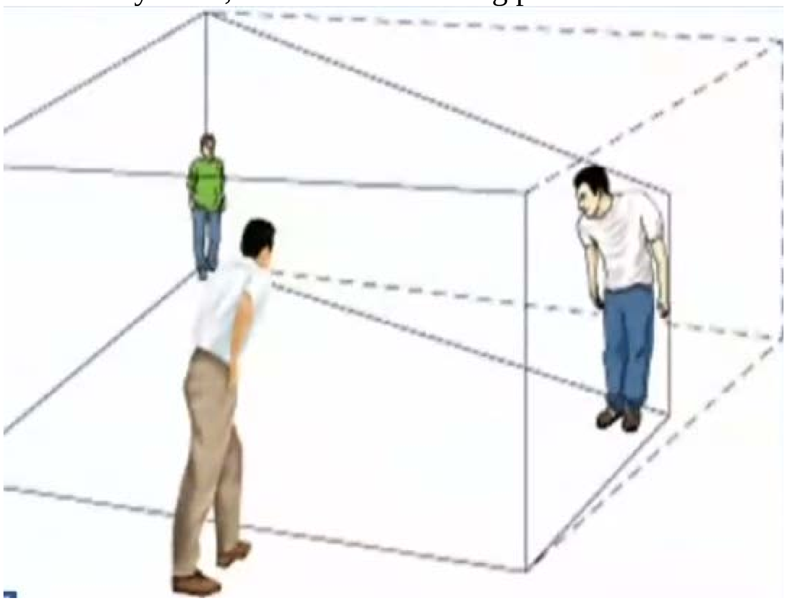
Learning theorists propose that an important result of experience and learning in perception is the development of assumptions about the world around us. For example, youngsters might look at cool and handsome heroes in movies. They might develop the assumption that cigarette smoking will make us look like a man, it will make us look stylish. Due to this assumption, they might start behaving similarly and start to smoke. However, this assumption is quite misleading, and usually such youngers end up facing the hazardous effects of smoking on health.

These assumptions facilitate our interactions with the environment because they save us time and effort in coping with new stimuli. That is, we do not have to approach new situations as though we have never encountered them before. We assume that many elements of the situation are similar to those of situations that we have previously experienced. For example, our responses to the word mother are usually like “Mothers are kind” or “I love my mother”. In other words, they are our assumptions about the word mother. Gestalt Psychologists also embrace the concept of assumptions because according to them perception process is very fast. Sensory receptors might process information in seconds and they believe that it is assumptions that make perception process faster.

Thus, we bring to the present situation learned assumptions in the form of expectancies about what is likely to happen. These expectancies are usually correct, making for easy processing of information and adaptive functioning. In these ways, learning theories are not unlike those of the functionalists.

Our assumptions, however, can sometimes be misleading, resulting in misperceptions or illusions (e.g., Ames, 1951). For example, look at the following image:

By looking at this picture you might think that the person on the right-hand side is bigger and the person on the left-hand side is a kid. However, this is just an optical illusion and has no contact with reality. Now, look at the following picture:



This is the actual reality of the people shown in the first picture. The room is designed in such a way that the backspace is a bit tilted. It naturally makes the person in the white shirt stand a bit forward and bent down so he appears bigger. And it also naturally makes the other person in the green shirt stand at the back, making him look smaller like a kid.

**Probabilistic Functionalism**

Given the complexity of environments, the goal stated earlier that environmental psychologists must account for the processes by which molar environments are perceived is an exceedingly difficult one to achieve. Therefore, one of the most important learning theories of perception, and potentially the most fruitful for environmental psychology, is Brunswick's lens model (1956, 1969). Brunswick's approach provides a model for mathematically describing individuals' perceptual processes when making judgments in response to molar environments containing multiple stimulus dimensions. Let's look at the following figure:

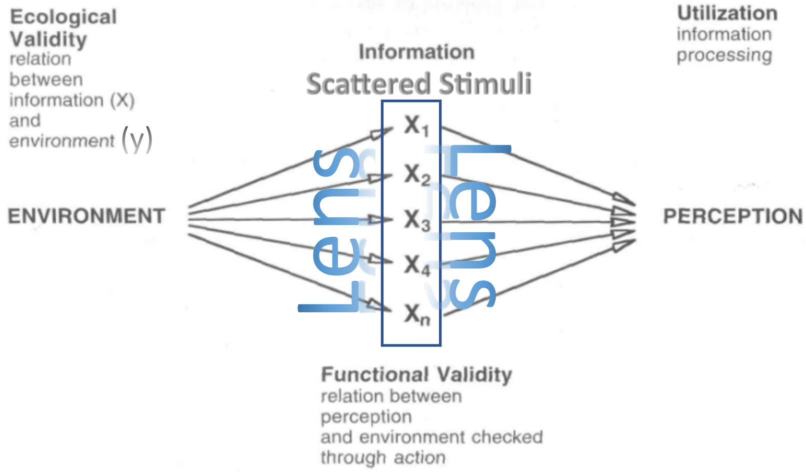



FIGURE: Brunswick's probabilistic theory illustrates one way of relating the information available from the environment to the way the individual perceives the environment. Let's break down a few concepts of this diagram.

- **Ecological validity:** It is what information we focus on. It shows the relationship between information x and environment y. It means in a certain environment; a certain stimulus is more important and our perceptual lens would automatically concentrate upon that particular information.
- **Functional validity:** The information on which our perceptual lens focuses has certain functional validity. It means the relation between perception and environment that is checked through action. For example, children learn to walk in a certain style or by using certain balance mechanisms. Children's' perceptual lens focuses on these techniques which have helped them to learn to walk.

Brunswick argues that complex stimulus patterns are processed as though through a "lens" (see Figure above), where the scattered stimuli are "focused" into a single perception of the environment. An example would be being stuck in traffic where the environment is giving a bulk of information and our perceptual lens is focused on the information that is important to us. In this way we manage to reduce the complexity of our environment by filtering the various available stimuli through the lens, discounting or ignoring some perceptual cues while emphasizing others. We do this not only to simplify judgmental tasks, but because we learn from experience that some sensory information is misleading (in Brunswick's terms, they are lacking in ecological validity), and other cues are of minimal value, (i.e., of low ecological validity) in correctly perceiving the true state of the environment.

**Environmental Cognition:**

All previous theories were about environmental perception. However, now we are going to discuss Environmental Cognition. Let's first look at the difference between the two:

- **Difference between Perception and Cognition:**

Perception is the process in which we use our sense organs to gather information about the environment and that information gets into our mind and helps us direct our behavior. For example, imagine you're walking in a garden and you see a rose. You notice its deep red color and its sweet fragrance. This immediate sensory experience of the rose is an example of perception. Cognition is a step ahead of perception which means that we are making a proper sense of the environment over here. The packets of incoming information here become a part of our working strategy, helping us build our insights. For example, Now, suppose you start thinking about the rose, recalling memories of roses you've given or received in the past, or considering the symbolic meanings of roses in different cultures. This process of thinking, understanding, and interpreting the rose is an example of cognition.

The term, cognition, refers to thought processes. Thus, environmental cognition concerns thinking about the environment (i.e., how individuals process information and organize their knowledge about the characteristics of their environment). Also, of interest is how this knowledge is acquired, or learned, as well as how individuals differ in their knowledge of the environment. Finally, environmental cognition concerns how variations in the environment affect the ability to understand the setting. For example, important differences in knowledge exist between familiar versus novel environments. We will consider novel environments first.

- **Responses to Novel Environments:**

From the participant's point of view, the environment is typically neutral and enters into awareness only when it deviates from some adaptation level. Your first few days on campus can be thought of as a non-neutral situation, one that from your perspective at the time deviated from the norm and made you self-consciously aware of your new surroundings. Ittleson, Proshansky, Rivlin, & Winkel (1974) claim that in situations of this kind, six different but interrelated types of responses occur: ***affect, orientation, categorization, systemization, manipulation, and encoding.*** While in the following paragraphs, we will discuss each of these in the order just presented, these responses are probably not made serially. Each tends to blend with the other as we attempt to bring the environment and ourselves into a state of equilibrium.

- ***Affect:*** The nature of our affective response to new environments will depend on many factors, some of which will be discussed later in this chapter. At a minimum, the effect will consist of a heightened degree of awareness and arousal occasioned by the need to know, predict, and therefore to feel in control of and secure in an unknown setting. Affect is our emotional response. For example, you might enter into a new building and say that I feel happy/secure here or I feel sad/insecure here. Aside from this general reaction, several other feelings may emerge because of the particular characteristics of a new setting. Discovering that the tennis courts are close to the dorm may lead to happiness, whereas finding that you live adjacent to the cemetery might lead to fear and nervousness. The thought of long walks to the library through a grove of oak trees might conjure romantic images and feelings of love, or graven images of rapists and muggers, leading to fear and anxiety. Such affective responses, both general and specific, may govern the direction that subsequent relations with the environment will take. First impressions (feelings) about places can and do have enduring consequences.

- ***Orientation.*** Individuals in a new setting actively seek to find their place, their niche. This is primarily a cognitive process and involves scoping out a place. For example, a student goes to the USA for studies. On his first day in university, the excitement will be followed by insecurity which prompts him to learn about his environment so that he can find out his role in it and adjust to it.

Six steps come under our responses to novel environments: ***affect, orientation, categorization, systemization, manipulation, and encoding.*** Affect and Orientation have been discussed in this lecture however all six processes will also be discussed in the next lecture again in detail.

## Lesson # 12

## RESPONSSES TO NOVEL ENVIRRONMENTS AND ENVIRONMENTAL COGNITION

### Responses to Novel Environments

A novel environment is an unfamiliar or new setting that a person encounters for the first time or after a long period of absence. Ittleson, Proshansky, Rivlin, & Winkel (1974) claim that in situations of this kind, six different but interrelated types of responses occur: affect, orientation, categorization, systemization, manipulation, encoding. While in the following paragraphs we will discuss each of these in the order just presented, these responses are probably not made serially. In fact, each tends to blend with the other as we attempt to bring the environment and ourselves into a state of equilibrium.

### Affect

The nature of our affective response to new environments will depend on many factors, some of which will be discussed later in this chapter. At a minimum, the affect will consist of a heightened degree of awareness and arousal induced by the need to know, predict, and therefore to feel in control of and secure in an unknown setting. Aside from this general reaction, a number of other feelings may emerge because of the particular characteristics of a new setting. Discovering that the tennis courts are close to the dorm may lead to happiness, whereas finding that you live adjacent to the cemetery might lead to fear and nervousness. Such affective responses, both general and specific, may govern the direction that subsequent relations with the environment will take. First impressions (feelings) about places can and do have enduring consequences.

### Orientation

Individuals in a new setting actively seek to find their place. This is primarily a cognitive process and to use a slang expression involves "scoping out" a place. You probably asked yourself on those first days: Where, relative to my dorm, is the dining hall, the recreation center, the library? Where do you go to buy books, to get athletic tickets, to pay parking fees? Which dorm is the one that your cousin Fred lives in? Ittleson, et al. (1974) describe the process of finding answers to these questions. Orientation, in short, expresses a person's desire to "know where one is" physically in relation to the total milieu. Here we see the beginning of environmental cognition, with the individual actively attempting to identify the location of important stimuli in the new environment in relation to where one is at.

### Categorization

In new situations, though, a person does more than just orient; he or she also *categorizes.* The individual evaluates the new environment and imposes his or her own unique meaning to various aspects of it. Not only do people ask where they can get a pizza, but they also ask where to find the best or the cheapest pizza. Categorization is therefore the process of extending the meaning of the environment by functionally relating its various aspects to one's own needs, predispositions, and values. Thus, categorization represents a more sophisticated understanding of the environment than simple orientation, in that the individual now knows several instances of stimuli in a particular category and is able to distinguish among them in terms of their relative utility to the satisfaction of one's needs.

**Systemization**

It is difficult to say where categorization leaves off and sys-temization begins, but at some point individuals organize their environments into more meaningful and more complex structures. They know, for example, the best time to go to the library, not only in terms of when it is the quietest, but also when it is easiest to find a place to park, or when the most helpful librarians are working, or when the latest issue of *Sports Illustrated* arrives.

**Manipulation**

Out of such systemization, individuals achieve a sense of order and understanding; they not only know the new setting but they can predict it and make it work to their benefit. If people have ordered their environment, they usually can manipulate it or control it to their advantage. For example, If the cheapest pizza parlor in town is closed, they know how to get to the second cheapest, or know their options with respect to having Chinese or Mexican food instead.

**Encoding**

Encoding represents the highest level of understanding about the environment, because the individual is no longer tied to concrete perceptions of the setting. People can function more effectively with this knowledge, and can also communicate with others about using shared symbols. Encoding allows us to "think-travel" through environments [redacted] and to prepare us for changes in our interrelationships with our environments.

**Characteristics of Cognitive Maps**

A cognitive map is a mental representation of the physical environment. It involves perceiving, storing, processing, and recalling spatial information using our brain’s cognitive abilities. It is a vital skill that helps us navigate and interact with our surroundings A common approach to studying spatial cognition is to ask people to draw "sketch maps" of environments. Research indicates that sketch maps are a reliable method of data collection (Blades, 1990). Lynch (1960) conducted one of the first comprehensive studies of the nature of cognitive maps when he asked residents of three American cities (Boston, Los Angeles, and Jersey City) to draw maps of their city environments. He analyzed these drawings for commonalities in the features of people's mental images of their cities. This resulted in the identification of five major characteristics:

1- Paths
2- Edges
3- Districts
4- Nodes
5- Landmarks

**Lesson # 13**

## CHARACTERISTICS OF COGNITIVE MAPS

A common approach to studying spatial cognition is to ask people to draw "sketch maps" of environments. Research indicates that sketch maps are a reliable method of data collection (Blades, 1990). Lynch (1960) conducted one of the first comprehensive studies of the nature of cognitive maps when he asked residents of three American cities (Boston, Los Angeles, and Jersey City) to draw maps of their city environments. He analyzed these drawings for commonalities in the features of people's mental images of their cities. This resulted in the identification of five major characteristics:

**(1) Paths:** Major arteries of traffic flow through the city (e.g., Main Street)
**(2) Edges:** Major lines (either natural or built) that divide areas of the city or delimit the boundaries (e.g., river)
**(3) Districts:** Large sections of the city that have a distinct identity (e.g., "Chinatown");
**(4) Nodes***:* Points of intersection of major arteries (e.g., the corner of Twelfth Street and Vine)
**(5) Landmarks:** Architecturally unique structures that can be seen from a distance and can be used as reference points (e.g., a tall building). Thus, the objective physical setting comes to be represented
as "cognitive space," organized and structured mentally in terms of distinct "regions" of the environment.
According to the **"anchor-point" hypothesis** (the regionalization and hierarchical organization of cognitive space is brought about by the active role of salient cues in the environment. For example, primary nodes or other reference points "anchor" distinct regions in cognitive space. These components or reference points provide the "skeleton" of the individual's map. As we shall see, the degree and accuracy of the detail of the remainder of the map is a function of both aspects of the environment and individual differences.

### Factors Affecting Cognitive Maps

**Environmental Differences:**
Environments differ from one another in the ease with which people can develop cognitive maps of them. Lynch (1960) coined the term *legibility* to refer to the extent to which the spatial arrangement of a city facilitates a clear and unified image in the minds of its inhabitants. For example, Boston provides a clear center, the Boston Common, around which people organize their cognitive maps. On the other hand, Los Angeles does not appear to have any central core, but sprawls out in all directions, which inhibits an organized mental representation. Milgram and his associates (Milgram, Greenwald, Kessler, McKenna, & Waters, 1972) argued that recognizable areas of an environment are important for developing accurate cognitive maps. They proposed the following formula for predicting the recognizability of an area:
$R=f\,(CxD)$.
This formula is read, "The recognizability of an area *(R)* is determined by its centrality to population flow (C) and its architectural or social distinctive-ness (D)." Thus, environments that have structures that stand out (such as a hilltop church) and are frequently passed by people facilitate a clear picture of the setting in the minds of the inhabitants.
Both environmental differences can be understood in terms of the "anchor-point" hypothesis in spatial cognition. That is, salient, objective physical cues in the environment facilitate the accurate organization of cognitive space, and the absence of such reference points inhibits accuracy.

**Errors in Cognitive Maps***:*

The physical characteristics of a setting are not the only determinants of the accuracy and detail of cognitive maps. People are prone to a number of cognitive errors in the process of developing cognitive maps. Downs & Stea (1973) point out that most of us form *incomplete* maps, leaving out both minor and major details; we tend to *distort* our maps, by placing some areas closer together than they actually are and others farther apart than they actually are; and, we sometimes *augment* our maps by
including elements that do not actually exist. Further, we often *give under prominence* to areas of the environment that are personally meaningful or important to us. For example, Saarinen (1973) asked students from different countries to draw maps of the world. He found that the students tended to draw their own countries in the center of the map, drawing them larger than countries that are larger than their own. Other recent studies have demonstrated errors in estimations of the differences in elevation between locations (Garling, Book, Lindberg, & Arce, 1990) and memory for turns of varying angularity encountered during pathway traversal (Sadalla & Montello, 1989).

The above errors are most likely due to limitations in human spatial cognitive abilities, rather than to objective environmental characteristics. As with any kind of information-processing task, the accuracy of our spatial knowledge of the environment is unlikely to ever be complete, and some tasks are easier than others. For example, Teske & Balser (1986) asked individuals to identify various destinations in a city and their strategic ordering (i.e., plan itineraries), and Veitch & O'Connor (1987) asked students to do the same on their college campus. Subjects found planning itineraries more difficult because this requires a higher-level cognitive organization. That is, while the former task requires knowledge only of the route from point A to B, the latter also requires knowledge of the route from B to C and the interrelationships among A, B, and C. Moeser (1988) has suggested such "survey maps" do not automatically develop in complex environments. She reported that student nurses failed to form survey maps of a hospital with a unique configuration, even after traversing it for two years. Further, these students performed worse on objective measures of cognitive mapping than did a control group of "naive" college students who were first asked to memorize the floor plans of the building.

Finally, Stanton (1986) investigated the relationship between "socio-spatial neighbourhood" (the perception of a street network without continuous boundaries) and the experience of "home ground" (defined as the mental form and geographical extent of those places that evoke a feeling of being near home). She reported that only residents of city blocks that were no more than 460 feet long were able to think of their home ground as an experiential network, concluding that there may be a mental time limit to such cognitions. Thus, these studies all suggest that general cognitive limitations of information processing • are involved in spatial cognition as in any other type of complex task, and these limitations are a major source of errors in cognitive maps.

**Individual Differences:**

Investigators have also suggested that some people seem to be better at forming cognitive maps than others. For example, gender differences have been reported. Appleyard (1970) reported that the cognitive maps of men are generally more accurate than are those of women. More recently, Ward, Newcombe, & Overton (1986) examined how men and women gave directions from maps that had been memorized. Male subjects exhibited higher levels of cognitive organization, such as using more mileage estimates and cardinal directions (i.e., east, west, north, and south) and made fewer errors of commission or omission than did female subjects. Antes, McBride, & Collins (1988) reported that distance judgments of women were more affected by a change in travel paths through a city occasioned by the construction of a new connecting street than were men. They suggested that women based their judgments on inferences from travel paths, while men approached the task in a more spatial manner. Orleans & Schmidt (1972) reported that women's maps were more detailed for the home and neighbourhood than were those of men, whereas men's cognitive maps were more comprehensive and complete for the larger surrounding environment. Finally, some investigators have reported socioeconomic differences, suggesting that the cognitive maps of people low in socioeconomic status are also less complete and accurate than are those individuals of higher socioeconomic status (Goodchild, 1974; Orleans, 1973).

Note that the individual differences listed above may not be due to differences in ability, but rather to differences in ***familiarity***. That is, there is much evidence that people draw more detailed and accurate maps of areas with which they have had more experience (and thus are more familiar to them) than areas where they have spent little time (e.g., Appleyard, 1970; Evans, 1980; Holahan & Dobrowolny, 1978; Moore,1974). Of course, it stands to reason that we would have better images of settings that are familiar to us than of unfamiliar places. Indeed, some of the studies on errors in cognitive maps discussed earlier also indicated that the extent of error can be moderated by experience.

Moreover, typically there are differences in ***mobility*** among the groups discussed above (i.e., in opportunity for travel through the setting). Thus, for example, if a husband works and the wife stay at home, it is not surprising that the husband would develop a better cognitive map of areas beyond the immediate neighbourhood while the wife would develop a more detailed map of the local environment. Similarly, people of higher socioeconomic status have much greater mobility to gain experience in the larger environment than do individuals of lower socioeconomic status. These suggestions are supported by the research of investigators who have controlled for mobility (e.g., Appleyard, 1976; Karan, Bladen, & Singh, 1980; Maurer & Baxter, 1972). While the individual differences reported above could be due to inherent differences in spatial ability, this seems unlikely to be the major reason. A study by Pearson and lalongo (1986) measured spatial ability and environmental knowledge independently. Spatial ability accounted for only 14 percent of the variance in environmental knowledge. Thus, learning brought about by relevant experience in an environment is likely to be a more important determinant of the accuracy of cognitive maps than are the individual or cultural differences discussed above.

**Lesson # 14**
# ENVIRONMENTAL EVALUATION

The question of how we come to evaluate an environment favourably or unfavourably is a complex one, yet it is an important one for predicting behaviour. At a very general level, people prefer and approach environments they evaluate favourably and avoid environments they evaluate negatively. Environmental psychologists have dealt with the first question using a variety of operational definitions. For example, a favourable evaluation could be viewed as a preference for certain configurations of stimuli in some environments over that of other environments, a cognitive judgment of beauty, or as a positive affective reaction to the environment. Each of these approaches has led to the identification of different, though related, aspects of the physical environment as determinants of evaluative responses. The answer to the question "What aspects of the environment lead to a favourable evaluation?" is going to depend on the answer to the question "What is meant by a favourable evaluation?" Consideration of each of these approaches, as well as their implications for the relationship between environmental evaluation and behaviour follows.

**Kaplan's Model of Environmental Preference:**

Would you rather be here or there? Kaplan (1975) developed a model for predicting preferences of some environments over others. This model provides a link between environmental cognition and evaluation, in that it assumes an important dimension of environmental preference to be the informational content of those environments. For example, Kaplan (1979) suggested that one basis for preferences is the ability of the individual to "make sense" out of the environment and the extent to which the environment involves the individual by motivating him or her to try to comprehend it.

**Basic Assumptions of Kaplan's Model:**

Individual base their preference for certain environments upon two factors:

- Can they make sense of it.
- Is there something in that environment that motivates them to learn more about that particular environment.

Kaplan & Kaplan (1978) identified four important factors influencing our preferences:

1- **Coherence:**
The degree to which the environment is organized and harmonious, with clear patterns and relationships among its elements.

2- **Legibility:**
The degree to which the environment is easy to understand and navigate, with distinctive features and landmarks that help orientation and wayfinding.

3- **Complexity:**
The degree to which the environment offers variety and diversity of elements, colors, shapes, and textures, without being overwhelming or chaotic.

4- **Mystery:**
The degree to which the environment invites exploration and discovery, with hidden or partially revealed elements that arouse curiosity and interest.

**Impact of the Four Factors:**

- People prefer environments high on coherence and legibility to those that are low on these factors.
- Greater degree of complexity and mystery produce a preference for a setting.

Kaplan (1987) has recently argued for a biological basis to such preferences.

**Biological basis to Preferences:**

There is a survival value to preferring environments that offer informational advantages over others.

The Kaplans' recent research points to the importance of mystery in predicting landscape preferences (Kaplan, Kaplan, & Brown, 1989), but others have applied their model to predict preferences for urban environments (Her-zog, 1989) and interiors (Scott, 1989).

**Environmental Aesthetics:**

Aesthetic is actually our sense to appreciate beauty. Is beauty in the eye of the beholder? Apart from the intuitive appeal of the notion that beauty is a subjective and relative concept, psychologists have attempted to identify objective dimensions of environments that lead to judgments of their aesthetic appeal. The assumption is that although there may indeed be individual differences in judgments of beauty, it is still possible to identify commonalities in what most people consider to be beautiful. For example, most people consider the Grand Canyon to be beautiful and a junk yard to be ugly. By researching a variety of settings, it is possible to identify dimensions along which different environments judged to be beautiful are similar.

We will discuss Environmental Aesthetics in detail in the next lecture

**Lesson # 15**
**Environmental Evaluation (Affective Bases of Environmental Evaluation)**

**Environmental Aesthetics:**
Is beauty in the eye of the beholder? Apart from the intuitive appeal of the notion that beauty is a subjective and relative concept, psychologists have attempted to identify objective dimensions of environments that lead to judgments of their aesthetic appeal. The assumption is that although there may indeed be individual differences in judgments of beauty, it is still possible to identify commonalities in what most people consider to be beautiful. For example, most people consider the Grand Canyon to be beautiful and a junk yard to be ugly. By researching a variety of settings, it is possible to identify dimensions along which different environments judged to be beautiful are similar.
Berlyne (1960) conducted one of the first important series of studies along these lines. He identified four basic collative properties of environments.

**Collative Properties of Environment:**
Characteristics of the environment that cause us to compare present settings to previous settings we have encountered.

1. **Complexity**
It refers to the degree of variety in the elements of the environment
2. **Novelty**
is the extent to which stimuli not previously encountered or noticed are present.
3. **Incongruity**
Concerns the extent to which the environment contains stimuli that do not seem to "go together" harmoniously
4. **Surprisingness**
It refers to the environment that contains elements that we do not expect to be present.

**Berlyne Arguments 1974:**

- Aesthetic judgments are most positive for environments at intermediate levels of each of these dimensions.
- we judge either too much or too little complexity, novelty, incongruity, or surprisingness of an environment.

**Wohlwill (1976)**
Wohlwill suggested that judgment of beauty increase consistently with increases in novelty and surprisingness, as well as with decreases in incongruity.

**Collative properties of Environment and Exploratory Behaviours**
There are two types of exploratory behaviours:

1. **Specific Exploration**

Specific exploration increases as the level of uncertainty in the environment increases. Presumably, uncertainty creates arousal, which leads to specific exploration to identify the source of the arousal.

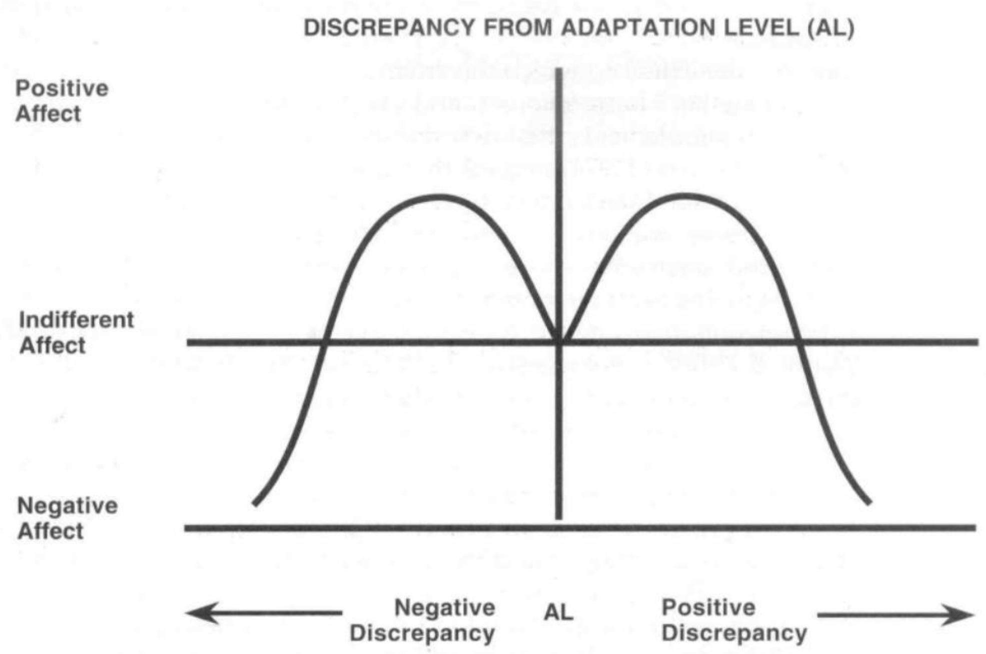
2. **Diversive Exploration**

Diversive exploration occurs when uncertainty-arousal levels are low. In other words, when the collative properties of an environment are at low levels, we feel under stimulated or bored, and we search the environment for ways to increase our arousal level.

**Affective Bases for Environmental Evaluation**

Affective responses to environment are determined by degree of discrepancy between current levels of stimulation and the adaptation level (i.e. the level of stimulation, we have become accustomed to).

**Wohlwill's Butterfly Curve Hypothesis:**




Wohlwill proposed the butterfly curve hypothesis predicting a curvilinear relationship between positive affect and discrepancies resulting in either increases or decreases in arousal from adaptation level (see Figure above). That is, we view moderate increases or decreases in stimulation from adaption level as pleasant. However, extreme deviations in either Direction result in negative affect. This approach is useful in accounting for individual differences in evaluations of a setting. Individual differences in adaption level result from living in different environments, leading to what might be called the "one person's ceiling is another person's floor" effect. That is, the same environment might be perceived as overstimulation to one person while under stimulating to another. For example, a person raised in a small rural town in Iowa might perceive a city of 20,000 people overwhelming, while the same city might be perceived as boring to a person raised in Chicago. Note, however, that both individuals would experience negative affect, because the discrepancies from their adaptation levels are extreme, even though in different directions.

**Mehrabian & Russell (1974)**

The work of Mehrabian & Russell (1974) has added considerably to our understanding of the relationship among stimulation levels, arousal, and positive/negative affective responses to environments. Their model accounts for the relationship between these variables and behavior. Mehrabian and Russell propose the concept of **information rate** to define environmental stimulation level. Information rate refers to the average amount of information impinging on the senses per unit of time. This concept can be used to integrate the various dimensions of environmental stimulation discussed above, such as complexity, novelty, incongruity, surprisingness, mystery, and coherence, in that all of these dimensions contribute to the information rate of a setting. Mehrabian & Russell(1974) present a great deal of research demonstrating that **arousal** is a direct correlate of information rate, and that approach behaviors in an environment (i.e., seeking out or desiring to remain in a setting) are greatest for intermediate levels of arousal. However, Mehrabian and Russell suggest that the curvilinear relationship between approach and arousal is moderated by the degree of pleasure a person experiences. Specifically, this relationship appears to hold only when pleasure is at an intermediate level. When pleasure is extremely high, approach behaviors strengthen with increases or decreases in arousal, and when pleasure is extremely low, both increases and decreases in arousal lead to avoidance behaviors.

We will discuss remaining part of Affective Bases for Environmental Evaluation in detail in the next lecture.

**Lesson # 16**
**Environmental Attitudes**

**Russell & Pratt Model of Emotional Reactions to Environment:**
Russell & Pratt (1980), proposed a model of emotional reactions to environments in which arousal and pleasure are viewed as independent dimensions. This is illustrated in Figure below. As can be seen, the model depicts all possible combinations of pleasure and arousal. Thus, environments high in arousal can be perceived as pleasant (i.e./'exciting" settings, such as a football game), as can environments low in arousal (i.e./'relaxing" settings, as a picnic in the park). Note that we are likely to engage in approach behaviors to both types of environments. The same can be said of avoidance behaviors to unpleasant environments high in arousal.


FIGURE: Circular ordering of eight terms to describe the emotional quality of environments (from Russell and Pratt, 1980).
("Distressing" settings, such as a final exam) and low in arousal ("gloomy" settings, such as a funeral). This circumplex model has been recently demonstrated to reliably differentiate between people's experiences in and preferences for suburban parks (Hull & Harvey, 1989).

**Environmental Attitudes:**
The relationships among the processes of environmental perception, cognition, and evaluation as well as their effects on behaviour can be best summarized in the notion of environmental attitudes.
**Attitude:**
Attitude involves the way we think about, feel about and behave towards an object. Attitude consists of a cognitive component, affective component, and behavioural component.

- **Cognitive component:**

Processes by which we perceive our environment and how the perceptual processes are involved in developing our cognition and understanding of the environment.

- **Affective component:**
  It refers to the emotional reactions or feelings an individual has towards an object, person, issue, or situation.
- **Behavioural component:**
  Particular characteristics of cognitions that have implications for one's overall attitude towards environment.

**Environmental Attitude Formation:**

Social psychologists have devoted considerable time and effort toward understanding the processes by which attitudes are formed. We will consider environmental attitude formation as a special case of the general process by which any attitude is formed. Most explanations of attitude development invoke learning principles such as classical conditioning, operant conditioning, and observational learning. Each of these principles will be discussed below, but we will first consider two important cognitive foundations of attitudes: **beliefs and values.**

**Beliefs:**

The way people feel towards things. Beliefs form the basis of our understanding and perception of the world.

**Values:**

Values are the characteristics patterns of behaving towards certain things. Values can include concepts like fairness, integrity, and respect for others, and they influence how people interact with the world around them

Daryl Bern (1970) has suggested that beliefs constitute the cognitive "building blocks" of attitudes.

**Types of Beliefs**

1. **Primitive beliefs**
2. **Higher order beliefs**

**Primitive beliefs:**

Primitive beliefs are non-conscious (i.e., they are accepted as givens, and are seldom consciously questioned). Primitive beliefs are either based on direct experience (e.g., the belief in the validity of our sense impressions) or on external authority (e.g., the belief that if Mommy says so, it must be true). The processes of sensation and perception discussed earlier are involved in the development of primitive beliefs. Because we have faith in our sensory impressions, if something smells or tastes bad, we will hold a negative attitude toward that thing. Alternatively, if Mommy tells us that we will get sick if we eat something, since we believe (at least as young children) that "Mommy is always right and never lies," we will also develop a negative attitude toward that thing.

**Higher order beliefs:**

These beliefs involve the insertion of a conscious premise in the thought process of arriving at the belief, i.e. the belief that air pollution is bad for health. For example, if I believe that air pollution leads to respiratory illness, it is unlikely that I arrived at that belief through direct experience or because Mommy said so. Rather, it is more probable that I heard a newscast reporting the results of a study conducted by the office of the Surgeon General warning of the health hazards of air pollution

We will discuss more about "Beliefs and its types" in next lecture.

**Lesson # 17**
**Environmental Attitude Formation**

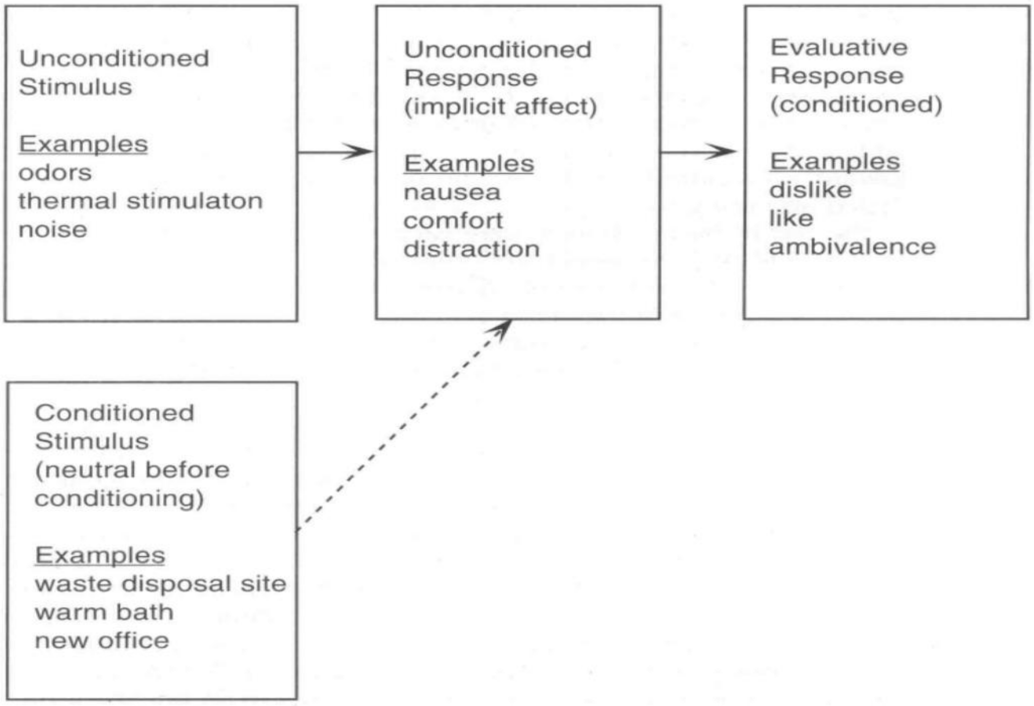
**Depth and Breadth of Beliefs**

**Depth:**
Many premises leading to the same conclusion.
**Breadth:**
Many syllogisms/ deductive reasoning leading to the same conclusion.

To complicate matters further, although the language of syllogisms, premises, and conclusions suggests that the process of arriving at higher-order beliefs is inherently logical and rational, this is not necessarily the case. Bern invokes the idea of psychological to account for irrational beliefs that are nonetheless internally consistent in the individual's cognitive structures. Further, it is possible to believe that something is bad for you but still evaluate the thing favorably. For example, I may know that smoking is bad for me, but simultaneously enjoy it (or alternatively, I may believe that exercise is good, but I may hate it anyway!).

**Values:**
Values can be defined as basic preferences for certain end states (see Rokeach, 1968). For example, **equality** is a value referring to an end state such as equal opportunity. Values serve as the functional basis for attitudes. That is, holding a particular attitude is a means for attaining a preferred end state. Thus, a person who strongly values equality is likely to be in favor of civil rights legislation. Similarly, if one values clean air, then that person is likely to evaluate actions to control pollution favorably. Indeed, Neuman (1986) has reported that values pertaining to environmental quality are positively related to beliefs about the efficacy and necessity of conservation and actual conservation behavior.

**Affective Bases for Attitude:**
Most social psychologists believe that the affective component of an attitude is learned. One important theory of attitude formation is The Byrne-Clore reinforcement-affect model (Byrne, 1971; Byrne & Clore, 1970). This model is based on principles of classical conditioning. Specifically, if an affectively "neutral" stimulus (called a conditioned stimulus) is paired with a stimulus that does elicit an affective response (called an unconditioned stimulus), and then the previously neutral stimulus will acquire the same ability to produce the response. The affective response to the unconditioned stimulus is referred to as an unconditioned response, because it occurs unconditionally (i.e., without the organism having to learn the response). The acquired response is called a conditioned response, because the organism must be conditioned to make the response in the presence of that stimulus (i.e., the organism must "learn" the response through association). This model can be readily applied to the conditioning of the affective component of environmental attitudes (see Figure below).




Do not think about the environment if the stimulation levels are within some optimal range). However, if some quality of the environment changes to deviate from the optimal stimulation level, a negative affective response automatically occurs. Through association, the entire environment comes to elicit the same negative affect. For example, when you first encountered your dormitory room, your affective response to the desk in the room (the "conditioned stimulus") was likely to have been neutral. However, if when you sat down to study at the desk, your neighbour turned rock music (the "unconditioned stimulus") on his or her stereo at full volume; you may have experienced negative affect (the "unconditioned response"). If this happened repeatedly, you would develop a negative affective response to your desk (the "conditioned response").

**Behavioural Bases of Attitudes**

**Operant Conditioning (Skinner, 1938)**

Any behaviour that is followed by a pleasant consequence will increase in frequency. Behaviours that lead to unpleasant consequences will become extinguished, that is, they will not be repeated. The expression of a particular attitude is a form of behaviour, and can therefore be operantly conditioned as any other behaviour. Thus, we learn to express attitudes that lead to favourable consequences for us, and we avoid attitudes that pose aversive consequences. For example, he used a specially designed box, known as the Skinner Box, to analyse the learning behaviour's of pigeons. In these experiments, pigeons were placed in the box and had to peck at a disc to receive food as a reward. Skinner manipulated various conditions to study the effects of different reinforcement schedules on the pigeons' behaviour.

We will discuss more about "Instrumental conditioning" in next lecture.

**Lesson # 18**
**Social Bases of Attitudes**

**Operant Conditioning (Skinner, 1938)**
Operant conditioning is a learning theory developed by B.F. Skinner that describes how behaviors are influenced by their consequences. It's based on the idea that behaviors followed by positive outcomes are more likely to be repeated, while those followed by negative outcomes are less likely to be repeated. Here's a detailed explanation:

**How Operant Conditioning Works:**
- **Reinforcement:**

This is any consequence that strengthens the behavior it follows. There are two types of reinforcement:

- **Positive Reinforcement:**

Adding a desirable stimulus to increase a behavior. For example, giving a child a treat when they clean their room.

- **Negative Reinforcement:**

Removing an aversive stimulus to increase a behavior. For example, buckling the seatbelt to stop the car from beeping.

- **Punishment:**

This is any consequence that decreases the behavior it follows. Like reinforcement, punishment can also be positive or negative:

- **Positive Punishment:**

Adding an aversive stimulus to decrease a behavior. For example, scolding a pet for misbehaving.

- **Negative Punishment:**

Removing a desirable stimulus to decrease a behavior. For example, taking away a teenager's gaming privileges for not doing homework.

**Social Bases of Attitudes**
Another important way in which attitudes are formed is through observation of other people's expression of attitudes and the consequences of their attitudes.

**Social Learning:**
The process of learning via observation of a model's behavior is known as social learning.

**Vicarious Learning:**
The means by which we are influenced by the consequences of the model's behavior is known as vicarious conditioning.

If an individual observes another person expressing a particular attitude, and also observes that the attitude led to favorable consequences, then the individual will imitate the model's behavior in anticipation of incurring the same favorable consequences. For example, if a person who works for an oil company observes a co-worker express opposition to a windfall profits tax,

and the co-worker is praised by the boss, then the individual is likely to imitate this opposition in anticipation of also gaining the boss's approval.
Another way in which social influences can impact attitudes is through the dynamics of interpersonal processes. Put simply, if my friends have pro-environmental attitudes, and I enjoy my friends' company, then I am likely to adopt similar attitudes. Manzo & Weinstein (1987) studied differences in active and nonactive members of the Sierra Club and reported significant differences in commitment to environmental protection as a function of club-related friendships. Thus, friends tend mutually to reinforce attitudes toward the environment.

**Attitudes and Behavior**

A major assumption of research on attitude formation is that attitudes mediate behaviors which are consistent with those attitudes. Thus, if I hold an attitude favoring energy conservation, it would be expected that this attitude would mediate behaviors such as walking instead of driving to work, or turning my thermostat down during the winter and up during the summer. Although this may seem like an obvious and reasonable assumption, in recent year's research has led social psychologists to question the degree to which attitudes do, indeed, reliably predict behavior (e.g., Wicker, 1969). These studies have suggested that there is much less consistency between measured attitudes and subsequent behaviors than had previously been assumed.

**Attitude-Behavior Consistency**

The major principle of consistency theories of attitude-behavior relationships (see Festinger, 1957) is that people strive to maintain logical consistency between cognitions about their attitudes and cognitions about relevant behaviors in which they engage. Thus, if a person has the opinion that air pollution should be brought under control, a logically consistent behavior would be to vote for measures requiring utility companies to install pollution-control equipment in energy plants.

**Cognitive Discomfort:**

Cognitive discomfort, often referred to as cognitive dissonance, is the mental stress or unease experienced when an individual holds two or more contradictory beliefs, values, or attitudes simultaneously. This state of tension arises when there is an inconsistency between what a person believes and how they behave.
In context to the above-mentioned example of air pollution, If this person knows that he or she voted against such a measure, the perceived attitude-behavior inconsistency would create a state of cognitive discomfort, which would motivate the individual to attempt to restore consistency. This would be accomplished either by changing the attitude (e.g., deciding that pollution control is not as important as controlling the costs of energy production), or by changing the behavior (e.g., by voting in favor of the next such measure).

**Lesson # 19**
**Impact of Environment on Individual (Personality Development and Individual Differences)**

**Personality:**
"Personality consists of all the relatively stable and distinctive styles of thought, behavior and emotional response that characterize a person's adaptations to surrounding circumstances." (Maddi 1976, Mischel 1976).
Personality can also be defined as a dynamic and organized set of characteristics possessed by a person that uniquely influences his or her cognitions, motivations, and behaviors in various situations. The word "personality" originates from the Latin word persona, which means mask. For example, someone might be described as outgoing, empathetic, and shy, indicating traits that make up their personality.

**Personality Development:**
Emergence of distinctive style of thoughts, feelings, and behaviour that make each human being a unique individual.

**Two questions for Personality Psychologists:**
1. Individual Differences

When several people encounter the same situation, why don't they all react alike?

2. Environmental impact

What is the relationship between human behavior and environment?

**Personality Research – Theoretical Frameworks**
- Trait Theory
- Murray's Theory of Personality
- Interactionist Perspective
- Metron's Notion of Self-Fulfilling Prophecy

**Trait Theories of Personality**
**Trait:**
Relatively enduring way in which an individual differs from others.
- Personality traits are relatively stable over time.
- Personality traits are consistent over situations.
- Intensity of traits.

1. **Gordon Allport Trait Theory:**

Allport & Odbert developed the first trait theory, and they categorized traits into three levels (1961).

**Cardinal traits**:
These traits dominate and shape a person's behavior, so much so that they seem to become synonymous with the person. These include the need for money, ambition, etc.
**Central traits**:
These traits are less dominating, and they are present in all human beings to different degrees. For example, "honesty" or "intelligence".
**Secondary traits:**
These traits are very specific behaviors or preferences that appear in certain contexts, say being impatient while waiting.

2. **Raymond Cattell**

Cattell reduced the size of Allport's list to make it more manageable and created a science of personality. In this way, he found out that personality consists of 16 distinct units, which he called source traits (1978).

3. **Hans Eysenck**

Hans Eysenck felt that even five factors were too many. So, he created a three-dimension model, which included extraversion, neuroticism, and psychoticism. Eysenck also claimed that individual differences are biologically rooted: introverts have central nervous systems that are more sensitive to stimulation and so avoid intense sources of excitement. (1967)

**Lesson # 20**
# Murray’s Theory of Personality Development And Interactionist Perspective

## Henry Murray and Psychogenic Needs
American psychologist Henry Murray (1893-1988) developed a theory of personality that was organized in terms of motives, presses, and needs. Murray described a needs as a, "potentiality or readiness to respond in a certain way under certain given circumstances" (1938)
Theories of personality based upon needs and motives suggest that our personalities are a reflection of behaviors controlled by needs. While some needs are temporary and changing, other needs are more deeply seated in our nature. According to Murray, these psychogenic needs function mostly on the unconscious level, but play a major role in our personality.

## Murray's Types of Needs

Murray identified needs of two types:
1. **Primary Needs**

Primary needs are based upon biological demands, such as the need for oxygen, food, and water.

2. **Secondary Needs**

Secondary needs are generally psychological, such as the need for nurturing, independence, and achievement.

## List of Psychogenic Needs
The following is a partial list of 24 needs identified by Murray and his colleagues. According to Murray, all people have these needs, but each individual tends to have a certain level of each need.

### 1. Ambition Needs
• Achievement: Success, accomplishment, and overcoming obstacles.
• Exhibition: Shocking or thrilling other people.
• Recognition: Displaying achievements and gaining social status.

### 2. Materialistic Needs
• Acquisition: Obtaining things.
• Construction: Creating things.
• Order: Making things neat and organized.
• Retention: Keeping things.

### 3. Power Needs
• Abasement: Confessing and apologizing.
• Autonomy: Independence and resistance.
• Aggression: Attacking or ridiculing others.
• Blame Avoidance: Following the rules and avoiding blame.
• Deference: Obeying and cooperating with others.
• Dominance: Controlling others.

### 4. Affection Needs
• Affiliation: Spending time with other people.
• Nurturance: Taking care of another person.
• Play: Having fun with others.

• Rejection: Rejecting other people.
• Succorance: Being helped or protected by others.

**5. Information Needs**
• Cognizance: Seeking knowledge and asking questions.
• Exposition: Education others.

**Interactionism: The interactionist perspective on the situation vs. person**
Traits and Situations interact to influence behavior - how else could it be? Its like the genetics vs. environment issue, one cannot exist without the other). So, the trait and situationist perspectives are too simplistic: reality is more complex. In reality, different situations affect different people in different ways. Some situations allow expression of personality; other situations provoke a narrower range of behavior. Thus, Behavior = personality x interpretation of the situation It is vital to appreciate that there are individual differences in the personality-situation relationship.
Research (Kenrid et al, 1990) has shown that a trait will show up only in a situation where it is relevant. So anxiety may show up as a predictor of behavior some situations, and not others. Also, some situations allow expression of personality; others provoke narrower range of behavior.

**Mischel's Contention (1968)**
Human behaviour across situations is unpredictable in terms of personal factors alone.
Walter Mischel's contention revolves around the interaction between individuals and their environments. He suggested that behavior is not solely a result of internal traits but is significantly influenced by the external situation and context. Mischel argued that an individual's behavior is better predicted by understanding the specific circumstances they are in, rather than relying on broad trait-based predictions.
Mischel's view challenges traditional trait theories that suggest personality traits are consistent across different situations. Instead, he posited that there is a dynamic interplay between the person and the situation, where the environment can shape behavior just as much as an individual's personality traits can.
It emphasizes the role of situational factors in shaping behavior and suggests that modifying the environment can lead to changes in behavior. Mischel's work encourages a more integrative approach that considers both personal characteristics and environmental influence.

**Behavior Contingency Units**
Behavior Contingency Units are the close association between a person's behavior and the stimulus or cue properties of the setting in which it occurs. Substantial proportion of behavioral variance is accounted for by situational variables.

**Bowers, Bem and Allen's Viewpoint**
- Situations are as much a product of the person as the person's behavior is function of the situation.
- The individual's capacity to shape the environment is evidenced quite clearly through behavioral processes such as art, architecture and community planning.
- Cognitive processes also play significant role in structuring the environment.

**Metron's Self Fulfilling Prophecy Self Fulfilling Prophecy**
Person's expectations about other people that lead him to act in a way that brings out the traits that he expects them to have. This is where interactionsit view of the environment and behavior

assumes a dynamic interchange between man and the environment in which people affect, and are affected by their settings.

**Lesson # 21**
**ENVIRONMENTAL CHANGES AND STRESS**

Humans are incredibly adaptable; when not satisfied with their lot, they have the intelligence and ingenuity to create new things, to adapt to what is available, and even to adjust or alter their living environment to make it more congenial. But this flexibility is daily being challenged. Forces from within the species (e.g., violent crimes, war, acts of terrorism, and genocide), widespread natural catastrophes (e.g., famines, floods, droughts, earthquakes, and volcanic eruptions), and ever increasing and dangerous technological developments (e.g., faster automobiles, increasing numbers of aircraft attempting to occupy the same air corridors, escalating numbers of chemicals, and the proliferation of nuclear devices) are coalescing to test the limits of human adaptability. Additionally, interpersonal forces demanding more material goods and greater and more efficient provision of services are increasingly straining limited physical and human resources.
Thus, over and over again in the course of daily living, we witness threats to human adaptability, feel pressure for increased ingenuity to provide protection from external forces, and struggle to reconcile material possessions with individual desires. The operation of these forces is inextricably tied to what has commonly come to be labelled **stress.**

**Technological Advancements:**
The rapid pace of technological progress, while offering numerous benefits, also introduces new risks. The increase in high-speed transportation options, such as faster automobiles and densely populated air corridors, has led to heightened potential for accidents and collisions. The chemical industry's expansion has resulted in a vast array of substances entering the environment, many with unknown long-term effects on human health and ecosystems. Furthermore, the spread of nuclear technology raises concerns about safety, waste management, and the possibility of catastrophic events.

**But what exactly is stress?**
Personal accounts of stress, scales for measuring it, instructions for coping with it, and personality tests to see how well one can endure it have found their way into print and are supplemented by TV and radio accounts featuring self-proclaimed "experts" in its early detection and management.

**Defining Stress: Theoretical Perspectives**
Stress has been defined as a state that occurs when people are faced with demands from the environment that require them to change in some way. Most researchers agree with this definition. What they do not agree on is whether stress is the demand itself or the person's response to that demand.

**Theoretical Models of Stress**

Several models have been proposed to explain the stress phenomenon:

• **The James-Lange Theory of Emotion**: Suggests that emotions arise from physiological responses to events. According to this theory, stress would be considered the physical response to a demand.

• **The Cannon-Bard Theory:** Posits that emotions and physiological responses occur simultaneously but independently. Here, stress could be seen as both the demand and the response.

• **The Schachter-Singer Theory:** Asserts that emotions depend on physiological arousal and the cognitive interpretation of that arousal. In this view, stress involves an element of personal interpretation or appraisal of the demand.

**Response Based Definition of Stress**

The response-based definition of stress emphasizes the physiological & psychological reactions that occur when an individual is faced with demanding or threatening situations.

- **Physiological Responses:**

Response of the body to any demand, whether it is caused by, or results in, pleasant or unpleasant conditions." According to this definition, stress is conceptualized chiefly in terms of the body's physiological reaction to any demand placed on it, like galvanic skin response, change in heart rate and blood pressure etc.

**Key Physiological Changes During Stress**

• **Cardiovascular System**: During stress, heart rate and blood pressure increase to supply more oxygen and nutrients to the muscles, preparing the body for action.

• **Respiratory System:** Breathing becomes more rapid to distribute oxygen quickly throughout the body.

• **Musculoskeletal System:** Muscles tense up, readying the body for physical activity.

• **Endocrine System:** Stress hormones like adrenaline and cortisol are released, triggering the body's alert systems

- **Psychological Responses:**

The psychological response to stress typically involves a range of emotional and cognitive reactions. These can include feelings of anxiety, worry, or fear, as well as difficulty concentrating, confusion, loss of individual control, and loss of self esteem.

**Emotional Reactions Under Stress**

• **Anxiety:** A heightened state of alertness that prepares an individual to face potential threats.

• **Worry:** Persistent concern over stressors that may be real or imagined.

• **Fear:** An intense emotional response to immediate threats, which can activate the fight-or-flight response.

**Cognitive Reactions Under Stress**

• **Concentration Difficulties:** Stress can impair the ability to focus, leading to decreased productivity and performance.

• **Confusion:** The overwhelming nature of stress can cause disorientation and indecisiveness.

- **Loss of Control:** A perceived inability to manage one's circumstances or outcomes.

- **Self-Esteem Issues:** Chronic stress can deteriorate an individual's confidence and self-worth.

**Objections to Response Based Definition**

A major problem with this approach is that these same responses occur as a result of very different stimuli (e.g., heart rate may increase as a result of physical exercise, viewing a horror movie, riding a roller coaster, or waiting for a blind date). Likewise, the psychological manifestations of anxiety, loss of felt control, and lowered self-esteem.

**Stimulus-Based Definition of Stress:**

Environmental events that impact on any of several response systems either immediately or as a result of prolonged exposure.

Other theorists have argued for a stimulus-based definition of stress. They argue for taxonomy of environmental events based on their covariation with systemic, psychological, and/or social disturbance, and include such events as noise, air, and water pollution, population density, odors, loss of loved ones, and changes in life style.

A stressful environment, for example, is one where noise level exceeds some specified decibel (dB) level, or where the carbon dioxide concentration equals or exceeds some agreed-upon value, or where environmental occupants exceed some number.

**Problems with Stimulus Based Definition**

It can be seen that a major problem with this view is that any stimulus may or may not be disturbing (stressful) depending on the personality of the individuals involved, the situation in which the event occurs, other competing behaviors, and the rewards and costs involved in dealing with the event. For example, a loud stereo can be enjoyable at a party, but very disturbing (stressful) if you are preparing for final exams; likewise, both high population density and isolation can be stressful or enjoyable depending on the amount of contact desired.

**Lesson # 22**
**Stress As Cause and Effect**

It is possible to think of stress as both something that is happening to a person and the person's response to what's happening. It involves environmental and psychological events, the interpretation of these events, and behavioral as well as physiological responding. Noisy environments, for example, may be related to physiological, psychological, and behavioral changes in those exposed to the noise. These responses, in turn, may change the nature and interpretation of the noise itself (i.e., noise changes neural activity in the reticular activating system, which subsequently changes the organism's perception of the noise; this altered perception, in turn, influences reticular activating system activity, and so on). **Stress, therefore, is neither the stimulus nor the response; it is a process involving both, and, as a process, it influences the ways in which environmental events are attended to, interpreted, responded to, and changed. It is also the process within which the responder also is likely to be changed.**

**<u>The Cyclical Nature of Stress</u>**

- **Environmental Stimuli:** Situations like noisy environments act as initial stressors.
- **Physiological Response:** These stressors trigger changes in the body's physiology, such as alterations in the neural activity of the reticular activating system.
- **Psychological Interpretation:** The individual's perception of the stressor is modified, which can either amplify or diminish the stress response.
- **Behavioural Adaptation:** The individual's behavior adjusts in response to the altered perception, potentially leading to a change in the environment itself.

While the specification of just what stress is has yet to be fully articulated, it is clear that such a thing actually exists, and it appears to involve physiological, psychological, and behavioral responding. It is also clear that at times it may even play a broader role by affecting human social systems. Confronted by environmental events which pose threat, challenge or danger, organisms respond physiologically, psychologically, and behaviorally. These responses not only are helpful in meeting the demands of the changing environment, but may even alter that environment, making it more benign (not always without cost to the organism).

**Monat and Lazarus's Definition of Stress (1977)**
They defined stress as any event in which environmental demands and/or internal demands (physiological or psychological) tax or exceed the adaptive resources of the individual, his or her tissue system, or the social system of which one is a part. Within this definition, change becomes stressful only when it strains the coping capacities of the entire organism or environment system to adapt to the change. This definition encompasses aspects of both the stimulus and the response, and includes the organism as an active participant in the process. It is therefore consistent with definitions utilized by most current researchers and is the one that will be used when referring to stress throughout the remainder of this text.

**The Role of Perception in Stress and Adaptability**
Perception plays a pivotal role in the relationship between organisms and their environment. Typically, the environment is perceived to be within the organism's "window of adaptability," allowing for a steady-state interaction where the organism can maintain equilibrium. However,

the definition of stress acknowledges that this balance is not always maintained and that states of disequilibrium can arise, challenging the organism's adaptive capabilities.

**Understanding Disequilibrium and Adaptive Resources**

Disequilibrium occurs when environmental demands tax or exceed the adaptive resources of an organism or its environment. This can lead to stress, which is a state of tension resulting from adverse or demanding circumstances. The organism's perception of these demands plays a crucial role in the stress response.

**Appraisal:**

Appraisal is the process by which an organism evaluates the demands of the environment and assesses its own resources to cope with these demands. This cognitive evaluation is fundamental in determining the level of stress experienced and the subsequent response to it. Appraisal involves two key components:

**1. Primary Appraisal:** The process of recognizing and assessing the significance of an environmental stressor.

**2. Secondary Appraisal:** The assessment of one's resources and options for coping with the stressor.

**Attitudes and Perception in Appraisal**

Attitudes towards potential stressors play a crucial role in appraisal. For instance, individuals who view nearby airports as economically beneficial are likely to be less bothered by the associated noise compared to those who do not recognize such advantages. This illustrates how positive attitudes towards certain elements can lead to a downplaying of potential threats.

**Types of Appraisals**

There are three fundamental types of appraisals

1. Harm or loss assessments
2. Threat Appraisal
3. Challenge Appraisal

**Harm or loss assessment:**

It typically involves analysis of damage that has already been done. The properties of a sudden event such as a tornado may predispose people toward such appraisal because damage is done very quickly and people are more concerned with the immediate consequences than the possibility of more. Harm-loss evaluation in the wake of the loss of a loved one, although when a loved one has been chronically ill for an extended period of time, bereavement may also occur in anticipation of loss. In addition to the actual loss of a loved one, we may be concerned with demands that will occur after death. Concern over these types of demands can be interpreted as challenges or as threats, that is, "Let's get on with life," or, "How will I get on with life?"

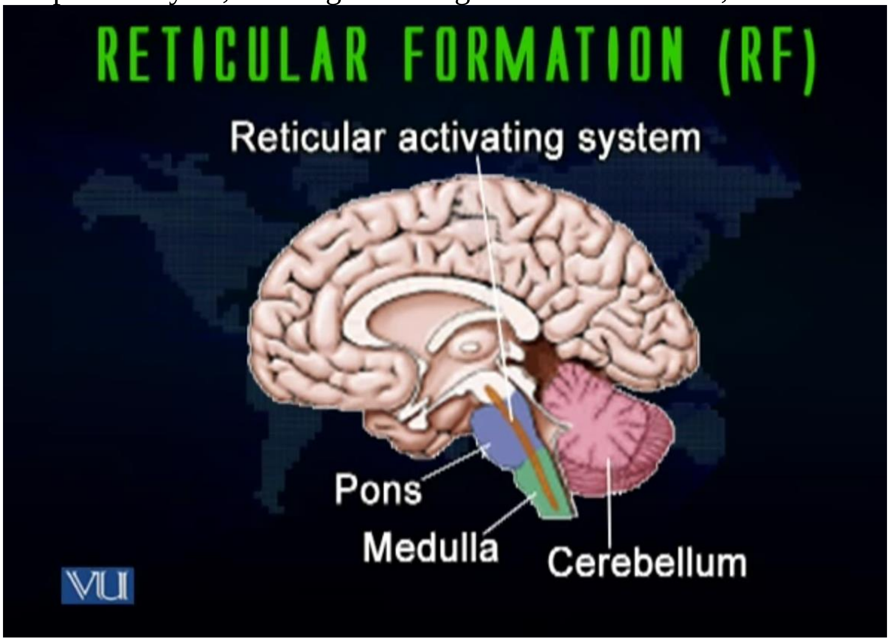
**Threat Appraisal:**

It is concerned with future dangers. If a tornado is sighted, it may initially be appraised as a threat and subsequently be appraised as something else. The stress of moving away to college, of learning to live with a roommate, and similar events is largely anticipatory as a student prepares to start school. Likewise, waiting to take an exam may be more stressful than taking it or even failing it. The ability to foresee problems and anticipate difficulties allows us to solve them or prevent their occurrence. At the same time, though, it may lead to the perception of threat and, thus, anticipatory stress.

**Challenge Appraisal:**
It focusses not on the harm or potential harm of the event, but on the possibility of overcoming the stressor. Some stressors may be beyond our ability to cope, but we all have a range of events with which we are confident of our ability to cope successfully. Stressors that are evaluated as challenges fall within this hypothetical range. The event may be seen as potentially harmful, but we feel that we can prevent the harm from occurring. A person may have just lost his or her job because of plant relocation. This stress can be seen as threatening (how are we going to make ends meet; how will we survive?) or as challenging (what else can I do to make a living; how can I make the best of a bad situation?).

**<u>Arousal</u>**
Arousal, refers to the physiological and psychological state of being alert and ready for action. It is a necessary component of the stress response, preparing the body to face potential threats or challenges.

**Reticular Formation:**
The reticular formation is a diffuse network of interconnected neurons located throughout the brainstem, extending from the spinal cord to the thalamus. It plays a vital role in controlling the overall level of consciousness and is involved in various functions such as regulating the sleep-wake cycle, filtering incoming stimuli to the brain, and maintaining alertness.




**Reticular Activating System:**
The Reticular Activating System (RAS) is a crucial neural network within the brainstem that extends to the midbrain and forebrain. It serves as a sentinel, filtering incoming stimuli and regulating wakefulness and consciousness.
The RAS performs two primary functions:

1. **Information Screening:**

It screens information on its way to the higher canters in the brain, blocking irrelevant information and allowing relevant information to pass upward where it can be processed and acted upon.

2. **Alerting the Cerebral Cortex:**

Activating the cerebral cortex into a state of alertness without specifying the nature of the alert.

**According to Beck (1983)** the reticular system works "something like a fire alarm that gets people into action but does not really say where the fire is". Through this system the organism is made vigilant and aware of what is happening in the environment, and is made ready for action. Included in this readiness are "increased metabolism of carbohydrates to produce more glucose and the release of fatty acids for greater energy, higher heart rate and oxygen consumption, constriction of blood flow to peripheral areas of the body with greater supply to the skeletal muscles, kidneys and brain" (Evans & Cohen, 1987; p. 576). This increased readiness, coupled with appropriate information about bodily needs and environmental demands, plays an important role in determining the ultimate expression of behavior.

## LESSON 23

## PHYSIOLOGY OF STRESS

We will begin lesson 23 by reviewing lecture 22 and further discussing the following topics related to Stress:

- Definition of Stress. (already discussed in detail in lecture 22)
- Cause and Effect. (already discussed in detail in lecture 22)
- Appraisal (already discussed in detail in lecture 22)
- Arousal from the perspective of the Physiology of Stress.

### Physiology of Stress and Arousal

Stress is a natural response to various stimuli in our environment, whether positive or negative. When faced with demands or changes in our surroundings, our bodies undergo physiological changes to adapt and respond effectively. These changes are orchestrated by complex systems within our brains and bodies. It is crucial to understand the physiology of Stress, as when we experience stress, our body undergoes various changes such as increased heartbeat and blood circulation. These bodily changes are reactions that prepare us to deal with environmental stressors. This process is commonly referred to as arousal. So, let’s delve into arousal from the perspective of the physiology of stress.

### Arousal

Arousal refers to a stirred-up state of mind in which we are inclined towards certain types of actions. In this state, we are either thinking about or being driven toward specific actions to address a scenario. Arousal can result from excitement, anger, or fear, signaling that our body is ready for action.

### Adaptation: Hormones and Biological Systems

Arousal serves as our defense mechanism to cope with changes in our environment, preparing us to deal with significant and drastic environmental changes. Therefore, there must be a system within our body that triggers this entire mechanism. This system includes various organs and components, such as:

- Different types of hormones.
- Different types of organs in the central, peripheral, and autonomous nervous systems.

The nervous system components involved in this process include the hypothalamus, cerebral cortex, reticular formation, limbic system, and autonomic nervous system. These organs and systems activate to prepare us to address environmental challenges.

### Reticular Formation

The reticular formation plays a crucial role in the physiology of arousal. Its primary function is to induce a state of readiness in the organism, indicating that the individual is prepared and quick to act. For example, before the start of a race, athletes assume their positions and await the starting signal. At this moment, they are in an increased state of readiness, with their bodies primed for action. Inside their bodies, the reticular activating system (RAS) functions are triggered, resulting in heightened metabolism rates and increased heartbeats to supply energy to the muscles. In such situations, blood flow is directed towards the muscles, while blood flow to internal organs like the kidneys or liver decreases. This physiological response prepares the athlete to address the scenario effectively.

**Levels of Arousal**

As arousal refers to the state of readiness or activation level of the body and mind in response to environmental stimuli. It can vary from low to high, with each level affecting our ability to perform tasks effectively. Understanding these levels is essential for optimizing performance and managing stress.

**Increased Readiness:** Extreme events or stimuli trigger a heightened state of arousal, characterized by increased heart rate, tremors, and heightened physiological responses.

**Medium Level:** This optimal state of arousal enables individuals to perform tasks efficiently and adapt to environmental demands without feeling overwhelmed or fatigued.

**Lowest Level:** A state of low arousal may lead to feelings of lethargy and disengagement, hindering performance and motivation.

**The Role of Arousal in Performance**

A crucial aspect of arousal is its impact on performance. While moderate levels of arousal can enhance performance by promoting alertness and focus, excessive arousal or stress can impair performance and lead to negative outcomes. Striking a balance between arousal and performance is essential for achieving optimal results.

**Examples:** Every time we experience stress, our body undergoes similar responses, entering a state of increased readiness. This heightened state prepares us to deal with actions in our environment. For instance, imagine driving along a main road when suddenly, a car speeds towards you, threatening a collision. Your reflexes kick in, and your body reacts swiftly to avoid the danger. In such moments, your heart races, and your body trembles. However, extreme readiness, akin to soldiers facing life-threatening situations, can also impair performance.

On the other side both extreme readiness and extreme lack of readiness negatively impact performance. Demoralized students lacking motivation represent one extreme, while soldiers facing imminent danger represent the other. Both scenarios hinder optimal performance. Consider a demoralized student who, with proper motivation, moves toward the middle of the readiness continuum. Here, motivation to study increases, leading to improved performance. Conversely, extreme readiness, akin to the state experienced during an imminent accident, hampers performance due to overwhelming emotion. Therefore, maintaining a balanced state of readiness is crucial for optimal performance, whether in academic or life-threatening situations. Maintaining a middle-level arousal is essential for peak performance, through maximum focus and concentration it can be achieved.

It is important to note that the Reticular Activating System (RAS) triggers these mechanisms, with readiness being both biochemical and psychological. On one side Motivating students to boost morale and move them toward the middle of the continuum is vital which is the psychological aspect. Additionally, mood significantly affects performance, highlighting the role of cognitive perception in behavior. On the other side, damage to the RAS leads to a comatose state, while drugs like amphetamines increase RAS activity and barbiturates depress it. So, it is important to study every aspect of arousal, whether it is neurochemical, psychological, or cognitive.

Hence, Arousal is closely related to stress and alerts organisms to changes in bodily needs and environmental demands. Understanding how arousal translates into goal-directed behaviors is also crucial for comprehending human behavior's influence by the external environment.

**Arousal and Goal-Directed Behavior**

We behave in certain ways to achieve our goals or objectives, and the environment plays a significant role in this process. For example, when entering a mall, the environment, such as billboards, induces an arousal state in individuals, directing them towards their goal, such as shopping.

**General Adaptation Syndrome**

Stress is not inherently negative; it is our adaptive reaction to the environment, enabling us to either change ourselves or the environment to establish a better equilibrium between the two.

**Structures of Cerebral Cortex**

The limbic system, housed inside the skull, comprises four important parts:

**Thalamus:** Relays information from sensory organs to the cerebral cortex.

**Hypothalamus:** Regulates feelings of fear, thirst, sexual drive, and aggression.

**Amygdala:** Influences motivation, emotional control, fear response, and interpretations of non-verbal emotional expressions.

**Hippocampus:** Plays a role in emotions, memory, and comparing sensory information to expectations.

In conclusion, our sensory organs process information, converting it into neurochemical energy that is transmitted to the relevant brain centers for processing. Additionally, glands, such as the pituitary gland, play a vital role in performing various bodily functions. Both systems are crucial for our body's functioning.

**Pituitary Gland:**

Situated within the brain, the pituitary gland serves as a crucial producer of human growth hormones and plays a pivotal role in regulating other hormone-producing glands. Referred to as the "master gland," the pituitary gland integrates incoming information from various sensory organs within the body. This information undergoes processing in the brain, which subsequently issues instructions. These instructions are then carried out with the assistance of glands, which produce different secretions to influence the functioning of internal organs.

The pituitary gland, as the master gland, not only activates other glands but also regulates the secretion of growth hormones, ensuring consistent growth from birth onwards. Growth is a fundamental physiological function that the pituitary gland oversees. Additionally, the pituitary gland controls the hormonal secretions of other glands, maintaining hormonal balance in the body.

The nervous system and glands work in tandem as a cohesive unit, preparing the body to respond to environmental changes and stimuli. This collaborative effort ensures that the body can adapt and function optimally in varying environmental conditions.

In the next lecture, we will study the General adaptation system in detail.

**Lesson 24**

**General Adaptation Syndrome**

In today's lecture, we'll delve into the intricate world of environmental psychology, focusing specifically on the physiological responses to stress and arousal considering general adaptation syndrome. Let's break down the key concepts that will be discussed in detail:

- Physiology of stress (Studied in lesson 23 in detail)
- General Adaptation Syndrome

**Stimulus-Response Pattern: A Closer Look**

When we encounter stimuli, whether positive or negative, our brain evaluates the information received through sensory organs. This evaluation triggers certain feelings and thoughts, leading to action. However, this process is not simplistic; it involves multiple stages, including stimulation, feelings, thoughts, and ultimately, action.

**Hans Selye's Adaptation Syndrome**

In 1976, psychologist Hans Selye introduced the concept of adaptation syndrome, proposing that stress unfolds in three distinct stages:

1. Alarm Reaction
2. Stage of Resistance
3. Stage of Exhaustion

**1. Alarm Reaction**

In response to any stressor, either physical or psychological, the hypothalamus is activated, mediating the secretion of large amounts of ACTH by the pituitary. This ACTH, in turn, stimulates the adrenal cortex to secrete increased amounts of adrenal corticoids. In general, these hormones activate the organism (as discussed in the previous section) allowing it to deal more adequately with its environment. This phase is called the alarm reaction.

**2. Stage of Resistance**

In the second stage, the organism recovers from the initial stress and begins to attempt to cope with the situation, mobilizing the body physically as well as psychologically to meet the demands of the environment. The organism, in a sense, is resisting the demands of the situation. This stage was, therefore, labeled by Selye as the stage of resistance.

**3. Stage of Exhaustion**

If the organism is unsuccessful in its attempts to cope, or if the stress persists, the stage of exhaustion is reached. At this stage the adrenal gland can no longer respond to the stress by secreting adrenal corticoids and the organism has exhausted its ability to cope with the stressor.

Important issues for consideration in this area are: how the stress of overpopulation is related to increased adrenal activity and to adrenal hypertrophy; how this has an inhibiting effect on gonadal functioning resulting in a decline in reproductive fitness, and ultimately to a decline in population and population density.

Evans and Cohen (1987) suggest that chronic exposure to a variety of environmental and/or social psychological conditions can in and of themselves be stressful, that the process of coping with them can be stressful, and that the " energy available for dealing with these conditions is limited. Thus, like Selye, they suggest that prolonged stress will eventually deplete the individual's adaptive resources.

Both approaches imply that stress produces a physical "wear and tear" on the system, and when coping abilities are exhausted, the individual becomes vulnerable to a variety of physical and psychological disorders. For further discussion and a comparison of the psychological and physiological models of stress see Baum, Singer, & Baum (1981), Cohen, Evans, Stokols, & Krantz (1986), and Evans & Cohen (1987).

Organisms, including humans, undergo stress in the context of the greater environment of which they are a part. At any given moment the environment/ organism system can be in a state of equilibrium on some dimensions, but in disequilibrium concerning others. It is therefore important to be able to identify potential sources of stress and to determine the levels of those sources likely to lead to disequilibrium.

There is growing awareness of the importance of psychological factors (i.e., cognitive, and emotional processes) in response to the environment. Additionally, such factors as beliefs, attitudes, and perceptual sets may, themselves, act as threats. For these reasons, researchers concerned with understanding stress are increasingly considering the impact of psychological variables and are incorporating them into stress theories as both mediators of physical stressors and as stressors (Frankenhaeuser, 1978).

**Psychology of Stress: Primary and Secondary Appraisal**

Moving on to the psychology of stress, appraisal plays a crucial role in shaping our response to stressful situations. There are two types of appraisals:

1. Primary appraisal
2. Secondary appraisal

**1) Primary appraisal: Interpretation of threat**

Primary appraisal involves interpreting the situation and assessing its potential threat. This initial evaluation influences how individuals perceive and react to stressors.

Appraisal of potential stressors depends on several factors including attitudes toward the source of noxious stimulation. Our attitudes and perceptions significantly influence how we perceive and respond to stress. Cognitive perspective also plays a crucial role, as our interpretation of a situation can either exacerbate or mitigate its perceived threat. For instance, individuals exposed to authoritarian parenting may adopt submissive attitudes towards authority figures, while those encouraged to express their talents and opinions may adopt assertive approaches.

**2) Secondary Appraisal: Selecting Coping behaviors and evaluating their effectiveness.**

Secondary appraisal occurs when individuals assess their resources and coping strategies to manage the stressor. They weigh the potential costs and benefits of different responses and choose coping strategies based on their past experiences.

Lastly, **our ability to cope with stress relies on various resources**, including:

- Psychological
- Social
- Genetic factors

These resources encompass our personality traits, social support networks, and genetic predispositions, all of which influence our resilience in the face of stress.

In the next lecture, we will discuss other areas of stress.

**Lesson 25**
**MEASURING STRESSORS**
In the last lecture, we discussed General Adaptation Syndrome and the Psychology of stress while in this lecture we will talk about:

- Hens' adaptation model (discussed in detail in lecture 24)
- Researching and Measuring Stress

## Researching Stress

Studying stress scientifically is crucial for gaining insights into its nature and effects. One of the primary objectives in this endeavor is to understand how to effectively measure stress. To achieve this, it is essential to first establish a clear definition of stress. By defining stress, we can identify various aspects or components that can be measured, thus facilitating accurate assessment of stress levels.
Stress in environmental context: Organisms undergo stress in the greater context of their environment. Environment impacts the individual, either its people around them or other environmental elements that affect them. Organisms stay in balanced condition with the environment. But this balance does not always stay as change is the only constant. This imbalance state is considered as stress/distress. In case of this disequilibrium of how we react, our primary and secondary appraisal helps us to measure it.

## Measuring Stress

There are two perspectives on measuring stress:

1. Qualitative differences
2. Quantitative differences

## Qualitative differences

Qualitative measurement of stress involves assessing the subjective experience and perception of stressors. Unlike quantitative measurements that focus on numerical data, qualitative measurements delve into the qualitative aspects of stress, including emotions, feelings, thoughts, and coping strategies.
To measure stress qualitatively it is important to classify the sources of stress. There are the following sources of stress:

### 1) Stressors

Stressors can vary in duration and intensity. Daily hassles, cataclysmic events, major life changes, and ambient stressors all contribute to the overall stress experience for everyone.

### 2) Conditions

One thing for a person can be pleasant in one condition and unpleasant/painful in other conditions/circumstances.
For example: Loud music is a great source of stress during studying and pleasant in the time of parties.

### 3) Chronic Stressors vs Acute/Recurring Stressors

Chronic stress persists over a prolonged period with lower intensity, while acute stress is characterized by higher intensity but shorter duration.

For example: Pollution is the source of chronic stress around us as it increases gradually with low intensity but for a prolonged time while in the parallel universe, this pollution is the source of acute stress for a person who belongs to hilly areas where there is no pollution and when they visited big cities it is the source of acute stress for them they perceive it with high intensity and for short period.

### 4) Controllable vs Uncontrollable

Few stressors can be controlled and there are few which cannot be controlled. For example, the temperature of the air conditioner can be controlled in your house but while traveling in a bus it cannot be controlled which causes flu or sore throat for some people.

So, stress can also be measured in the aspect of control as we can adjust the stressor according to our needs.

## Evans and Cohen's typology of stress

Evans and Cohen compiled a typology of stress in 1987. There are 3 dimensions of this typology:

- Duration
- The magnitude of the required response
- Number of people affected.

In light of these dimensions, there is a need to study a few daily life stressors that are the sources of stress.

### Daily Hassles

The category that includes some of the most chronic environmental stressors has been labeled *daily hassles.* These are present during most of our daily lives and include such conditions as job dissatisfaction, neighborhood problems, crowding, pollution, and noise (Glass & Singer, 1972). For example, the daily life of a student from going home to school and coming back comprises a lot of stressors every day but when we talk about whether these stressors are chronic or acute then we can say that these daily life hassles are chronic as they build up gradually but for persistent time and we get used to it.

### Cataclysmic Events

Great natural upheavals that turn down our lives are called cataclysmic events, for example, Earthquakes or Floods. Their impact is huge and affects every person.

### Major Personal Life Events

A third group of stressors includes those events powerful enough to individually challenge our adaptive abilities. These events include illness, death, or significant **loss (psychological or economic).**

**Ambient Stressors**

Environmental background conditions represent relatively stable, continuous, and intractable conditions of the environment and are labeled as ambient stressors such as work overload, job, poverty, family conflicts, and air pollution. They represent the relatively continuous, stable, and intractable conditions of the environment (Cambell, 1983).

**Noticing Stress**

The main question in the stress part is now when people start noticing stress. Stressors may go unnoticed until they interfere with some important goal or directly threaten one's health.
For example, environmental pollution.
Stress and Predictability
The predictability of stressors also plays a significant role in how individuals respond. Regular stressors allow for better preparation, while unpredictable stressors can intensify the stress response.

**Quantitative differences**

Measures stress quantitatively physiological responses, help as our body helps to deal with the situation, and in that case physiology of our body changes. To measure stress quantitatively 4 elements are most important.

- Intensity
- Duration
- Rate
- Controllability

**Intensity:** The intensity of the stressor refers in some sense to the "power" of the stimulus. Intensity can be "measured," for example, in terms of the magnitude or frequency (loudness or pitch) of sound, the concentration (parts per million) of a population, the velocity of the wind, or the temperature or relative humidity of the atmosphere. It can also be measured in terms of its physical effects; for example, the number of homes destroyed in *a* flood, the number of persons killed by a tornado, the number of victims hospitalized because of a nuclear accident, or the number of birth defects resulting from water contamination.
It is generally assumed that the greater the intensity of the stressor, the greater the resulting stress response.

**Rate.** While some stressors occur once, others may be recurring. Their periodicity can be regular or irregular, predictable or unpredictable, and short-phased or long-phased. For example, the temperature of every month in these we can find a pattern that summer comes once a year.
Duration. Independent of the intensity of the stressor is its impact concerning time. The immediate presence of a tornado, for example, is relatively short (perhaps only a few minutes), but its intensity is great. On the other hand, an airborne pollutant (such as asbestos) might have a low concentration yet be ever-present. Duration and intensity, it can be seen, are independent dimensions. All other things being equal, we can expect stressors of long duration to have a greater effect on the stress response than stressors of short duration. However, the accumulative effect of low-level stressors

over long periods may result in deceptively severe consequences. Any consideration of the potential effects of an identified stressor, therefore, must consider both the duration of the stressor and its intensity.

**Controllability.** Stressors vary concerning the degree of control humans have over them. It is possible to exert some control over the temperature of our indoor environments, over the noise level of our offices, and even over the traffic we must drive in. We do this by turning the thermostat either up or down, by putting sound-damping or sound-absorbing equipment in our offices, and by choosing the times of the day and the routes to take to avoid traffic. It is not possible to exert control over other stressors. We cannot stop hurricanes; we cannot prevent earthquakes. There are still other stressors where it is possible, theoretically, to exert control, but practically we are unable to (e.g., the loud stereo in the adjacent apartment or the litter in the streets below). Researchers have shown that the lack of control over environmental stressors where it is possible to have it can exacerbate the stress response.

In our discussion of environmental stressors throughout this text, we will, again and again, be looking at potential stressors in terms of their intensity, duration, rate, predictability, and controllability, and we will be calling for additional means of measuring these characteristics.

In the next lecture, we will discuss the immediate response and long-term Response.

**Lesson 26**

## Measuring Stress

In this lecture, you will study about following topics:

- Psychological and Somatic Responses
- Measuring Organisms Immediate and long-term Responses
- Models of Adverse Health

### Psychological and Somatic Responses

Psychological responses can pertain to our feelings, attitudes, cognitions, and behavioral patterns because psychology is the study of mind and behavior. Moreover, psychological responses affect somatic patterns. When we pass through stress our body must go through many changes to prepare itself and many physiological reactions take place. And if the stress is short-term, then the physiological reaction will be for a short period. Moreover, if these responses are for the long term, then they can hamper activity and create cause kind of danger to the body.

### Measuring Immediate Response to Stress

It means to measure galvanic skin responses, the level of hormonal secretions, and measure level of secretion in blood regarding adrenaline and non-adrenaline (which are two hormones that help us to deal with emotional states).

### Long term Response

We have little control over long-term stress and must live with it for a longer time. For example, people living in refugee camps.

When the condition of stress persists then these physiological reactions also continue for longer periods and this leads to certain kinds of physiological diseases.

### Bio-medical Model

To understand disease bio-medical model tells us that there are 3 ways for a person to get ill.

- Illness results from a specific physiological dysfunction.
- Invasion of foreign substances.
- Some diseases do not fit the traditional bio-medical model and are named diseases of lifestyle (heart diseases seem to be correlated to the patterns of lifestyle).

## Adverse Health Models

### Dohrenwend & Dohrenwend Victimization: Model A

Dohrenwend & Dohrenwend (1981) have conceptualized the possible processes where stress induces adverse health changes (see Figure 5-4). In this figure, Model A contains no intervening processes. Rather, it postulates a simple and direct effect of stressful events on adverse health changes. It further suggests that the effects of stressful life events are cumulative, thus accounting for studies of extreme situations, such as combat or incarceration in concentration camps. Other severe stresses over which an individual has no control, such as the death of a loved one, are also accounted for by this model. This model is called the *victimization hypothesis.*

**Model B: Stress and Strain Hypothesis**

Model B, the *stress-strain hypothesis,* postulates that psychophysiological strain mediates the impact of life events on subsequent health and illness. Evidence in support of this hypothesis comes from studies that show that if the effects of symptoms of psychophysiological straining are eliminated, correlations of stressful life events scores and measures of illness are significantly reduced (Bloom, 1985). This model is most closely related to the conceptualization of stress in this text.

**Model C: The vulnerability!**

The *vulnerability hypothesis,* suggests that there are preexisting personal dispositions and social conditions which moderate the causal relation between stressful life events and health. It is this model that suggests to researchers a search for such mediating factors as the strength of social support systems, and personal conditions.

**Model D: *additive burden hypothesis!***

By contrast, postulates that personal dispositions and social conditions make independent causal contributions to the occurrence of pathology. This model suggests that personality variables and/or social conditions are a potential source of added burden to the individual in the precipitation of illness and is thus called the *added burden hypothesis.*

**Model E: Chronic burden!**

Proposes that transitory life events have no role in precipitating illness, but rather that stable personal characteristics and social conditions by themselves cause adverse health changes. This model is called the *chronic burden hypothesis.*

**Model F: Event Proneness Hypothesis**

Finally, Model F portrays stressful life events as exacerbating already existing health disorders. It is called the *event proneness hypothesis* because stressful life events are thought to characterize individuals who are already ill.

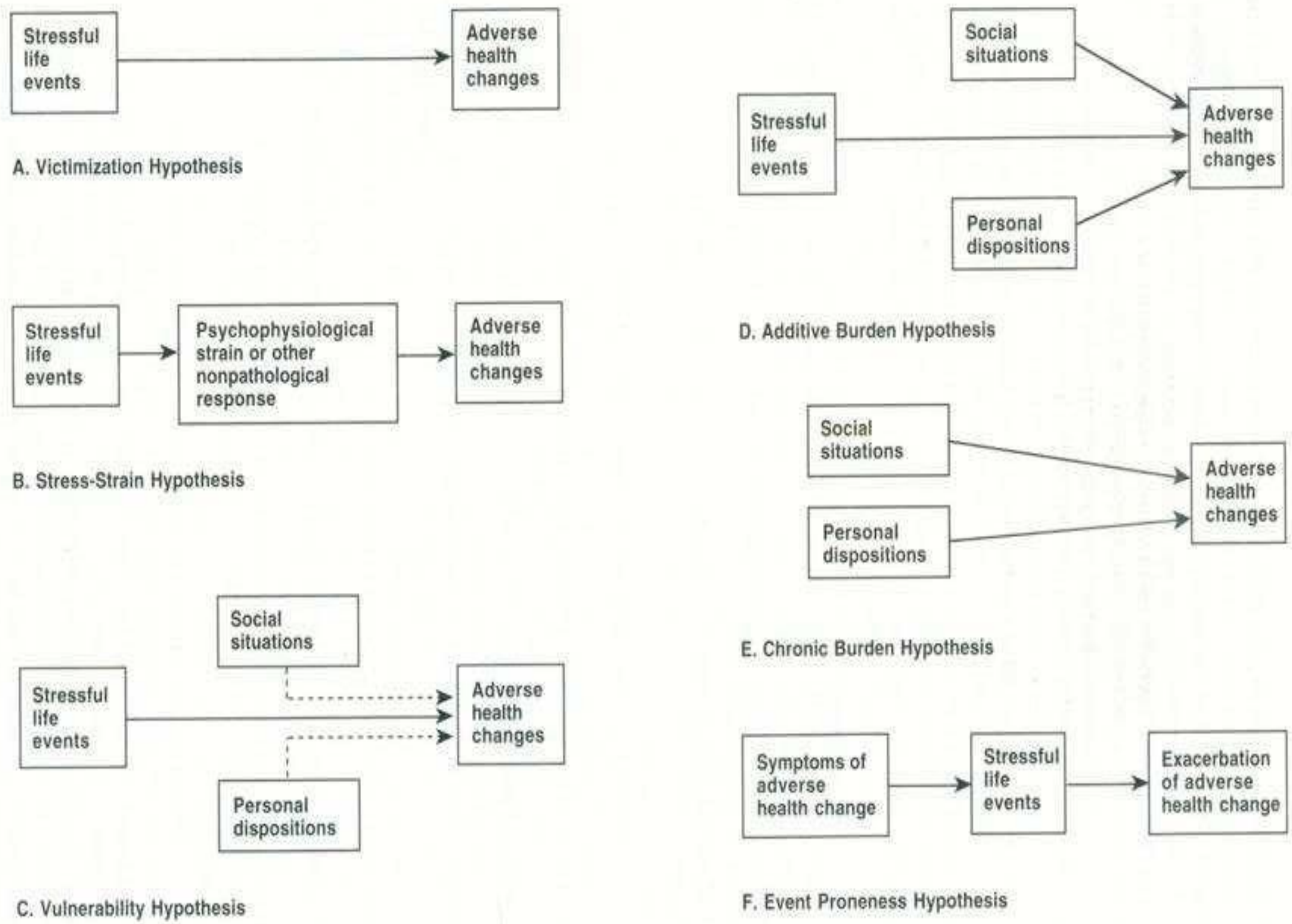



FIGURE 5-4 Six hypotheses about the life stress process.
From Dohrenwend, B.S., and Dohrenwend, B.P. (eds.), *Stressful Life Events and Their Contexts*. Neale Watson Academic Publications, Inc. (1981).

**Measuring Stress Psychologically**

When we talk about psychological assessment, we note down all factors that contribute to behavioral patterns. Psychological measurement of the stress response has focused on various psychiatric symptoms and has utilized a variety of standard psychiatric symptom inventories. The common denominator among the various scales seems to be the extent to which respondents perceive themselves to be in an intractable situation marked by negative affect and a degree of uncertainty.

## Measures of Coping

The level of stress a person experiences and the extent to which deleterious effects occur because of exposure to a stressor will depend, in part, on how well the individual copes with the stressor. Whereas Behavioral coping responses can be categorized according to individual differences. Every individual will cope with stressful events differently.

In the next lecture, you will study moderators of stress response.

**Lesson 27**

## MODERATORS OF THE STRESS RESPONSE

After a review of the previous topics, lecture 27 will comprised of the following:

- Moderators of Stress Response

### Moderators of Stress Response

By response, it means that a person is pitched against a stressful scenario and what type of response comes out of him. This response pattern is important to understand and study for psychologists to find out how they can build up a person's capacity even further to deal with the stressors. The extent to which a stressor produces a response is moderated by many factors:

- Attitude toward the source of stress
- Control
- The Hardy Personality
- Social support
- The Relaxation Response

All these factors will increase or decrease a stressor's impact, we turn now to a discussion of them.

1) **Attitude toward the source of stress**

As mentioned earlier, responses to stressors are necessarily related to how a stressor is perceived. Attitudes toward the source of the stress are important psychological factors and act as filtering devices that moderate the perception of the stressor.

**Example:** Let's consider the case of a student who may not perform well in his mid-term exams. This suggests that they are struggling to align their study routine with their academic goals. Despite making efforts, they find it challenging to adapt. At this juncture, it's crucial to engage with the student and understand their perspective on their academic routine. Sitting down with them, we can inquire about the difficulties they face in their studies. Perhaps they find attending morning classes challenging, or they feel overwhelmed by peer pressure, or they perceive the course material as too difficult. By assessing various stressors that students encounter at school, we simultaneously delve into their perceptions of these stressors. Do they believe they can overcome these obstacles, or do they feel incapable?

The student's perception holds significant weight. If they express a belief that they can't meet academic requirements, it's essential to connect with them on an emotional level and understand their underlying concerns. Recognizing their mental state, we can work on bolstering their confidence and resilience.

Once the student develops a positive outlook towards their academic challenges, we can then identify specific problems and work towards solutions. Now Let's also discuss phobias in the light of perception. **Phobias** are instances where individuals feel intense and irrational fear toward specific objects, situations, or activities. These fears can vary widely among people, but certain phobias, such as the fear of traveling, are more commonly observed.

It's important to note that phobias stem from irrational fears, and individuals' perceptions of these fears play a crucial role. For example, someone with a fear of flying may harbor negative thoughts and beliefs about air travel, leading to avoidance behaviors. When faced with their phobia, individuals may experience physical symptoms like rapid heartbeat, shortness of breath, and nervousness. These reactions stem from the negative perception they hold towards the source of their fear.

Addressing phobias involves shifting these negative perceptions towards a more positive outlook. Psychologists often employ techniques to gradually expose individuals to their fears in a controlled environment, helping them build resilience and overcome their phobias. For instance, individuals may be guided through relaxation exercises while imagining themselves in a feared situation, gradually desensitizing themselves to it. This process aims to move individuals from extreme fear (10 on a scale) to a more manageable level (1).

Therapists may also accompany individuals on real-life exposures to help them confront their fears directly. By addressing the negative perceptions associated with phobias, psychologists work towards empowering individuals to regain control over their lives.

So, to these above examples, people’s perception of stress is very important, and it is also vital to control it.

**2) Control**

Another crucial aspect to consider is the perceived control over the stressor. The level of control one feels they have over a stressful situation plays a significant role in how they respond to it. When individuals feel they have some control over a stressor, they are better equipped to cope with it. However, when they perceive the stressor as beyond their control, it can become overwhelming.

A poignant **example** can be found in the experiences of prisoners of war during World War II. Held in Nazi concentration camps, these individuals faced unimaginable hardships and lacked control over their circumstances. This lack of control contributed to chronic stress and various psychological disorders among them.

Similarly, in discussing phobias, the perception of control is crucial. If someone believes they can control their fear, it's possible to gradually build their resilience to it. Conversely, if they feel powerless/uncountability against their fear, intervention is necessary to address this perception.

Moreover, an **individual's overall physical fitness** level plays a significant role in their ability to withstand stressors. Physically fit individuals possess higher energy reserves, enabling them to endure challenging situations for longer periods. Conversely, those lacking physical fitness may find themselves more vulnerable to stressors, as their energy and resilience diminish.

Note: Understanding stress from multiple angles is essential. However, it's important to note that discussing stress does not aim to make individuals more vulnerable to it. Instead, it underscores the importance of resilience in navigating stressful circumstances. Ultimately, how one responds to stressors determines their success or failure in overcoming them.

**The Hardy Personality**

In the realm of leadership, the concept of a hardy personality emerges as a significant factor. Think about your favorite leaders. They could be historical figures like Muhammad Ali Jinnah, renowned poets like Allama Iqbal, or even sports icons who led their teams to victory. What sets these individuals apart as heroes?

When we envision a hero, we often picture someone who confronts daunting challenges with unwavering determination. Regardless of the adversity they face, they remain steadfast in their commitment to their goals. This resilience stems from their strong character, refusing to succumb to the pressures of their circumstances. Many heroes throughout history faced setbacks and failures, yet they persisted in their pursuit of their objectives. Some even sacrificed their lives before achieving their goals. Their heroism lies not in their immediate success, but in their unwavering resolve to overcome obstacles.

Consider the example of those who fought for independence or social justice. Despite enduring hardships and setbacks, they remained dedicated to their cause, inspiring future generations to continue their struggle. In this context, a hardy personality emerges as a beacon of hope amidst the challenges of stress and adversity. Hardy individuals possess a sense of resilience and determination that enables them to navigate difficult circumstances without losing sight of their objectives. The key components of a hardy personality include a strong sense of commitment, a perception of challenges as opportunities for growth, and a belief in one's ability to exert control over their life. When faced with setbacks or failures, it's natural to experience feelings of self-doubt or frustration. However, reframing these experiences as opportunities for learning and growth can help mitigate negative feelings.

Next time one encounters a setback, consider redefining it as a learning opportunity rather than a mistake. By adopting this mindset, we can cultivate resilience and develop the resilience needed to overcome challenges. Consider reframing mistakes or challenges as opportunities for growth. Every problem you encounter presents a chance for personal development. While it's natural to feel daunted by challenges, approaching them as opportunities allows you to harness your intellectual and mental capacities. View each problem as a canvas upon which you can paint your journey of growth. By tackling challenges head-on, you not only overcome obstacles but also enhance your skills and resilience. Reflect on your experiences and mistakes as opportunities for learning. Embrace the equation: Mistake plus learning equals growth. Every error becomes a stepping-stone towards personal development when accompanied by insights gained from experience.

Conversely, failure to learn from mistakes stuns growth. Remember, mistake minus learning equals stagnation, a state antithetical to personal progress. Understanding the concept of hardiness is essential. It encapsulates resilience and determination in the face of adversity, forming the cornerstone of personal growth. As we conclude this discussion, remember to approach life with an open mindset, embracing challenges as opportunities for growth.

In our next lecture, we'll delve into our next topic.

## Lesson 28

## Environmental and Cultural Variances

In this lecture, we're delving into how the environment and culture influence human behavior, and how human behavior, in turn, shapes the environment and culture. Here's an overview of what we'll cover in this chapter:

- First, we'll examine the intricate relationship between behavior and environment, summarizing how they interact with each other.
- Next, we'll explore phenomena like conformity and obedience, including a case study to illustrate these concepts.
- We'll then analyze the factors influencing conformity and individual behavior within groups.
- We aim to understand how individuals conform to certain ideas, become obedient, and adapt to different environments.
- Moving on, we'll discuss attributing behavior to both personal dispositions and situational factors. This section will help us grasp how people interpret and attribute behavior.
- A detailed exploration of group behavior follows, covering topics such as group pressure, conformity, and the reasons behind group influence and mob behavior.
- Our final focus will be on culture. We'll investigate how the environment shapes culture and how individuals learn about and adapt to the cultures they inhabit. Key processes like enculturation and acculturation will be explored, along with the roles of language, symbols, and rituals in cultural formation.
- We'll view culture as a dynamic force, constantly evolving and influencing individuals and society. Finally, we'll touch on the measurement of culture and briefly discuss subcultures.

### Relationship between environment and human behavior

We've come to grasp the intricate relationship between human behavior and our surroundings. Our environment can significantly shape our actions, sometimes to the extent that we might attribute our behavior solely to external influences. Yet, we mustn't overlook the undeniable impact of the environment on our choices and actions. This dilemma echoes the age-old debate: is it the environment or human free will that predominantly governs our behavior, or is it a complex interplay of both? There are instances where individuals act in ways they never anticipated, highlighting the profound influence of environmental factors. It's a nuanced interaction between the environment and individual agency that ultimately molds human behavior, posing a perpetual conundrum for psychologists and philosophers alike.

Human behavior often defies expectations, showcasing a remarkable adaptability to various situations and environments. Consider individuals who typically exhibit calm and peaceful demeanors suddenly becoming aggressive under certain circumstances, or those who are naturally aggressive displaying unexpected submissive behavior. Moreover, there are instances where introverted individuals become talkative and extroverted in specific situations. These examples underscore the significant role played by environments and situational factors in shaping behavior. Yet, amidst this complexity, the question persists: is it ultimately the exercise of human free will that guides behavior, or does external influences/environment wield greater control?

To understand this concept more deeply a case study will help us.

**My Lai Massacre Case Study**

In March 1968, three platoons of American soldiers, non-collectively as Charlie Company, swept into the South Vietnamese village of MyLai, and killed several hundred unarmed civilians, women and children, and old men. Most of the soldiers joined in shooting those who did not fire, not one attempted to stop the slaughter. Some of the men of the Charlie Company reported that they followed the orders of their leader, a lieutenant of the US Army.

Others said they simply followed the example of others. The gruesome, isn't it? The question is, why would the soldiers behave so savagely? There are certain other details. The soldiers of the Charlie Company had joined in Vietnam about two to three months before this incident took place. They mentioned that they were being constantly attacked. They were fighting a guerrilla war over there because they were in the enemy area, and there was insurgency taking place over there. They were under extreme stress, of course. And they received the orders that in the My Lai village, the American army suspected the armed men of the Vietnam army, and they had to launch an operation to go there and get those men. And they simply went in there. Just like one of the soldiers said, “I was just fulfilling my orders over there”.

However, the types of pictures that we have seen are extremely gruesome, because there were few men, and all those men were old men, and there were ladies and children. If the troops had to enter that village, the question is, why would they behave so savagely?

We understand that the training of the army is very, very extensive. This is not only physical training, but this is also a lot of mental training. Soldiers are raised to become civilized citizens. Thou, their job is to fight wars, is to defend against the enemies of the country. However, their training is made sure that they learn a lot about how to become civilized citizens. The question is that any person who enters such sort of a situation would be behaving the way the American soldiers behave. Because normally, ordinarily, normal people do not tend to see these types of situations. However, when we come to think of it, can you say that we can forgive those soldiers because they were under extreme stress? See, it is a question that relates to the same debate. Was it the environment? Was it the situation? Or was it the free will of the people? Students, we have some more information about this my life massacre. However, before that, we would like to tell you what one of the Charlie Company soldiers said. He said, I looked around and saw everyone shooting. I did not know what to do, so I started shooting. This sounds more like this soldier was under the grip of the circumstances. The environment was impacting upon him very, very heavily. And what he did, he did what the rest of the people were doing. Environment and behavior. Social environment and behavior. This means the social environment and the social dynamics of that place at that time just prompted the soldier to do the same way as the other members of the group were doing. So, is it a justification?

Let us give you some questions to think about this massacre. The question is, is the way people think and act the production of their inner dispositions? If you take this question to understand the My Lai massacre, the question would be, do you think what people did over there? What Americans would have stood over there was based upon their inner disposition because they were the people like that. Or because they were in that situation where somehow the circumstances made them behave in a certain way. The next question is, does the situation play a role? Was it only because of the situation that soldiers behaved savagely the way they did? The next question is, do environmental factors contribute to the way people think, feel, and act? The way those soldiers were thinking, feeling, and acting, was because of the environment. Was it because of the situation that made them do that? And finally, the question is direct, why did the soldiers in my life behave savagely? Why did the people behave very, very savagely over there?

Of course, as psychologists, we can look at this phenomenon and we can bring in all the technology, the behavior technology that we have learned to understand this entire fiasco of dreadful human behavior. However, let us give you some more information on the My Lai massacre. The remaining information is something like this. While the massacre was taking place, a helicopter happened to be passing by that village. This is what this all. A helicopter pilot from an aerospace team witnessed many dead and dying civilians as he began flying over the village. He landed in his helicopter and asked the surgeon he encountered there if he could help get the people out of the ditch and the surgeon replied that he would help them out of their misery. Later, he told his crew that if the US soldiers shot at the Vietnamese while he was trying to get them out of the bunker, they were to open fire at these soldiers. Now, students, this is entirely antithetical behavior to what was going on from the soldiers of the same army. From the same army, few soldiers were engaged in the massacre while from the same army, there came a few other soldiers who did not want those soldiers to engage in that massacre and they were so charged for that that they would even fight against their fellow soldiers. These soldiers were later given medals for their bravery.

Once again, the question to you is? these two divergent types of behavior Is the environment playing upon these people to behave in two different ways or is it the people themselves that are trying to interact with the environment or they are trying to control the environment in different ways? Once again, let me remind you what that soldier said. I looked around and saw everyone shooting, I didn't know what to do so I started shooting. Is that enough reason for that? Of course, this is a question that you would be thinking. At this point students and based upon this soldier's interview and what he said in that, I would like to give you the definition of conformity.

### Conformity

Conformity means a tendency to shift one's opinions or actions to correspond with those of other people because of implicit or explicit social pressures. Conformity is a phenomenon here. Conformity is pervasive in society, influencing career choices, among other things. For instance, in our country, many gravitate towards careers like medicine or engineering, following societal norms.

**Why do people conform?** Often, it's to align with prevailing standards. We may sacrifice our true desires to fit in. How many aspire to less conventional paths, like philosophy or poetry? Society molds our aspirations, nudging us towards conformity.

Ultimately, conformity reflects the tug-of-war between societal norms and individual autonomy. It's a complex dynamic that shapes behavior in profound ways.

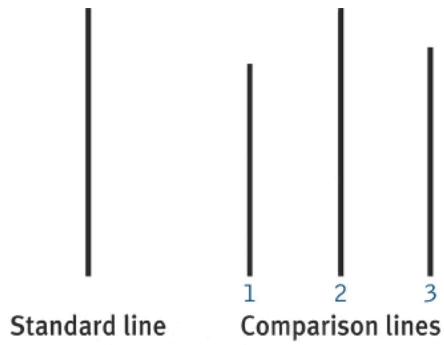
Let's revisit the My Lai case and the question of why the soldiers behaved savagely. Similarly, let's consider why young people often conform to societal norms when choosing their careers.

It's tempting to label the soldiers as bad or the conforming students as lacking in courage.

### Asch's Experiment

But let's examine how people conform to social norms through a simple experiment.

In this experiment, subjects were shown two sets of lines and asked to judge their lengths while working in a group. Seven subjects participated, but only one was real—the rest were confederates instructed to give obviously wrong answers. Despite the subjects being able to clearly see the correct answer, 37% still conformed to the group's incorrect responses.




**Factors Influencing conformity.**

There are the following factors that influence conformity:

Agreement amongst participants

Dissenters

Anonymity

1) **Agreement among participants**

Why did they conform? This experiment, known as the Asch experiment, sheds light on the factors influencing conformity. One key factor is the agreement among participants. When everyone else in the group gives the same wrong answer, individuals often succumb to groupthink, even if it contradicts their perceptions.

For 37% of the participants, the pressure to conform outweighed their ability to trust their judgment. This highlights the powerful influence of social norms on individual behavior.

Let's revisit the scenario of the American soldier in Vietnam who said, "I looked around, everybody was shooting, so I started shooting." Similarly, imagine a conversation with a young person today who chooses a career path based on societal influence, like, "My dad said computer scientists make great money, so I'm going for it. Plus, lots of my friends are studying computer science." In both cases, individuals are following the crowd, conforming to social norms.

2) **Dissenters**

Now, let's examine this from a different angle. Think about dissenters—those who go against the group's thinking. Picture a scenario where a group is unanimously leaning towards one option, like going to a certain place, but one person suggests an alternative. This dissenter challenges the group's consensus, opening new possibilities. Research, including the famous Asch experiment, shows that dissenters can significantly reduce conformity rates compared to groups without dissenters.

3) **Anonymity**

Another factor to consider is anonymity. When individuals feel anonymous, they may express their true opinions more freely. In experiments where subjects remained anonymous in aloof rooms, they were less likely to conform, indicating the impact of social pressure on conformity.

4) **Self-confidence vs self-doubt**

Additionally, self-confidence versus self-doubt also plays a role. Confident individuals are more likely to resist conformity, while those plagued by self-doubt may succumb to group pressure more easily.

**Reasons for conformity**

What is the reason that moves people to confirm with the group? One of the great reasons is often involves people outwardly agreeing with the group while inwardly retaining personal opinions. This internal conflict can be stressful, leading individuals to eventually adopt the group's viewpoint to reduce discomfort—a process known as internalization.

**Process of Conformity**

- In confirming people first go through the process of Identification.
  ***Identification*** means they want to go along with the group.
- In the second step, they go through the process of Internalization.
  ***Internalization*** comes when they start believing in the group's point of view. They kill their own point of view.

Now, Consider historical examples like World War II and the rise of the Nazi party. Despite initial resistance, many individuals eventually conformed to Hitler's ideology due to social pressure and the suppression of dissenting voices due to group norms. It was also because dissenters were not treated rightly.

Ultimately, the interplay between environmental factors and individual agency shapes behavior. While social influence can be powerful, individuals with strong convictions and confidence may resist conformity, paving the way for dissent and change. So, is the environment more powerful, or is human will? It's a question worth exploring scientifically, as we seek to understand the complexities of human behavior.

In the next lecture, we will study Attribution Theory.

**Lesson 29**

**Attribution Theory**

In this lecture, after reviewing the previous lecture you will study the following topics:

- Attribution Theory
- Attribution Errors
- Internally vs Externally caused Behaviors.

**Attribution Theory**

Our perceptions and judgments of people's behavior are heavily influenced by our assumptions about their internal state, a concept known as attribution theory. For instance, we might come to start thinking that this gentleman is a person who is a bad guy because he is trying to demean us, or he is trying to embarrass us in front of the crowd. Or a certain person comes to us, and he just becomes friendly, gives you a nice shake, introduces himself, and starts talking to you. You might say that he is a nice guy because he is behaving like a nice guy. And when we look at both people, we try to see what is the state that is inducing this type of behavioral pattern. If you come to find out that this type of thing is being done by the person because he internally actually plans to do that to you, then you would consider that person bad. However, if you come to think that that person was doing, just throwing a joke to make everybody laugh at the party and make you laugh also at the party, then you might come to interpret the same thing from a very different point of view.

Attribution theory is a theory that helps us to understand 2 things:

1. How do we explain people's behavior
2. How do we come to attribute people's behavior to their internal states

Let's consider an **example**. Syra, a student of business administration, seems shy in class but talkative with friends. Is she inherently shy, or is her behavior situation-dependent? If we observe her only in class, we might conclude she's shy. But if we see her with friends, we might see her as confident. This illustrates how our judgments can be influenced by the context in which we observe behavior.

Attribution theory has been developed to explain why we judge people differently based on the meaning we attribute to their behavior. We attempt to determine whether behavior is internally or externally caused. It's crucial to consider whether someone's actions stem from their personality or the situation they're in.

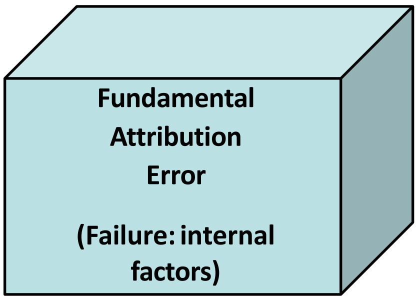
When someone's behavior is internally caused, we hold them responsible for their actions. However, in some cases, external factors can influence behavior. For example, someone might behave differently in stressful situations. It's essential to consider these factors when making judgments about people's behavior.

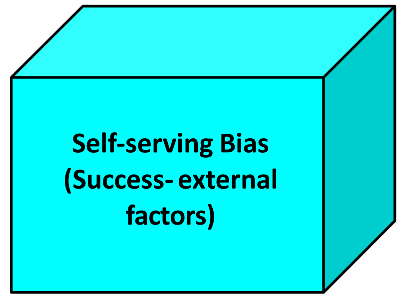
However, as we discussed in the last lecture, sometimes it happens that people in certain situations behave in a queer sort of way. In one situation they are cool and calm. And when you capture them in an aggressive mode, you might come to say that this gentleman is very, very aggressive. However, you need to see that aggressive behavior is internally caused or externally caused. If that behavior is externally caused according to your perception, then you might say that this gentleman is behaving aggressively only in this situation. However, if you come to think that this behavior is internally caused, then you would say that this gentleman is aggressive. In either case, our perception might be wrong because you do not have complete information on that.

**Attribution Errors**

There are 2 types of attribution errors.

- Fundamental Attribution error
- Self-Serving Bias




- **Fundamental Attribution Error**

Fundamental attribution error means that we are always making wrong judgments based on our assumptions. Understanding the influence of external factors and overestimating the influence of internal factors. That is the fundamental attribution error.

For example, assuming that a person is late all the time because he or she is not interested in the work rather than finding out that the shift starts when the parking is full. We assume that something is wrong with this person as he is always late which means that the behavior is internally caused. However, there could be a major reason for that person to be late every day.

However, instead of finding out why this person comes late every day and helping this person out, we might simply build up an assumption without getting enough information that this person is a habitual latecomer. This person is lousy, he just wants to drift away from the responsibility. So, fundamental attribution error is an error that we commit when we try to underestimate the influence of external factors and overestimate the influence of internal factors. So, we underestimate and overestimate the influence of internal and external factors. This is the fundamental attribution error that we are constantly making.

- **Self-serving bias**

Self-serving bias means that we attribute success to internal factors and failure to external factors. Like they say, nothing succeeds like success and failure has many excuses. When a person is successful, that person is successful and that means that success becomes part of that person and we do not say that this person is successful because of certain external factors. We simply say that that person is successful, and we come to think that that person might be very, very intelligent. So self-serving bias is attributing success to internal factors and failure to external factors. We might come to say that I failed because I faced these difficulties and if you are successful, we would say I am successful because I worked very, very hard for it. So, we try to tailor the opinion according to our choice. We try to tailor our perception of the situation in such a way that we appear to be heroes in that situation and this attribution fundamental attribution error is called self-serving bias.

**Internally VS Externally Cause Behaviors**

We believe that internally caused behaviors are under an individual's control. Externally caused behaviors are motivated by external forces. So, when we say that someone has done something because of his intent, then we tend to attribute the entire blame to that person. We would say that this person behaved in this way because this person is bad, or this person is good, or whatever that person is. However, when we find out that there are reasons for a certain behavior that is externally caused, we attribute the behavior to external factors, and this is the time that we usually try to forgive people.

So, we are trying to determine and one of the most important things out of these is that when we are trying to find out that the behavior is internally caused or externally caused is distinctiveness. As, how we determine the source of behavior is determined by three factors.

- Distinctiveness.
- Number two is consensus.
- And number three is consistency.

**Distinctiveness**

Distinctiveness means How much a person's behavior is unique. How much a person's behavior is distinctive? “Distinctiveness refers to whether an individual displays different behavior in different situations”. If the distinctiveness is high, if the behavior is very, very distinct, then your attribution towards that behavior would be very high towards external factors. If the distinctiveness is low, then your attribution towards the external factor would also be very, very low because this behavior is not distinct at all. As we discussed in a previous **example**, if the employee comes late one day, then that behavior is very, very distinctive because it does not happen all the time. If the employee is coming late every day, then that behavior is not distinctive at all because this is something that happens every other day. And if that happens every other day, then that means you might come to say or attribute his behavior to internal factors. So, if we see that the person's behavior is distinctive, then we may say that this is externally caused. If this person's behavior is not distinctive, we may say it is internally caused. And internally caused means the person is penalized. **Consensus**

“If everyone who is faced with a similar situation responds in the same way, we can say the behavior shows consensus”.

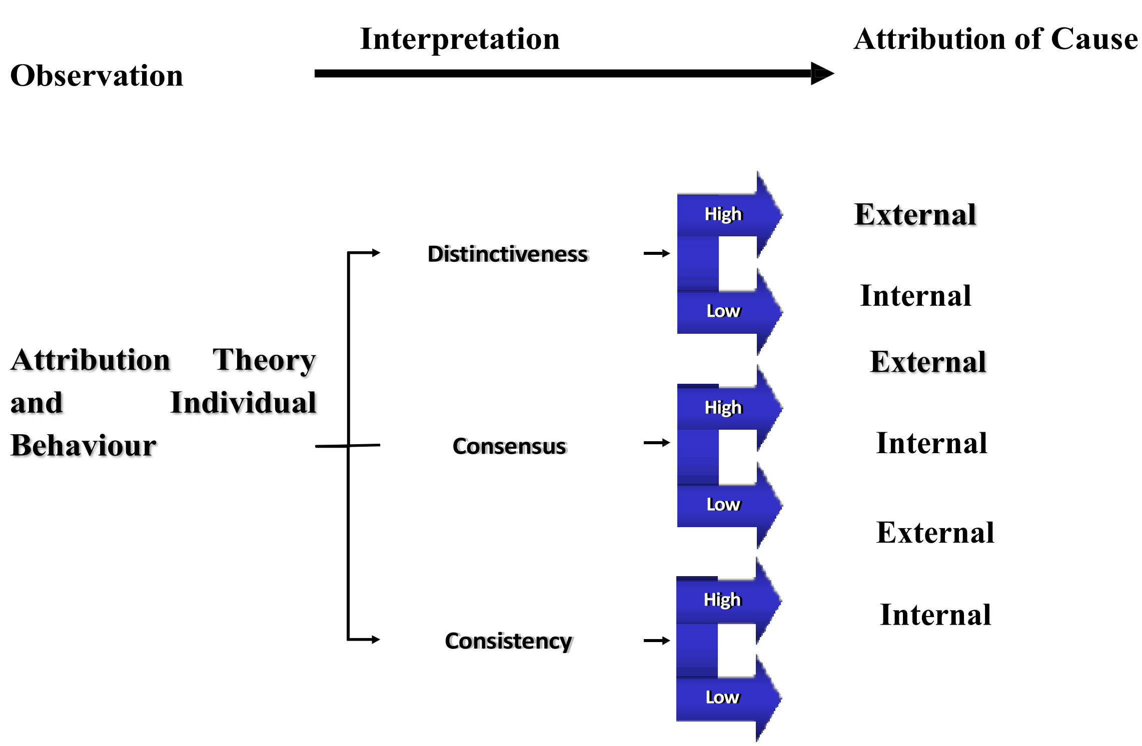
**Example:** If most of the students in a class let us say 90% of the students in a class fail in mathematics. Who is responsible for that? Is it the students or the teachers? Just make a general understanding of it. Just pass on the judgment right now 90% of the students in a class fail in mathematics. Is there something wrong with the students, or is there something wrong with the teacher? Based upon the consensus point of view, we will say that since 90% of the students are failing in the class, then that means something is wrong with the teacher because everyone is failing in this class. Hardly anyone passes. So, what are the teachers teaching in this class? Maybe there is something wrong with their teaching methodology. We see a consensus in most people's behavioral patterns, and then we say that this is understandable. You might have heard a lot of people who come say that I could not progress in life because I was born into a very poor family and my parents could not send me to a good school. And based on the poverty, I could not progress in life. And then you would say that most of the people who are in poverty are not able to progress in life. So, is it poverty that is responsible for their failure, or is it the people themselves who are responsible for their failures? Who is at the helm of the affairs? Poverty or the people themselves? Is it poverty that makes people fail, or is it the failed people who produce poverty? This is along the bay. However, based on the consensus principle, we might say that this is the poverty that is making those people poor. However, there might be some leaders who might say that this is not poverty. Another leader might say that it is the people who are responsible because they are poor. Because I was born in the same society, I was born in the same locality, I come from the same country, and I am not poor anymore. I worked hard in my life. And these people who did not work hard in their lives are poor. However, if we go by the principle of consensus, we will find out that if the behavioral pattern is common with most people out there, we might say that this behavior is externally caused, this behavior is not internally caused. An external cause of this behavior is poverty, not the people who are failing.

**Consistency**

When we observe people, we look for consistency in their actions. That people behave the same way in many different situations. That someone's behavior is consistent over time. When someone's behavior is consistent over time, it means that they have established this behavior. Because they are consistently behaving in the same fashion.

**Example:** There is a student who fails in every class. He has been failing in every class for the last five years. He hardly passes any exams. And every time the exam comes, this student is upset about it because he knows that he's going to fail it. Who do you think is wrong? Is it the student who's wrong? Or is something happening with the student that's making him do the wrong, which is failing in the exam? Who's wrong here? If we look at the student's behavior, we can say that his behavior is consistent over time. He has been failing for the last three years. And there is something wrong with this chap. So, you say that this chap is poor. You might start commenting upon his intellectual ability, upon his intelligence. You might say so many things to find out the reason inside that person. However, let me tell you that most of the great scientists that have a great name in history were bad students, were the students who were failing consistently. So based upon this perceptual thought, we always say that when a behavior is consistent over some time, then that person is responsible. We do not try to find out the reason in the external environment in the external scenarios.

These are the three important points of attribution theory and individual behavior.



And the three processes that we discussed are distinctiveness, consensus, and consistency. If the distinctiveness is high, then we would attribute the cause to external factors. If the distinctiveness is slow, we will attribute the cause to internal factors. Similarly, if the consensus is very high, we would attribute the cause to external factors. And if consensus is slow, we will attribute the cause to internal factors. And the same is the case with consistency. If someone's behavior is consistent over a long period, then we would say that this behavior is caused by external factors. And in other situations, we would say that this behavior is caused by internal factors.

**Judgmental Shortcuts**

We make certain shortcuts in making judgments. Now, what are those shortcuts that we make when we judge people? These are called judgmental shortcuts. There are the following 5 judgmental shortcuts:

- selectivity
- zoomed similarity.
- stereotyping
- halo effect.
- Self-fulfilling prophecy.

Studying these judgmental shortcuts is very, very interesting in the sense that we try, we understand that most of the time that we are trying to learn about people, we are trying to explain their behavior, then we are making certain shortcuts that help us understand their people. But these shortcuts are not always right. These shortcuts give us extremely misleading information about the people. And since that misleading information is going into our minds, and we are building up our opinions and attitudes accordingly, chances are that we would negatively react to our environment in a way that is probably not very conducive for our adjustment over there.

In our next lecture, we will talk about all these judgmental shortcuts.

**Lesson 30**

**Judgmental Shortcuts and Culture**

In this lecture, we will continue the last lecture from judgmental shortcuts and will also discuss culture.

There are 5 types of Judgmental Shortcuts.

1) **Assumed Similarity**
   In this assumed similarity shortcut, we try to find out about others through our own perception of ourselves.
   Example: When delivering a presentation, it's common to experience nervousness, a feeling shared by most people. During our initial presentations, this nervousness can be particularly intense, often causing physical shaking. However, as we gain experience, confidence typically grows. Similarly, if we observe someone exhibiting physiological signs of anxiety before a presentation, we may assume they are nervous, reflecting a perceived similarity based on our own experiences.
2) **Stereotyping**
   We start perceiving a group of people in a certain way.
   Example: If a person is heightened and muscular we think about him in a particular way that they will be good at sports and think about chubby people in other ways. These judgments can be right or wrong.
3) **Halo Effect**
   We pick one characteristic of a person and start thinking that all characteristics of a person will be good.
   Example: In an interview by seeing the color of the tie interviewer makes the judgment that the person will be materialistic and bad for the company.
4) **Self-Fulfilling Prophecy**
   When our beliefs or expectations about a person influence their behavior, it's known as a self-fulfilling prophecy. For instance, if you initially perceived your friend as struggling in mathematics, but later believed their skills improved after tutoring, you might seek their assistance. Consequently, they may also feel more confident in their abilities and willingly assist you, aligning their behavior with your beliefs and expressions about them.
5) **Selectivity**
   When assessing others, we tend to prioritize traits or interests that resonate with us personally. For instance, if you enjoy playing golf, you might inquire whether someone shares this interest. If they don't, you may subconsciously view them as less compatible or not meeting your standards.

## Culture

The study of culture is a challenging undertaking because its primary focus is the broadest component of social behavior – an entire society.

To comprehensively study society, it's essential to grasp the collective behavioral patterns it comprises. Culture can be dissected into three primary components:

1) The component of beliefs
2) The component of values
3) The component of customs

**Belief Component**

In the Pakistan movement, the driving force was a belief component that a different country for Muslims. Muslims are those who have a certain set of beliefs and spend their life accordingly.

**Belief and Value Relationship:** The beliefs and values component of our definition refers to accumulated feelings and priorities that individuals have about things and possessions as well as ideas.

**Values Component**

Values are culturally acceptable behavioral patterns. Values are those preferences according to which we conduct or monitor our behavior.

**Customs Component**

Overt modes of behavior that constitute culturally approved or acceptable ways of behaving in specific situations. Customs consist of the everyday routine behavior of the people.

**Example:** Talking to elders respectfully is our custom.

## How Culture is Learnt

Culture is learned by making use of two processes:

1. Enculturation
2. Acculturation

**Enculturation**

Learning one's own culture is called Enculturation. Learning about our own culture helps us to adjust. The more we learn the more we get appreciated and adjusted.

There are two ways of learning our own culture:

- Formal (parents taught us)
- Informal (Through literature or media)

**Acculturation**

The learning of a new or foreign culture is called **Acculturation.** Culture is a dynamic process, and it evolves and changes. Nowadays a lot of information comes through computers, which helps with acculturation, and we learn about other cultures.

Example: From 1857 to 2009 a lot of changes have been made in our culture. For instance, before that, there was no microwave oven but now in 5 minutes, we can cook food. It has changed our life and culture to evolve like this persistently.

In our next lecture we will study Population and its impact on our Culture.

**Lesson 31**

**Population and Culture**

- In our last lecture, we discussed that many factors affect culture like; New technologies, Population shifts, Resource Shortages, Wars, Accidents, Natural Disasters, Changing Values, beliefs, and Customs borrowed from other cultures.
- In Today's lecture, we will talk about population in detail.

But before that, we talk about a culture that preserves itself as it is without any change.

**Amish People**

They have preserved their values, beliefs, and customs without any change. They believe worldliness can keep them from being close to God and can introduce influence that could be destructive to their communities and to their way of life. They do not permit the use of tractors in their fields. They do not own or operate automobiles. They also do not use electricity.
They wear plain clothes without designs so named as plain people.

But except for them, 99% of the time in the entire world culture is constantly changing and there is one force behind it and that is the force of population. We are experiencing a population explosion in our cultures.

## Population and culture

**World Population trends**

Current trends in world population point out that by 2030 the world population is likely to be 10 billion and by the end of the 21st Century the numbers may rise to 30 billion. Populations in Sub-Saharan Africa and The Himalayan regions of Asia have already exceeded the immediate capacity of the area to sustain life.

**Pakistan Population trends**

Pakistan's population is likely to be doubled by 2050.

| | |
|---|---|
| Population Mid-2007 | 169,300,000 |
| Births per 1,000 Population | 31 |
| Deaths per 1,000 Population | 8 |
| Rate of Natural Increase (Percent) | 2.3 |
| Projected Population, 2025 | 228,800,000 |
| Projected Population, 2050 | 295,000,000 |
| Projected Population Change 2007-2050 (%) | 74 |

Our population is increasing because the birth rate is high, and the death rate is less. So, the growth trend increases.



In the above diagrams, life expectancy and density are given for Pakistan's population.

**Factors Affecting Earth**

1) **Environmental Indicators and population**

Based on the population more people use more resources, use more gadgets, and as a result carbon dioxide emissions increase. Stats are given below for these trends.


Vehicles are the biggest danger to the environment. More cars mean more carbon dioxide, so it means that our land fertility decreases and intense damage to the earth. The level of raising up crops decreases.


2) **Resource depletion and environmental deterioration**

Natural resources start depleting slowly. Resource depletion means the erosion of agricultural soil, Stalinization of highly productive irrigated farmland, deforestation, and Lake Acidification (more prevalent in the USA). This includes extensive loss of tropical forest, permanent soil degradation (Amazon Rive Basin) regional water shortages, and deteriorating water quality.

### 3) Public Policy and the environment

Public policy must Preserve the carrying capacity of the earth, major policy changes coupled with government business and individual actions can do much to alleviate these problems. Important measures need to be taken such as mandating reforestation after cutting, detoxification of chemical by-products before disposal, judicious soil management, energy and material conservation, and industrial and household recycling.

**Important facts about Pakistan need to be taken under consideration:**

**1) Pakistan's urban population to equal rural by 2030: UNFPA**

Pakistan's urban population is likely to equal its rural population by 2030, according to a report titled 'Life in the City: Pakistan in Focus', released by the United Nations Population Fund (UNFPA) here on Wednesday.

**2) Rural-urban migration (females)**

According to the report on Pakistan, the proportion of females is lowest in rural-to-urban migration and highest in rural-to-rural migration. The same pattern has been observed in India. In the rural-urban stream, the share of females is 51 percent in Pakistan. A relatively large fraction of rural-urban migrants' cross provincial boundaries. The perception that "the urban migrant is invariably a male" is incorrect; females make up a considerable proportion of migrants.

**3) Slum Population**

At least one in every three city dwellers in Pakistan lives in a slum. Many migrants, who move to cities to find jobs and have a better life, may not find jobs in the formal sector or any kind of decent shelter with a minimum of basic amenities. The informal sector employs most migrants, and they gravitate to squatter colonies where they build some kind of shelter for themselves. As a result, slums and marginal human settlements have spread in most urban localities, particularly in urban agglomerations.

**4) Increase in Urban Population**

The report also shows the share of the urban population increased from 17.4 percent in 1951 to 32.5 percent in 1998. The estimated data for 2005 shows the level of urbanization as 35 percent. The level of urbanization in Pakistan is the highest in South Asia.

More than 60 percent of the population of urban **Sindh** lives in Karachi and this concentration has increased over time. Approximately three-quarters of the total urban population of Sindh is concentrated in just three urban centers: Karachi, Hyderabad, and Sukkur.

In **Punjab**, 22 percent of the urban population lives in Lahore, and half of the total provincial urban population lives in five large cities.

**Peshawar** has a population of approximately one million without counting the Afghan refugees, which is 33 percent of the urban provincial population. The share of **Quetta** in the total urban Baluchistan population was 37 percent.

More than half of the total urban population of Pakistan lived in 2005 in eight urban cities: Karachi, Lahore, Faisalabad, Rawalpindi, Multan, Hyderabad, Gujranwala, and Peshawar. Between 2000 and 2005, these cities grew at the rate of around 3 percent per annum, and it's projected that this growth rate will continue for the next eight to nine years.

By 2015 it was estimated that the population of Karachi will exceed 15 million, while Lahore and Faisalabad will cross eight million and three million respectively.

This process is called **“Urban Blight”** The population is very Fast moving from rural areas to urban areas.

In the last part of the video lecture lesson 31, interviews with a few Officials from the society have also been conducted for your better understanding in the light of Pakistan and population.

**Lesson 32**

**Impact Of Environment on Its Incumbents**

In this chapter following topics will be discussed:

- Firstly, we'll delve into the relationship between human population growth and the environment.
- Nature and Environmental Changes in Rural and Urban Areas
- Urbanization
- The city as an unnatural habitat
- Crowding
- Rural Areas
- Impact of Environmental Changes on Industrial and Geographical Development
- Catastrophes and Human Adjustment

**Human Population Growth and Environment**

The human population of 6000 B.C. was estimated to have been about 5 million people. By A.D 1650 our numbers reached 500 million and in the succeeding two hundred years we had doubled our numbers to 1 billion. In 80 years, by 1930, we had doubled them again.
The current rate of population doubling is approximately 35 years (Ehrlich, 1968); that is, at current growth rates, we can expect the number of human beings inhabiting the earth to double every 35 years and to quadruple within the expected life span of any given individual.

**Growth rate = Total Births – Total Deaths**

Put differently, the world's population of humans is currently growing (total births minus total deaths) by an average of over 100,000 people per day (that's approximately one new city of Chicago each month). By the year 2030, barring unforeseen catastrophes, the number of humans vying for space on our planet will exceed 10 billion. The problem is humans do not scatter but make clusters and live in the same place rather than scattering on Earth evenly.
People move toward **urbanization** as they move towards cities more. So, population growth follows the trend of Urbanization. The number of cities with populations over 100,000 has quadrupled in the last 20 years and is expected to quadruple again in the next 20 years.

## URBANIZATION

As noted in the chapter's introduction, most of our population lives in or near large cities. Thus, it is not surprising that considerable effort has been expended toward describing and understanding the experience of urban life.

**Pakistan Situation**
Pakistan's urban population is likely to equal its rural population by 2030.

*Life in the City: Pakistan in Focus', United Nations Population Fund (UNFPA)*

**Urbanization in Pakistan**

**Sindh:** The share of the urban population increased from 17.4 percent in 1951 to 32.5

percent in 1998. The estimated data for 2005 shows the level of urbanization as 35 percent. The level of urbanization in Pakistan is the highest in South Asia. Sindh is the most urbanized province with 49 percent of the population living in urban areas.
**NWFP:** is the least urbanized province with only 17 percent of its population living in urban areas (according to the 1998 census).
The urban population in **Punjab and Baluchistan** in 1998 was 31 and 23 percent respectively. The urban population in Baluchistan and Islamabad has been increasing at higher rates of 5.1 and 5.8 percent respectively.

**Urban Centers of Pakistan:** More than 60 percent of the population of urban **Sindh** lives in Karachi and this concentration has increased over time. Approximately three-quarters of the total urban population of Sindh is concentrated in just three urban centers: Karachi, Sukkur, and Hyderabad. In **Punjab**, 22 percent of the urban population lives in Lahore, and half of the total provincial urban population lives in five large cities.
**Peshawar** has a population of approximately one million without counting the Afghan refugees, which is 33 percent of the urban provincial population. The share of **Quetta** in the total urban Baluchistan population was 37 percent.

**Urban Population Concentration**
More than half of the total urban population of Pakistan lived in 2005 in eight urban agglomerations: Karachi, Lahore, Faisalabad, Rawalpindi, Multan, and Hyderabad. Gujranwala and Peshawar between 2000 and 2005, these cities grew at the rate of around 3 percent per annum, and it's projected that this growth rate will continue for the next eight to nine years.

**Urbanization Growth Rate**
By 2015 it is estimated that the population of Karachi will exceed 15 million, while Lahore and Faisalabad will cross eight million and three million respectively. According to the UNFPA global report, more than half of the world population, around 3.3 billion, will be living in urban areas by 2008 and the number will swell to around five billion by 2030.

Country representative of UNFPA, Dr France Donnay said that the growth of cities would be the single largest influence on development in the 21st century. But little was being done to maximize the benefits of urban growth or reduce its harmful consequences. "Between 2000 and 2030, Asia's urban population is to increase from 1.36 billion to 2.64 billion and Africa's from 294 million to 742 million".

Urbanization results in the formation of slums and affects cities' infrastructures and resources.

## Slum Formation

At least one in every three city dwellers in Pakistan lives in a slum. Any migrants who move to cities to find jobs in the formal sector or any kind of decent shelter with a minimum of basic amenities. The informal sector employs most migrants, and they gravitate to squatter colonies where they build some kind of shelter for themselves. As a result, slums and marginal human settlements have spread in most urban localities, particularly in urban agglomerations.

### “Katchi Abaadi” or Slum Statistics

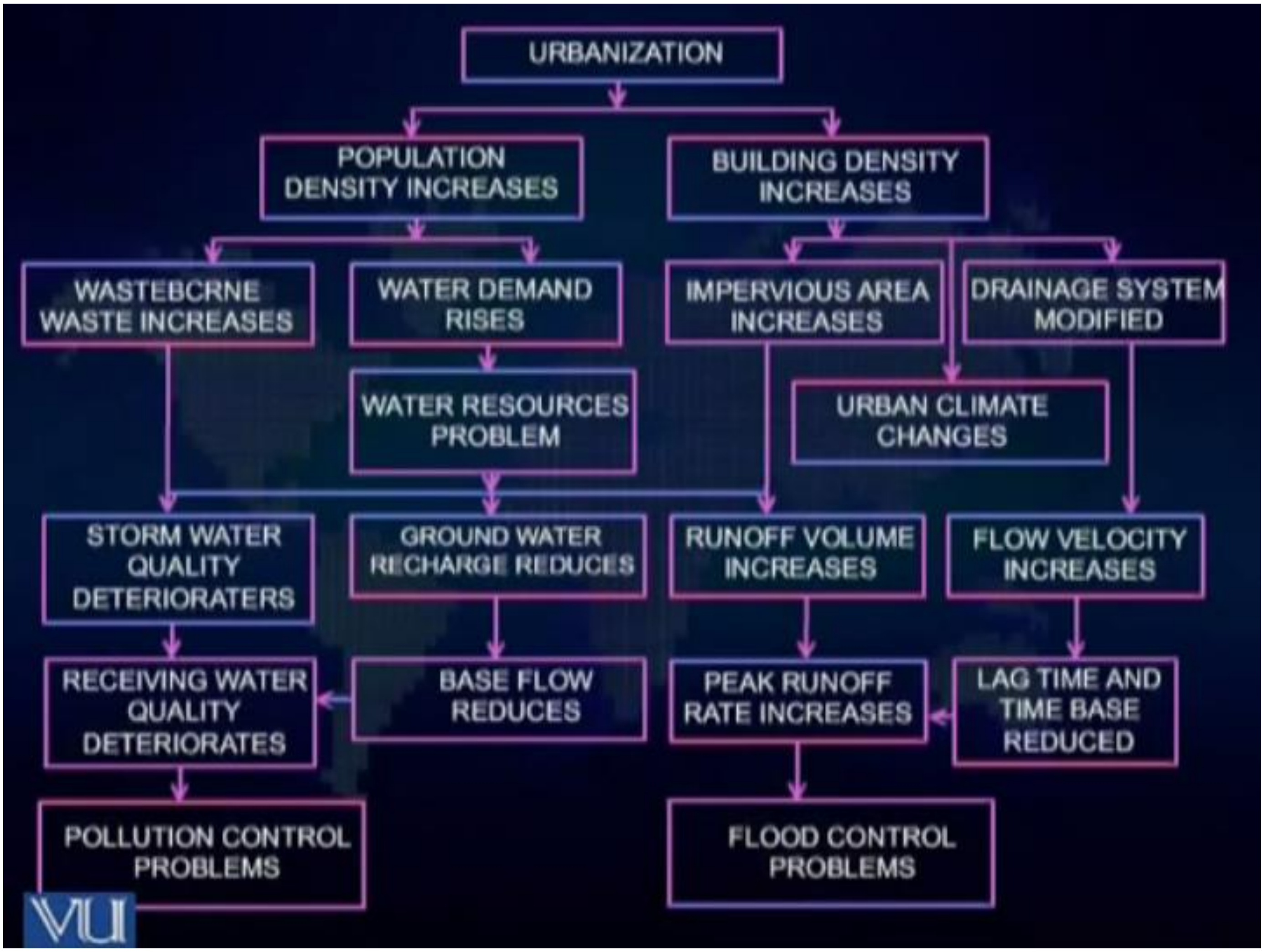
In Pakistan, the urban population living in “*katchi abadis*” varies between 35 and 50 percent. The growth of these informal settlements in the two megacities, Karachi, and Lahore, has particularly been massive. In Karachi, these settlements increased from 212 in 1958 to more than 500. In Lahore, there are more than 300 katchi abadis in Faisalabad, at least 40 percent of the population lives in these abadis.

### Effects of Urbanization

There can be the following effects of urbanization:

- Physical Effects
- Social Effects
- Psychological Effects

When urbanization increases, people start moving to the suburbs of the city and all of this remains unplanned, and this urbanization keeps moving like a vicious cycle. As a result, following **physical effects** occur:



**Psychological Effects**

In cities, people become busy and don't find time to become friends with others, start feeling lonely and problems increase.

Anonymity increases as people don't know each other and hence loneliness increases as well as the crime rate increases, and privacy is compromised here.

**Lesson 33**

**Effects of Urbanization**

In this lecture, we will continue the psychological and physical effects of Urbanization.

**Physical and Psychological Effects of Urbanization**

Then there are problems related to crowding. **Crowding** means that more people are sharing a certain area. That means people might be closer to each other than this is comfortable for them. An example could be one gentleman who has been living in probably some rural area and moves to Karachi and starts living in an apartment building, he finds out that people are very close for comfort. And they are scooped up into small little dingy apartments where everybody is living their lives.

So, these people might find out that their privacy is being confiscated.

Secondly, because of urbanization **pollution** also rises. When pollution rises, then some physical diseases and disorders are also in their eyes.

There would be more food required and the supply chains for such food probably might not be very well in third-world countries.

So, **starvation** might also be one of the negative outcomes of the big cities. With this **drug and alcohol abuse** has been found more in the bigger cities than in the rural communities. And there has been **family disorganization**. It's been said that in Western countries, the institution of families is deteriorating, and families are disintegrating. People living as a family are much fewer than they were used to in the past. This has been seen that as people move into the urban areas, some kind of family disintegration starts taking place because families go too far for the comfort of the people. They are constantly working; they are constantly taking different types of pressures. So, family disintegration becomes one of the major negative outfalls of urban lives. However, it does not happen so frequently.

So, students, these were some of the physical and psychological factors that people suffer from in urban areas.

**Problems of Population/Urbanization**

There were two main problems:

- One is that the population is growing very fast,
- And second is population is following the trend of urbanization.

Now we will understand these problems under multiple theoretical frameworks.

**Theories of Urban Effects**

1. Wohlwill's Adaptation Level Theory
2. Milgram's Information Overload Theory
3. Proshansky, Ittleson, Rivlin- Behavior constraint Theory

**1) Wohlwill's Adaptation Level Theory**

Adaptation level theory means that human beings function best when their level of arousal is on an intermediate scale.

This means that they are not too much aroused, that they are not too under-aroused. When we apply this theory to city life then we find out that there is a lot of population in the city. There are so many entertainment places.

Then so much is going on all the time in the city. A gentleman who comes from a far-off village to a city in Karachi or Lahore might say that this city never sleeps. Even at two o'clock at night, you could find cars on the roads, you could find out the food hangouts are open, and a lot of activity is going on throughout the night. So, apply the adaptation level theory. You will find out that in the cities, people are constantly stimulated. Low-income, constant stimulation will be here.
Or if stimulation will be here constantly, that means the arousal level is up. And when the arousal level is constantly up, then what happens?
An arousal level increase means that a lot of hormonal activity is going on. People might bump into a fight-or-flight reaction. So, you can find out the problems like low concentration, and low performance level.

### 2) Milgram's Information Overload Theory

That means if more information is coming our way, then we can handle it, then our behavior, and we simply start missing out on that information. This means that we are so loaded with information that we don't know how to find a way out of it, how to analyze it, or what to do with this. According to the information load theory, the quantity and rate of stimulation that urbanites are exposed to exceeds their capacity, resulting in information load. So, another way to understand the city would be that concerning information load, you will find a lot of people who complain about information load. Because the city is constantly stimulating them, the city is constantly providing them with something.

### 3) Proshansky, Ittleson, Rivlin - Behavior constraint Theory

The demands of city life limit an individual's freedom, leading to behavioral constraints that can produce feelings of loss of control, leading to behavioral constraints. For instance, a boy living in the village moved to the city because he must study at a university now. When he comes to the university, he finds out that situations are so complicated, he must work very hard to find his way into it and way out of it. Eventually, he bumps into behavioral constraints, which means that for some time, he would start feeling that he's kind of losing control over the situation. And when people are losing control over the situation, what kind of psychological feeling do they go through? That means they're feeling frustrated. That means they're feeling depressed about the entire thing. Because the entire situation, the entire environment becomes so complicated for them, and it is charging so heavily upon their senses that they find it hard for themselves to kind of control it. And that type of feeling creates frustration, anger, and remorse, dejectedness, depression.

Now, can you correlate these feelings with the incidence of crime somehow? Do you think that a satisfied person who's happy about himself, do you think a person who's calm about things and who has control over the things around himself, has some inclination towards crime, the possibility is very less? Compared to people who are going through extreme psychological feelings, the incidence of crime in such a state of mind might be more prevalent. There's more possibility for the crime to take place when a lot of people are going through such a mental state. So, we can apply this theory also.

At this point, it is also interesting for you to note that a gentleman, a psychologist whose name is Baker, said cities are overmanned. In cities, there are so many people living together that the city's infrastructure chokes. And when the infrastructure chokes, then people bump into negative behavior patterns. And when people bump into negative behavior patterns, that might be devastating not only for themselves but also for the people around them. These are some of the theories that we can apply to understand the urban effects.

**Urban VS Rural**

Number one is urban-rural comparisons. Number two is social behavioral effects. Number three is beneficial effects.

Find out a friend of yours or a cousin of yours who lives in a big city and ask him what charges him. Ask him about his routine. You will find out how urban and rural life differs. Urban life seems to be more complicated. People seem always to be on the run. Urban life can be the lowest income in the years of the year. When you get into the villages, you find out that rural life is kind of simple. This is why if you have spent a long time in the city, you probably might point out yourself to be very peaceful in village life. You have different facilities that you are so used to using in the cities that are not available to you in the village. You might bump into a different type of reaction. You might not actually like that place. However, environmental psychology is the subject. This study the effect of the environment on human behavior. In this very comparison, we can find out that the environmental features in urban life and rural life are two distinct and different types of features. And they impact human behavior differently. So, it is very natural that we can always compare urban life with rural life.

And for the start of it, when people compare both lives, they bring up their own points of use into that. They might say that rural life is very good, calm, and serene. And then there might be people who might say that urban life is wonderful because you have all the facilities and entertainment available to you. We start our comparison with the physical and health diseases that are found prevalent in the people of rural areas and the people of urban areas.

1) **Physical and Health diseases**

Statistics show that there is probably not uniformly worse physical and ornamental health in cities compared to rural life. There is a higher rate of respiratory diseases in the cities. That is understandable because in cities there is a lot of pollution. The minimal differences in the incidence of stress-related illnesses such as heart disease and hypertension. This research was carried out in old research (Hay & Wantman, 1968).

Then it is seen that alcoholism and drug addiction are more common in cities than in rural areas. It has also been found that the incidence of psychosis is higher in urban areas than in rural areas. This research was conducted by Doran Mann and Doran Mann. So, students, when we compare urban and rural areas concerning the health and disease factor we find out that there is not much significant difference over here. Certain diseases are more specific to city life and there might be certain diseases that could be more specific to urban life.

2) **Social and behavioral effects**

Differences between urban and rural settings can be observed concerning two important psychological factors. Number one is **affiliation** and number two are **pro-social and anti-social behavioral patterns**.

**Affiliation**

When it comes down to affiliation we find out that urbanites are less affiliative towards strangers than are people in the rural areas. This research was conducted by Macaulay, Coleman, and Defuso in 1977, then Milgram and Newmann, and Macaulay once again studied it in the same period. They also found out that urbanites tend to avoid eye contact with strangers and are less likely to reciprocate friendly gestures than rural dwellers. That means people living in the cities are very bad because they have a lesser affiliation towards other people than the people living in the rural areas. Well, these are the statistical findings. This is some of the research that has been conducted and these are two different behavioral patterns. It does not mean that people living in the city are bad or people living in the rural areas are good or bad either. It means that both people are exposed to two different types of environments and since they must survive that environment, different behavioral tactics work in one place and different behavioral tactics work only in the other place. In rural areas, few people know each other, and usually, the threat to the community or the fabric of the community is very, very strongly woven. In the cities, the threat or the fabric of the community is not very strongly woven because there are people from all over the country who are living in the cities. As we discussed in our last lecture, people also might be suffering from the feelings of anonymity and people might be feeling that there are a lot of strangers living in this place, they might have been using this type of behavioral pattern as a protective measure to survive in the city. This is exactly what these two researchers found out.

**Pro-social Behavior**

It has been observed that urbanites are less likely to help a stranger. This research was conducted by Galphand, Hartman Welder, and Page, then Kurt and Kerr as well as Melkram. Urbanites are less likely to help a stranger. What does it mean? Urbanites are selfish people or people living in urban areas are not good ones. It could mean so many different things. But it has been found that since in the city, a lot of people are living together and people are from different areas, the level of anonymity is high. The level of feeling stranger is kind of very high. And then simultaneously in the city, the stimulation level is also very high. People are constantly aroused. There could be the application of behavioral constraint theory. People tend to lose control over the scenario. And since they are already losing control over the scenario, they might not use extra energy for the strangers. If you apply the stimulus adaptation level theory, you might find out that so many stimulants are available to the people in the city that little energy is left for the stranger. Then of course in the city a lot of trainers are living and then people are from so many different places, you don't know about them. You probably would not be able to trust them so easily. So, this is one type of behavioral pattern that is found concerning pro-social behavior.

**Anti-social behavior**

Anti-social behavior is the exact opposite of pro-social behavior. Pro-social means that you are pro-people. You are towards the people. You affiliate with them. You connect with them. You relate to them. And what is anti-social behavior? Anti-social behavior means that you are against people. You don't want to connect with them. Or you want to take advantage of them. So, when we talk about anti-social behavior, you will start a behavioral pattern that will be used in your crime.

**Statistics:** Clear differences have been reported between herbal and ruined areas in the incidence of crime. This research was conducted by a courtsman, and Levy, fisher, and Zimbardo. They have found out that in urban areas, the incidence of crime has been quite high. And in the rural areas, the incidence of crime has been quite low. Now what can be the reasons? The factor of anonymity might play a role in the crime. Then if you apply the adaptation level theory, we understand that people are constantly stimulated. Their arousal level is constantly high or constantly low. They are not performing very well. And when they are not performing very well, probably their efforts towards achieving certain objectives are also not very successful. That leads to frustration and dejection that might lead to the incidence of crime also. Then if you apply the behavior constraint theory, loss of control over things. A young boy comes to the city and wants to be a rich man. He probably does not find good opportunities. He gets so frustrated and upset. And he thinks that I don't know anyone here. And this is why so many different people are living. It might decide to bump into a crime. So, in city life, it has been found that the incidence of crime has been higher than in rural areas. And there are some understandable facts about that also.

There is everything negative about the cities. Is that so? Now do not mean to say that. There is, it is not like that that everything is negative about the city. The city life is kind of different. Otherwise, if you ask most of the people, they will not say that they want to live in the cities or they want to live in the urban centers.

**Beneficial effects of city:** There is an interesting fact to note, which was researched by Malton and Hargrove says the city provides you with all the infrastructure where you could become a more sophisticated person in your career. There are so many different options available in the city, the business is going on. In the city, all the opportunities are available to you. So that is why cities are places where people find more opportunities. When they find more opportunities, they can express their inner potential and their inner talent. And once they can express their inner potential and talent, probably that might be a satisfaction for them. And of course, people make more money in the cities. So, they say that rural scenes are substantially more likely than urban scenes to be described as pleasant turns. Rural inhabitants are more likely to be perceived as friendly. This was research that was conducted by Malton and Hargrove in 1987.Looks like this research once again is telling us that cities are bad places because a lot of people find out that the rural scenes are more beautiful. They find out that the rural people, rural inhabitants are more friendly than the city inhabitants. How do you understand this research? This research is simply a comparison that compares people from rural or urban areas concerning behavioral patterns. So, as we discussed when you look at the city from the positive point of view, we find out that much of the research starts with a biased and pessimistic assumption that cities are bad. Most of the research has probably been carried out with a certain bias in the mind of the researcher that the city is something bad. It's a bad place to live. And this fact was researched by Friedman and that's also quite old, way back in 1975.

According to a Gallup opinion index, most of the respondents indicated they preferred living in cities. So, there are some positives also attached to the cities. Life in the city makes an individual more versatile and adaptable and gives a broader perspective on life than is afforded in rural life. So, these are some of the ways rural areas and urban areas can be compared.

## Crowding

Considering the above-mentioned topics, we will study crowding in detail under the following topics:

- Animal Studies
- Human Correlational/survey studies

- Human Experimental Studies
- Social Affective Responses
- Crowding in everyday setting

Crowding means a lot of people are living in little space. When more people are living in little space, then that means their privacy has been hampered. That means they are probably not feeling comfortable with each other. So crowding is an artificial state where people are too close for comfort. Crowding means that you are too close to comfort with the people. This means the density of the population is ticking and people are so close for comfort with each other that they are not feeling comfortable or peaceful with each other anymore.

**Animal Studies**

Naturally, we can use many ways to do our studies or our research. However, it has been found that whenever we are doing psychological research, since it involves human beings, sometimes conducting experiments becomes very, very dangerous. So, the experiments are conducted on animals. Huge caution needs to be taken from animal study skills that are generalized to human beings. Secondly, animal studies have a direction to do this. It gives us a strong hypothesis that is almost there to be proven that this is the way human beings can also actually act. Animal studies are very difficult to understand. We can use them in 2 lines to understand the research.

**Naturally occurring population cycles:** This is how we can naturally do population cycles.

**Experimentally controlled population cycles**: This means the population is experimentally controlled. There are certain conditions. There are experimentally controlled species of a certain animal. There is another group that is not being applied to any kind of independent variable, which means that no changes have been imposed upon this group's growth of population. They let it happen very, very naturally.

1) **March to the Sea Study (Natural Population Cycle Study)**

This research was conducted by Dubos in 1965. Whenever we want to understand what happens concerning a population of human beings, this becomes very, very interesting research to code. You can see an animal that closely resembles a rat. This is a Norwegian Lemming. Norwegian Lemmings are small rodents resembling the field mouse but having short tails and covered feet. They live primarily in the Scandinavian Mountain regions. Lemmings can be used by the natives that about every 3 or 4 years they appear to migrate or march to the sea with many of them drowning as a result. This is very, very strange behavior. Lemmings are growing up, and up to a certain extent they grow, and then they certainly decide to commit suicide. They simply move towards march towards the sea and drown themselves up so that they can balance their population. So, what exactly is this? Do you think Lemmings committed suicide? Because they are too many in number and some of the people, some of the Lemmings decide to sacrifice themselves. What is happening over here?

When we do a closer observation, we find the Lemmings when they are marching toward the sea, this moment is not very orderly. This is rather full of frenzy. So, what is going on? Is it suicide or an accident? They find out that certainly Lemming's bump into a frenzied activity or helter-skelter on the Lemmings are just madly running towards the sea or in some Lemmings simply drown themselves in the sea. So closer observation reveals that this is not an organized thing. This is a disorganized thing. This is some kind of madness that the Lemmings are experiencing, or the

Lemmings are going through. Lemmings are prolific reproducers. Every three or four years they reach considerable numbers. This increased population density acts to influence brain and adrenal functioning which in turn is overly manifested in non-directed activity. They found out that when Lemmings reach a certain level of population then in Lemmings a very heightened level of adrenal activity starts taking place. Adrenal activity is very closely related to frenzied behavior. So biological pre-programming means that they simply bump into frenzied because of the adrenal gland glandular functioning and then in that frenzied, they don't know where they are going, and they don't know what they are doing, and they simply commit suicide, or they simply die and drown themselves into the sea.

The question is do human beings do the same thing? Human beings are taken as lemmings and the population becomes more and they are jumping into the sea. Are there any Lemmings humans? Does the population create some impact on us? Well, we can give you another interesting study. That will take the story forward. And this interesting story is called Dear Die Off on James Island.

**Dear Die, Off on James Island**

This research was conducted by Christian Figer and Davis in 1960. In 1916 4 or 5 deer were released on James Island in Chesapeake Bay off the coast of Maryland. By 1955 this small herd had grown to 280-300 deer. In 1958 half of the herd died. This was something very alarming what happened with all these deer? How come they die? They had crossed their population level up to a certain extent and then they simply started dying. What happened to them? They found out that the herd stabilized on the threshold of 80 members. From 1958 onwards they maintained the level of threshold of 80 members. So, what was happening with these deer? A closer examination was conducted by Christian Scrope and this is what they found. Scientists John Christian Fliger and Dave Devas studied 18 after surviving deer. They performed a detailed histological examination of the deer's adrenal glands, thymus, spleen, thyroid gland, kidney, liver, heart, lungs, and other tissues. The care cast was found to be in excellent shape throughout. The only abnormal finding was an increase in the size of the adrenal glands. In most severe cases, adrenal glands were found to be 10 times the size of the base rate samples. They found out the deer were in excellent shape. There was nothing wrong. Their heart spleen, they were, everything was functioning fine, and they had the right amount of fat in their body which shows that they were healthy deer, but they found only one abnormality, and that abnormality was died der had 10 times bigger adrenal glands. The **adrenal gland** is the gland that controls the thinking activity that controls the behavioral activity of an animal. So, if the adrenal gland is 10 times thicker than the normal adrenal gland which is found in most deer, then what is the reason for that? You would find out that the Christian's crew came up to be forming a very interesting conclusion. They said they determined that while they were in good nutritional condition, their physiology indicated hyperstimulation from the stress of crowding and attributed the many deaths to this. You remember Wahlbl's theory, adaptation level theory, that when people perform best at the normal level of arousal, at a medium level of arousal. In this case, the deer is overstimulated, which means he is not performing at the normal level. Deer are overstimulated because of the overcrowding. They are too close to each other probably. And that creates stress. That stress leads to these biological deformities and eventually, the deer dies.

We find out one thing about the animals, that the animal’s instinctually controlled behavior is usually very, very huge. However, when we come down to human beings, we find out that there is reason and logic. That is why human beings are more adaptable, they adapt more and more. They change more and more. They just want a human being that is surviving. So, however, these studies point out one thing that more population, if there is more population, that is some kind of discomfort.

How are we going to deal with this? We can talk about this in the next lecture.

**Lesson 34**
**Problems Related To Crowding**

Crowding-related issues are significant topics of investigation within the field of environmental psychology. Research on animal behavior in crowded conditions provides valuable insights into the impacts of population density on various aspects of physiological and behavioral functioning.

**Animal Studies of Crowding**

Animal studies concerning crowding can be categorized into two primary streams of inquiry. Firstly, investigations focus on naturally occurring population cycles, while secondly, researchers experimentally control population concentrations.

a. **Naturally Occurring Population Cycles**
Examples of it include studies such as the lemmings' phenomenon of migrating towards the sea, as observed by Dubos (1965), and the die-off of (Sika) deer on James Island, as documented by Christian, Flyger, and Davis (1960).

The migration of Norwegian lemmings, small rodents resembling field mice, every three to four years, often culminates in many of them drowning. While Scandinavian folklore attributes this behavior to a deliberate suicidal tendency ingrained in their biology, closer examination reveals it as a frenzied, rather than orderly, movement. This phenomenon suggests that increased population density influences brain and adrenal functioning, leading to non-directed activity.

Similarly, Christian, Flyger, and Davis (1960) noted abnormal adrenal functioning in a herd of deer experiencing increased population density. Their observations on the herd's health, particularly the enlargement of adrenal glands, indicated heightened stress levels due to crowding as a probable cause for the observed physiological changes.

b. **Experimentally Control Population Concentrations**
In contrast, controlled experiments, such as those conducted by Calhoun (1962) with Norway rats, provide further insights into the consequences of overcrowding. In this study, Calhoun provided Norway rats with an environment characterized by abundant resources, including unrestricted access to food, water, and nesting materials. Notably, the only limiting factor imposed was space, as the physical environment remained constant throughout the study. Over time, the rat populations expanded, leading to a phenomenon where dominant individuals claimed more spacious territories, while others were confined to increasingly crowded areas within the confined space. This confined area was termed a "behavioral sink" by Calhoun.

Within the behavioral sink, observable behavioral and physiological abnormalities emerged among the rat population. For instance, nesting behaviors among females became incomplete or poorly executed, resulting in neglected pups and disrupted nursing behaviors. Additionally, juvenile males exhibited pansexualism and instances of cannibalism, while varying levels of aggression and withdrawal behaviors were observed among different individuals. Furthermore, increased rates of infant mortality and aborted pregnancies were noted. Physiological abnormalities included the development of tumors in mammary and sex glands among females, as well as anomalies in the kidneys, livers, and adrenal glands in both sexes.

Calhoun's research prompted inquiries into the potential association between population density and the prevalence of pathological conditions in humans. The pathologies identified by Calhoun, including morbidity, mortality, fertility issues, ineffective parenting, and psychiatric disturbances, or their related manifestations such as infectious diseases, juvenile delinquency, and adult criminality, have been tentatively linked to escalating human population densities, albeit with varying degrees of success. This body of work underscores the complex interplay between environmental factors and behavioral and physiological outcomes, providing valuable insights into the dynamics of crowded environments and their potential impacts on human well-being.

Additional effects of increased population concentrations have been reported subsequent to Calhoun's seminal work. For example, it has been shown that population density leads to the production of fewer sperm by males, to later onset of estrus, less frequent and shorter duration estrus, and consequently less frequent births and smaller litters (Crow & Mirsikowa, 1931; Snyder, 1966; 1968) among females. There is also evidence for an increase in alcohol consumption and performance decrements on complex tasks (Goeckner, Greenough, & Maier, 1974) for rats raised under high-density conditions.

### Challenges in Extrapolating Findings to Human Behavior

Similar effects have been observed across a diverse range of species including swine, chickens, cattle, elephants, and various rodents (Greenburg, 1969; Hafez, 1962; Marsden, 1972; Theissen, 1966). However, caution is warranted when extrapolating these findings to humans:

- Firstly, the behavior of non-human animals is often more biologically determined and less influenced by learning and cultural factors (Swanson, 1973).
- Secondly, humans typically have opportunities for temporary relief from high-density environments, unlike animals studied under controlled conditions (Evans, 1979).
- Lastly, humans have demonstrated superior capacities for adaptation and adjustment compared to other animals.

### Value of Animal Research in Understanding Human Behavior

Despite the difficulties in using findings from studying animals to understand humans, this research is still really important for learning about human behavior. Animals can help us come up with ideas about how people might behave, especially when it's not ethical to test on humans directly. For example, if we want to see how living in crowded places affects future generations, it's better to use animals that have shorter reproductive cycles than to use humans. Also, with new technology, we can monitor animals' behaviors and bodies without hurting them, which helps us avoid ethical problems we might face with human studies. But it's also important to note that we do research with humans too, and there's a lot more to learn in this area.

**Lesson 35**
**Impact of Population Concentration among Humans**

Population concentration among humans has been a subject of significant interest in environmental psychology, particularly concerning its effects on crime, mental illness, and social disorganization. This section explores early research and contemporary perspectives on the impact of population density on human behavior.

**Historical Perspectives**
Early investigations focused on the correlation between population concentration and various social indices, such as crime rates and mental health indicators. Studies have consistently shown higher rates of violent crimes and worsening mental health conditions in densely populated urban areas compared to rural regions.

**Allure and Pessimism**
The idea that increased population concentration could be responsible for these effects presents both pessimistic and alluring implications. While acknowledging concerning trends, researchers are intrigued by the possibility of understanding human behavior through mechanisms similar to those affecting other animals.

**Defining Density**
Investigators have utilized various measures of population concentration, often interchangeably referring to them as crowding or density. However, consensus on the appropriate measure remains elusive.

**Schmitt's Measures**
Schmitt (1957), for example, assessed the relationship between five semi-independent measures of density and adult crime and juvenile delinquency.
They used the following five measures

- Average household size
- Proportion of married couples without their own home
- Proportion of dwelling units in structures with four or more living units
- Population per acre
- Percentage of occupied dwellings with 1.51 or more persons per room

Drawback of the above measures is that only the last two showed strong positive relationships with delinquency and adult crime.

**Inside vs. Outside Density**
In subsequent work Schmitt (1966) has distinguished between inside density and outside density. That is, Schmitt noted the difference between the number of people per unit of living space and the number of people in the larger community (e.g., persons per acre). The importance of these contrasts becomes apparent when the following possibilities are considered: low inside-low outside (suburbs); high inside-high outside (urban ghettoes, barrios, and others); low inside-high outside (luxury inner city apartments and condominiums); high inside-low outside (rural farm areas).

Drawback of Relying solely on inside density would assume uniform effects across different environments, while considering outside density provides more nuanced predictions. These insights underscore the importance of precisely defining the term density.
A second challenge highlighted by the earlier considerations pertains to the differences in socioeconomic, ethnic, and social structure variables among the four areas under study. These variables might independently contribute to degenerative or pathological effects, but their co-variation with density can lead to misattribution of causality.

**Galle, Gove, and McPherson Study**

Recent research has aimed to address this issue by adopting more explicit measures of density and carefully accounting for confounding variables related to social structure. For instance, in a study conducted by Galle, Gove, and McPherson (1972), correlations were observed between persons per acre and public assistance rate, juvenile delinquency, and admission rates to mental hospitals. However, these correlations disappeared when controlling for social class and ethnicity, suggesting that density might not be directly related to social disorganization once these factors are considered.
Given these findings, Galie's group undertook a reanalysis of their data. In this subsequent analysis, density was defined differentially, encompassing factors such as:

- the number of residential structures per acre,
- the number of living units per structure,
- the number of residents per living unit,
- and the number of persons per room.

Utilizing multiple correlation techniques, they demonstrated that these refined measures of density significantly correlated with various pathologies, even after controlling for class and ethnicity.
Their findings indicated that the number of persons per room was the most significant contributor to various pathologies, except for admissions to mental hospitals. Additionally, factors such as the percentage of people living alone and the number of persons per living unit emerged as strong predictors of certain pathologies.
Drawback of the above studies was pointed out by Kellett (1989), whom has questioned the adequacy of density as a concept for investigating the relationship between health and housing, suggesting that further scrutiny is warranted.

**Lesson 36**
**The Distinction between Density and Crowding**

Recent research by scholars such as Altman (1975), Loo (1973), Proshansky et. al (1970), and Stokols (1972a, 1972b, 1976) has highlighted the crucial difference between physical density and psychological crowding. Stokols (1972a) notes the tendency to interchangeably use the terms density and crowding in human research, emphasizing the need for distinct definitions. While density refers to physical limitations of space, crowding represents an experiential state influenced by personal and social factors in addition to spatial constraints. Thus, Stokols proposes that density is a necessary but not sufficient condition for crowding. As Stokols views it, the potential inconveniences of limited space such as the restriction of movement, the preclusion of privacy, and other disadvantages of space limitation must reach some degree of saliency and be viewed as aversive before people experience crowding.

**Diverse Definitions of Crowding**
Various scholars, including Desor (1972), Zlutnick and Altman (1972), and Van Staden (1984), have proposed different definitions of crowding. For example, Desor (1972) has defined crowding in terms of excessive stimulation from social sources; Zlutnick and Altman (1972) have defined crowding in terms of an individual's inability to adequately control interactions with others; Van Staden (1984) emphasized perception of large numbers of people, spatial restriction, and an aversive experience of the situation as relevant components of the crowding construct. In this regard crowding may be thought of as a psychological concept referring to a subjective experience, which may or may not be adequately reflected by population density measures such as the amount of physical space per person or number of people per unit of living space. In short, there seems to be general agreement that high density is seldom a sufficient and probably never a necessary condition for the experience of crowding.

**Complexity of Density and Crowding**
Density, particularly areal density, encompasses multiple components such as residential average, structures per area, dwellings per structure, rooms per dwelling, and persons per room. This complexity highlights the variation in neighborhoods with similar population densities but distinct land use patterns, affecting the experience of density differently. Operationalizing crowding proves challenging, with current approaches relying on correlational/survey methods. However, limitations in controlling extraneous variables and definitively linking density to pathology hinder conclusive findings.

**Laboratory Studies for Enhanced Understanding**
To address these challenges, environmental psychologists have turned to laboratory studies to manipulate physical and social parameters of crowding more effectively. For instance, Arkkelin et al. (1982) developed simulation techniques allowing independent manipulation of density components and found that interpersonal distance interacts with social variables to influence subjective discomfort more significantly than the absolute number of people present. Their research suggests that crowding is characterized by heightened arousal, displeasure, and a sense of loss of control. Subsequent sections will review additional studies exploring density manipulation and its impact on affective and social-behavioral responses.

## Crowding & Density Human Experimental Studies

Conducting carefully controlled experimental studies of the effects of population density on human behavior involves many difficulties. Nonetheless, a number of studies have been conducted and reported. Some research has utilized rather interesting, but naturally occurring, high-density situations like prisons, commuter trains, playgrounds, and shopping centers. Others have purposely manipulated either numbers of persons within a setting while holding setting size constant (i.e., manipulated social density) or have maintained the same number of respondents while manipulating the setting size available to them (i.e., manipulated spatial density; see, for example, Loo, 1972; McGrew, 1970; Pedersen, 1983; Saegert, 1973, 1974). In keeping with the previous chapters, we will look at what is known about the effects of density on physiological processes, on task performance, and on such social affective behaviors as attraction, altruism, and aggression.

### Physiological Reactions

One of the first researchers to report on the physiological effects of density was D'Atri (1975). He observed the blood pressure levels of inmates in a prison who were confined to either single or double occupancy cells and found those inmates in the latter cells to exhibit higher levels. Paulus, McCain, and Cox (1978) have also found blood pressure to be positively related to increased density, and McCain, Cox, and Paulus (1976) have reported that inmates in high-occupancy dormitory settings complain more of illness than do those in lower-density cell blocks. This finding may occur as a result of real illness brought on by physiological disturbances or by a desire on the part of the inmate to be taken from the high-density dormitory to the lower-density infirmary. In either case, the differential density seems to be the reason for the discrepancy in the number of complaints.

### Task Performance

Early research on the effects of density on task performance utilized relatively simple tasks like psychomotor tasks, problem solving, or anagram solving in laboratory contexts where subjects either knew or could easily discern the interests and hypotheses of the experimenter. Saegert, Mackintosh, and West (1975) found support in an interesting and provocative field study for the notion of increased density leading to performance decrements.

Saegert and her colleagues tested subjects in a socially dense department store and in a busy railway terminal. Among the tasks subjects were asked to perform was to provide a cognitive map (see Chapter 4) of their environment. The researchers concluded that tasks which involved knowledge and/or manipulation of their environments were impeded by increased density. The fact that increasing density is accompanied by a corresponding decrease in the clarity of one's mental image and knowledge of the immediate environment may also explain the findings of Glassman, Burkhart, Grant, and Vallery (1978) and Karlin, Rosen, and Epstein (1979). These researchers showed that social density was related to grades attained in two separate university settings. Additionally, Karlin et al. (1979) showed that once conditions of high social density were removed (i.e., students were reassigned to less crowded quarters) grades improved significantly. Thus, increased density may overload one's information-processing ability, resulting in impaired performance on tasks that require higher-level cognitive skills.

**Social-Affective Responses**

It is not difficult for any of us to recall situations of high density that have made us feel uncomfortable. Crowded elevators, bargain basement sales, the first day of classes in a large lecture hall, bus depots and airport terminals where we are hurrying to make connections are but a few. Without much difficulty we could probably also recall occasions when this discomfort has led to feelings of tenseness, anxiety, or even anger. On the other hand, parties, rock concerts, and crowds at athletic contests also represent situations of high density that may have evoked very different, perhaps even very positive, feelings. Hence, our feelings are obviously related not only to density but to the circumstances under which density occurs.

A number of studies have investigated the effects of density on social behaviors such as attraction, altruism, and aggression. Let us examine these findings.

- **Attraction**

  The first laboratory study to show that interpersonal attraction (liking) is reduced under conditions of high social density was conducted by William Griffitt and Russell Veitch (1971) at Kansas State University. Subjects in that experiment were exposed to one of two conditions of social density and one of two levels of ambient temperature. After being given time to become inured to these conditions, participants were asked to make evaluations of strangers and to indicate their degree of probable liking for them. Subjects who were exposed to the high-density conditions gave significantly more negative evaluations and expressed a lower degree of probable liking than did those exposed to low-density conditions.

- **Aggression**

  It might be expected that if differential density leads to diverse emotional responses, then to the extent that aggression is related to mood, aggression might be expected to vary as a function of density. Density does not automatically lead to a specific emotional response. For this reason, and following the above logic, aggressive responses would be expected only to the extent that density elicits negatively toned affective responses. This elicitation results not only from the density of the environment but by the personal and social conditions of the setting. For example, Rohe and Patterson (1974) have found that high density leads to aggression in children when the play situation does not provide ample toys for each. As long as there are plenty of toys to go around, increases in density did not affect children's aggressive behavior. Thus, it is likely that density, like temperature, acts only to moderate aggressive responses and is not the direct cause of them.

- **Altruism**

  Altruism studies suggest that density doesn't directly impact helping behavior, but rather influences it indirectly based on other factors. Factors like environmental affect, fear for personal safety, and positive mood states play roles in determining helping behavior. Discrepancies in helping behavior between rural and urban areas may be due to differences in attentional processes or varying lifestyles affecting available time for assistance.

## Crowding In Everyday Settings

In the field of environmental psychology, researchers have explored how living spaces affect people's well-being. However, their findings have not always been consistent.

**Residence and Density**
Researchers studying environmental psychology have explored how living spaces affect people's well-being, particularly in relation to density. Sweaney et al. (1986) found that as the number of people in a living space increased, children tended to feel more crowded. On the other hand, Duckitt (1983) discovered that dissatisfaction with living spaces could occur in both high-density and low-density environments. In a related study, Gabe & Williams (1986) noted that students living in rooms meant for three people but occupied by two had lower self-esteem than those in rooms designed for two people. However, Ronchi & Sparacino's findings did not show significant differences in how people rated the pleasantness or arousal levels of different living densities.

**Why Inconsistent Results?**
The inconsistencies in research findings could be attributed to several factors. One major factor is the differences in how researchers conducted their studies and defined terms like "density" and "crowding." These variations in methodology may have led to differing conclusions about the relationship between living density and psychological experiences. Additionally, it's possible that there isn't a consistent relationship between living density and psychological responses, highlighting the complexity of this area of study.

**Social Variables and Density**
Some studies have suggested a link between living density and suicide rates, but this relationship may be influenced by social and cultural factors. For instance, perceptions of population density vary between different cultures, such as the differences observed between the United States and China. People from different cultural backgrounds may also exhibit varying levels of tolerance for high-density living. Gillis, Richard, and Hagan (1986) noted that Asians tended to be more tolerant of high-density living, while British individuals were less adaptable, and Southern Europeans fell somewhere in between these extremes.

**Cultural Adaptation to Density**
Long-term exposure to different population densities could contribute to varying levels of adaptation across cultures. Gillis, Richard, and Hagan's (1986) study highlighted these differences, with Asians showing greater adaptability to high-density living compared to their British counterparts. This suggests that cultural factors play a significant role in shaping individuals' perceptions and responses to living density, influencing their overall well-being in residential environments.

In the next lecture we will elaborate about the environmental changes and rural areas.

**Lesson 37**
**Environmental Changes in Rural Areas**

## Rural Areas and Their Characteristics

Rural areas are characterized by their strong connection to community and nature. However, it's argued that no truly untouched natural places exist anymore; human influence has shaped all rural landscapes.

- **Rurality and Nature:** Historically, there's been a close link between nature and society, with rural areas often associated with a purer and nobler form of living compared to urban spaces.
- **Romanticized Association:** Rural landscapes are often seen as natural, and rural activities are perceived as working harmoniously with nature. Rural inhabitants are often seen as more attuned to and respectful of nature.

## Factors Contributing to Environmental Changes

Following factors contribute to the environmental changes in rural areas:

- **Agricultural Practices:** The use of pesticides, chemical fertilizers, and the removal of hedgerows have led to increased yields but also contributed to habitat destruction and reduced plant diversity.
- **Urbanization:** The expansion of urban areas into rural spaces has led to loss of open space, increased pollution, demand for infrastructure like roads and drainage systems, and light and noise pollution.
- **Forestry and Primary Production:** Deforestation, afforestation of open areas with non-native species, mining and quarrying, and flooding for reservoirs have all impacted rural environments.

## Degradation of Natural Environment

The causes of the degradation of the natural environment are as follows:

- **Modern Agriculture:** Intensive farming practices have led to declines in plant and animal species due to habitat destruction and chemical use
- **Tourism Impact:** Tourists attracted to rural landscapes have contributed to erosion, pollution, and construction-related land loss.
- **Global Environmental Change:** Rural areas also face challenges from global warming and other environmental shifts.

## Approaches to Dealing with Rural Environmental Change

The two main approaches to deal with the rural environmental change are given below:

- **Utilitarian Perspective:** Some argue that nature can adapt to a certain level of change, and not all environmental alterations are problematic.
- **Natura-Ruralistic Perspective:** Others believe that environmental changes have already caused irreparable damage, and urgent action is needed to prevent further harm.

## Course of Action and Challenges

Following course of action to be taken in order to deal with the effect of environmental changes occurring is rural areas are:

- To protect wildlife and habitats, farming practices may need extensive regulation.

- Utilizing alternate energy sources like wind power can significantly impact rural environments.

Despite various measures and programs, finding the right response to rural environmental changes remains a significant question influenced by political and economic factors.

**Lesson 38**
**The Built Environment and Human Adjustment**

In environmental psychology, the built environment plays a crucial role in shaping human behavior and adjustment. The design of institutions such as hospitals, prisons, and educational environments significantly impacts individuals' experiences and interactions within these spaces. Let's delve into how institutional design influences human behavior and adaptation.

**Understanding Queuing Theory: The Never-Ending Wait**
Lines or queues are a ubiquitous aspect of daily life, from grocery stores to fast-food restaurants. The experience of waiting in line can be frustrating and time-consuming, leading to questions about the design of these spaces.
Vile's Laws of Advanced Linesmanship humorously capture the nuances of queuing behavior, highlighting challenges like line length, unexpected delays, and social interactions in queues. Here are funny rules that explain how people behave while waiting in lines:

- If you are running for a short line, it suddenly becomes a long line.
- If you are waiting in a long line, people behind you are shunted to a new short line.
- If you step out of a short line for a second, it becomes a long line.
- If you are in a short line, the people ahead of you let in their friends and relatives and make it a long line.
- A short line outside a building becomes a long line.
- If you stand in one place long enough, you make a line.

These laws humorously reflect the frustrations and unpredictability often associated with waiting in queues or lines. They highlight common experiences and observations people make when navigating through lines in various settings.

**The Role of Good Architecture in Public Spaces**
Architectural design shapes our experiences in public spaces such as supermarkets, airports, and hospitals. Good architecture minimizes human discomfort and maximizes functionality, yet achieving this balance requires understanding users' needs and preferences. C.M. Deasey's insights underscore the importance of aligning design with human nature and demands for change in public attitudes toward architecture.

## General Hospitals

The hospital environment encompasses a variety of people, including patients with diverse needs and medical staff with specialized roles. However, conflicts can arise between the needs of patients and staff, highlighting the importance of balancing efficiency with patient well-being. For instance, during surgery, patients require a specific atmosphere, while staff members are under stress and may need different environmental conditions.

**A Mental Hospital – Example of Design**
Robert Sommer's observations in a mental hospital shed light on the impact of design on patient well-being. The installation of new amenities without consulting patients led to a detrimental arrangement that hindered social interaction. This case underscores the need for inclusive design practices that consider user input and promote positive experiences.

**Key Factors in Hospital Design**

Examining hospital design reveals critical factors influencing human behavior, such as proximity, privacy, and control. Design choices like the location of nurses' stations, room layouts, and environmental controls significantly affect staff efficiency and patient satisfaction. Understanding these dynamics can lead to improvements in hospital design that enhance both functionality and user experiences.

Hospital design should aim to maintain behaviors, encourage healthy behaviors for rehabilitation, and provide freedom for individuals to operate and realize their potential. The environment must compensate for individual failures and enhance performance levels where possible, striking a balance between efficiency and patient care.

**Hospital Design and Its Effects**

As environmental psychologists, studying institutional design allows us to understand how physical spaces shape behavior, emotions, and interactions. By analyzing real-world examples and research findings, we gain insights into creating environments that promote well-being, productivity, and positive social dynamics. Institutional design serves as a lens through which we examine the intricate relationship between humans and their built surroundings.

Effective hospital design directly influences user behavior and experiences. Observing how the physical layout of a hospital affects daily activities can reveal opportunities for improvement. For instance, assessing the accessibility of facilities, signage clarity, and ease of navigation can lead to design suggestions that enhance user experience and efficiency.

**Conclusion: Bridging Design and Human Experience**

In conclusion, the built environment profoundly influences human behavior and adjustment in institutional settings. Through a multidisciplinary approach that considers user needs, feedback, and empirical research, environmental psychologists contribute to designing spaces that optimize human experiences and outcomes. By studying institutional design, we uncover valuable insights into fostering environments that support well-being, functionality, and meaningful interactions.

In the next lecture we will talk about the prisons.

**Lesson 39**
**Prisons**

In the realm of correctional facilities, there exists a complex interplay of conflicting goals and varied philosophies regarding the purpose and design of prisons. This handout delves into the intricacies of imprisonment, exploring different models, architectural ideologies, and their impact on inmates' experiences.

**Conflicting Goals of Imprisonment**
Prisons face conflicting goals and perspectives, resulting in disparities in sentencing and rehabilitation opportunities.
**Diverse Perspectives:** Various stakeholders, including corrections employees, judges, legislators, and inmates, hold conflicting ideas about the goals of prisons.
**Sentencing Disparity:** This diversity of perspectives leads to significant differences in sentencing, parole criteria, and rehabilitation opportunities.

**Philosophical Models of Imprisonment**
Let's explore different philosophical models of imprisonment and their impact on prison design.
**Sommer's Models:** Robert Sommer delineates several models of imprisonment, such as deterrence, rehabilitation, retribution, and integration, each with its unique implications for prison design. Each model represents unique challenges in prison design and management, influenced by its underlying philosophy. These conflicting viewpoints contribute to sentencing disparities, parole variations, divergent vocational and psychological support, and discrepancies in actual time served for similar offenses, termed as a "model muddle" by Siegler and Osmond (1974).
**Architectural Adaptations:** The design of prisons varies significantly based on the model being followed. For instance, an integration-focused prison may include vocational workshops and educational facilities, while a deterrence-oriented one may prioritize security features.

**Life inside Prisons**
Exploring the dynamics of life within prison walls unveils intriguing analogies between prison environments and hospital settings, highlighting the contrasting philosophies and approaches to design.
**Hospital vs. Prison Analogies:** The physical design of prisons symbolizes containment and control, akin to hospital settings but with stricter regulations and limited personal space.
**Historical Architectural Alignment:** Traditional prison architecture aligns with deterrence and retribution philosophies, resulting in impersonal and restrictive environments for inmates.
**Modern Approaches:** Contemporary perspectives advocate for prison designs aligned with re-education, reform, and integration philosophies. Examples like New Jersey's Leesburg prison showcase attempts to create more humane and rehabilitative environments through innovative architectural features.

**Empirical Insights**
The empirical findings presented in the following studies are as follows:
- Gilbert's study in 1972 emphasized that limiting inmates' movement within the prison environment is more crucial than the actual size of their cells.

- Sommer's research in 1972 highlighted that factors such as crowding, lack of privacy, and sensory deprivation contribute significantly to the challenges of confinement and restriction faced by prisoners.
- Luxenberg's analysis in 1977 indicated that modern prisons have made efforts to mitigate the negative effects of designing solely for maximum security, achieving varying levels of success in this endeavor.
- Paul Paulus and colleagues found that inmates in single or double occupancy cells perceive less crowding compared to those in cells with larger numbers. Consequently, prisoners in denser cells report more health issues and spend more time in the infirmary.
- Schaeffer, Baum, Paulus, and Gaes discovered in 1988 that partitioning cubicles in prison cells led to a reduction in urinary catecholamine levels, which is an indicator of stress among inmates.
- Wener and Keys' research in 1988 revealed that increased population density within prison cells is associated with higher rates of perceived crowding, more sick calls, and increased isolated behavior among inmates.
- Eckland's study in 1986 suggested that violence in prisons, often linked with high density, is less about cognitive confusion and tension and more about issues of control within the prison environment.

However, challenges persist in fostering a cooperative and rehabilitative prison environment, emphasizing the ongoing need for research and progressive design strategies.

**Case Study**

Charles in City Prison: Analyzing individual experiences like Charles' incarceration sheds light on how prison design may influence inmate behavior and rehabilitation outcomes. Charles, a 34-year-old individual, was arrested for driving 41mph in a 25 mph zone. Despite having a clean record regarding alcohol and drugs, he was found with a small quantity of cocaine in his car. As a result, he faced additional charges for possessing an illicit drug. Charles was sentenced to two years in city prison.

During his time in city prison, Charles was housed in a single cell designed for two people. He encountered a series of cellmates who had received sentences ranging from three days to life imprisonment. Throughout his incarceration, Charles learned about new drug dealers and methods of drug use that he had not been aware of before. This experience raises questions about the potential influence of prison design and environment on inmates' exposure to criminal behaviors and their subsequent behavior post-release.

**Conclusion**

Reflecting on questions about the relationship between design, punishment philosophy, and inmate behavior encourages critical thinking and exploration of reformative approaches. Prison design is not merely about physical structures but also about embodying philosophical ideals and fostering environments conducive to rehabilitation and integration. Understanding these complexities is essential for informed discussions in environmental psychology and criminal justice studies.

In the next lecture we will talk about the education environments.

**Lesson 40**
**Education Environments**

When discussing the educational environment, the classroom holds significance within that context. This lesson highlights how the physical design and layout of classrooms influence various aspects of the educational experience, such as class participation, student behavior, and academic performance.

The built environment refers to the physical structures and spaces that humans create for living, working, and learning. In the context of education, the built environment encompasses classrooms, lecture halls, libraries, dormitories, and other educational facilities. These physical spaces play a crucial role in shaping the overall educational experience and can significantly impact students' learning outcomes.

## Traditional Classroom Arrangements

The traditional classroom setup, characterized by rows of students facing an instructor, reflects the modal arrangement in educational environments. This layout, with its genesis in utilizing outside light for illumination, serves several purposes such as ease of surveillance, attendance tracking, and maintaining control by the instructor.

## Seating Arrangement

Seating positions within the classroom can affect class participation, attentiveness, and academic achievement.

- **Impact on Student Behavior:** Studies by various researchers have shown the influence of seating arrangements on student behavior. For instance, Sommer's research indicates that students in the front row tend to participate more, while Koneya's findings suggest that low verbalizers may participate less regardless of their seating position.
- **Attention and Academic Performance:** Further studies by Schwebel, Cherlin, and others highlight that students in front-row seats are more attentive and often achieve higher grades, even when seating positions are assigned rather than chosen by students.
- **Designing for Interaction:** In smaller classrooms or seminar-type settings, seating positions directly influence participation levels. For example, students seated opposite the instructor tend to participate the most, emphasizing the importance of seating proximity in facilitating interaction.

## Color and Environmental Factors

Research by Kuller and Wilson suggests that colors can influence psychological processes, mood, and attitudes, thereby indirectly impacting students' learning experiences. Blue influences feelings of security. Red is stimulating and leads to excitement. Colors influence blood pressure and respiration rate.

Class room behavior, performance and learning will be influenced by such environmental conditions as:

- **Temperature**: Temperature plays a crucial role in creating a comfortable and conducive learning environment. Extreme temperatures, whether too hot or too cold, can distract students and affect their focus and concentration. Maintaining an optimal temperature in the classroom helps students stay attentive, engaged, and more receptive to learning.

- **Noise:** Similarly, noise levels in the classroom can have a profound impact on student performance. Excessive noise, such as chatter from neighboring classrooms, outdoor sounds, or mechanical noises, can disrupt concentration and hinder effective communication between students and teachers. On the other hand, a quiet and controlled acoustic environment allows for better concentration, improved comprehension, and enhanced classroom interactions, leading to more productive learning experiences.
- **Crowding:** High levels of crowding in educational environments have a profound impact on students, affecting behavior, performance, and learning outcomes. This crowding often results in discomfort, limited personal space, and heightened stress levels among students, which can disrupt their focus and engagement in academic activities. Furthermore, it hampers social interactions and collaborative efforts, thereby impairing peer learning and teamwork.
- **Light (illumination):** As illumination increase, visual activity increases. With more light we are able to recognize and detect smaller visual details. Effects of changes in illumination are more pronounced on difficult tasks than on easy tasks. Greater illumination allows for more accurate and quick discrimination. At very high levels of illumination performance decrements are likely because additional light can act to suppress some information cues such as visual gradients.

**Open Space Classrooms**

Exploring innovative classroom designs, open-space classrooms offer flexibility and increased interaction among students and teachers. However, studies also highlight challenges such as increased noise levels and visual distractions, emphasizing the need for careful design considerations.

**Optimizing Learning Environments**

Effective classroom design involves balancing factors like illumination, noise control, and seating arrangements to create conducive learning environments. By addressing these design elements, educational settings can maximize learning potential and student engagement.

**Considerations for Dormitory Design**

Beyond classrooms, dormitory design also plays a crucial role in students' educational experiences. Research indicates that factors like social density, noise levels, and personal control within dormitories can significantly impact students' well-being and academic performance.

**Conclusion**

In summary, the physical design of educational environments, including classrooms and dormitories, can significantly influence learning outcomes, student behavior, and overall academic success. By understanding and addressing environmental factors, educators and designers can create spaces that enhance the learning experience and promote student.

**Lesson 41**
**Institutional Design Reconsidered & Queuing Theory**

In this lecture, we delve into the critical aspects of institutional design within the realm of environmental psychology. We explore the complexities involved in designing spaces that cater to the diverse needs and behaviors of their users, emphasizing the importance of considering human welfare in the built environment.

**Understanding Institutional Design**

- **Complexity of Design**
  - Highlighting the limitations of oversimplified design paradigms.
  - Emphasizing the need for comprehensive data analysis to inform design decisions.
- **Interaction of Physical and User Characteristics**
  - Discussing how the performance of a built environment is influenced by its physical attributes and user requirements.
  - Acknowledging the varied needs of multiple users in designed environments.
- **Evaluation Beyond Prototypes**
  - Stressing the importance of evaluating designs based on actual user experiences rather than theoretical models.

**Challenges in Design Process**

- **Technical Focus of Architects and Designers**
  - Exploring the primary concerns of architects and designers in terms of technical and structural aspects.
  - Recognizing the potential gaps in understanding user cultural dynamics and ecological relationships.
- **Cultural and Ecological Considerations**
  - Discussing the significance of factors like privacy, territoriality, and adaptation in design.
  - Addressing cross-cultural nuances that impact design and social change.

**Steps in Design Process**

- **Preparation and Problem Statement:** Defining the problem statement and design objectives clearly.
- **Data Gathering and Analysis:** Emphasizing the importance of gathering relevant data for informed decision-making.
- **Outcome Evaluation and Creativity:** Defining desirable outcomes and exploring cost-effective design solutions.
- **Selection and Implementation:** Decision-making on design alternatives and transforming plans into physical structures.
- **Reevaluation and Adaptation:** Assessing design performance and refining structures based on user feedback.

**Considerations for Design Criteria**

- **Performance Standards:** Discussing criteria such as profit, quality, performance, competition, compatibility, flexibility, elegance, safety, and time in setting design standards.
- **Human Factors and Physical Limitations:** Exploring design prescriptions based on human physical and anatomical characteristics.

**Queuing Theory and Its Applications**

Queuing theory, focusing on waiting dynamics, is vital for efficient design in various contexts, from retail settings to transportation systems. Understanding traffic intensity and optimizing service processes can significantly enhance user experiences.

- **Queue and Traffic Intensity:** Explaining the concept of queue and traffic intensity in service environments.

  Example: Illustrating queue dynamics with an example of customers at a grocery store. This example demonstrates how queuing theory can be applied to analyze and optimize customer service in a grocery store setting. Store managers can use this data to anticipate customer arrival intervals, optimize service efficiency, and make informed decisions such as opening more registers when required.

  Recommendations for Store Managers is to use strategies for anticipating customer arrival intervals and optimizing service efficiency.
- **Other Uses of Queuing Theory:** Queuing theory has utility in any situation where all potential users of a facility cannot utilize the facility simultaneously. It goes beyond service industries and can be applied to various scenarios to improve system efficiency and user experiences.
  - **Accessible Environments for Persons with Disabilities:** Revisiting discussions on creating inclusive environments.
  - **Energy Use and Behavioral Change:** Analyzing energy demands and the need for behavioral shifts. Discussing grassroots efforts and community-based initiatives for energy conservation.
- **The Future Challenges:** Energy use and abuse, environmental costs of energy and material usage, strategies for saving energy, alternate energy resources, behavioral solutions to environmental problems, environmental attitudes and behaviors, integration, and looking to the future are some of the challenges that lie ahead. Understanding these challenges and working towards sustainable solutions is crucial for a better future.

## Lesson 42
## Energy Use in Homes and Commercial Buildings

### Determinants of Energy Use at Homes
Heating and cooling systems play a crucial role in determining energy consumption in homes. For instance, using a natural gas furnace for heating is more energy-efficient than relying on electrical baseboard heating. This efficiency stems from reduced energy loss during the conversion of primary energy sources like coal, fuel, oil, or natural gas into electricity. Additionally, lifestyle factors significantly influence energy use at homes. The number and types of electrical appliances, such as color TVs versus black and white TVs, frost-free refrigerators versus conventional ones, air conditioners versus window fans, and incandescent lights versus fluorescent lights, all impact energy consumption levels.

### Savings for Consumers – An Example
Considerable economic and environmental benefits arise from choosing energy-efficient appliances and practices. For example, if 6.6 million refrigerators/freezers sold in 1970 were energy-efficient, it would lead to substantial savings in electricity and dollars. This would not only benefit consumers economically but also result in significant reductions in environmental pollution, including sulfur dioxide, nitrogen oxides, and reduced land use for power plants and transmission lines.

### Home Construction and Energy Use
The construction of homes is a critical factor in determining energy consumption. Key considerations include wise use of insulation, effective shading in summer, and utilization of sunlight in winter. Implementing these strategies can result in substantial energy savings without compromising comfort. For instance, well-insulated homes with strategic shading can retain heat during winter and keep interiors cool during summer, leading to reduced energy needs for heating and cooling.

### Energy Use in Commercial Buildings
Commercial buildings often exhibit excessive energy consumption due to design choices such as extensive glass usage and tall structures. For example, buildings like the Sears Tower in Chicago consume more electricity than entire towns. Implementing energy-efficient design principles and modifying appliance usage can lead to significant energy savings in commercial buildings.

### Environmental Costs of Energy and Materials Usage
The use of energy and materials has far-reaching environmental impacts throughout extraction, processing, and usage stages. Oil drilling activities, for instance, can result in blow-outs, spills, and refinery pollution, causing harm to marine life and ecosystems. Addressing these environmental costs requires adopting sustainable practices and reducing reliance on non-renewable resources.

### Strategies for Saving Energy and Promoting Sustainability
Efforts to save energy and promote environmental sustainability include using and producing less, extending the lifespan of products through reuse and recycling, and choosing energy-efficient materials and appliances. Embracing renewable energy sources like solar energy and

implementing energy-saving measures in construction and daily practices are crucial steps towards a more sustainable energy future. However, challenges such as initial investment costs and the availability of existing energy sources need to be addressed to facilitate widespread adoption of sustainable energy practices.

**Lesson 43**
**Conservation as Energy Source**

**Three Types of Conservation**

- **Curtailment:** Curtailing energy use involves reducing the overall consumption of energy resources. This can be achieved through conscious efforts to use energy-efficient appliances, adopting energy-saving practices, and minimizing wastage.
- **Overhaul:** Overhauling refers to making dramatic changes in lifestyle and societal norms to reduce energy consumption. An extreme example of this would be implementing laws against suburbanization, which can significantly impact energy demands.
- **Adjustment:** Adjustment involves making specific changes to enhance energy efficiency. This includes insulating homes, improving the efficiency of automobiles and home appliances, and capturing waste heat for productive use.

**Imagine Your Life 50 Years from Now**

Visualize a future where energy conservation and efficiency are ingrained in everyday life, leading to sustainable and eco-friendly practices across all sectors.

**Behavioral Solutions to Environmental Problems**

- **Tragedy of Commons:** The concept of the tragedy of the commons highlights the conflict between immediate self-interest and future public interest. It underscores the need for collective action and responsible environmental stewardship.
- **Education to Preserve Environment:** Educational initiatives play a crucial role in promoting environmental preservation. However, simply telling people what to do may not be effective. Instead, efforts should focus on changing attitudes and behaviors gradually.
- **Applied Behavior Analysis and Intervention:** Applying principles of behavior modification can lead to positive changes in environmentally relevant behaviors. Strategies include:
    - **Baseline Period**: Observing naturally occurring behaviors related to energy use.
    - **Treatment Period:** Introducing interventions like rewards for desired behaviors and punishments for non-target behaviors.
    - **Follow-Up Period:** Continuing observations post-intervention to assess effectiveness.

**Intervention Strategies**

- **Antecedent Strategies:** Using prompts and cues, such as written messages encouraging eco-friendly actions, can trigger desired behavioral responses. Specific prompts near the intended behavior location and polite messaging are more effective.
  Examples of prompts:
  "Help keep your pool clean."
  "Do not litter."
  "Please avoid littering."
- **Consequent Strategies:** These strategies involve presenting or removing pleasant/unpleasant consequences following behaviors to reinforce or discourage them.
  Types of consequences include:
    - **Reinforcement:** Increasing the frequency of desired behaviors.
    - **Punishment:** Decreasing the frequency of undesirable behaviors.

Types of reinforcement:

- **Positive Reinforcement:** Offering pleasant consequences for desired behaviors.
- **Positive Punishment:** Imposing unpleasant consequences for undesirable behaviors.
- **Negative Reinforcement:** Removing negative consequences for desired behaviors.
- **Negative Punishment:** Removing pleasant events for undesirable behaviors.

**Integration and Look to the Future**

Our collective behaviors significantly impact environmental quality. By adopting energy-efficient practices and promoting sustainability, we contribute positively to long-term public interest and mitigate environmental degradation caused by excessive energy consumption.

**Lecture 44**
**Understanding Urban Sprawl: Challenges and Solutions**

**Further Plans – The Cities**

- **Urban Sprawl:** Urban sprawl refers to the expansion of low-density residential areas, shopping centers, and industrial facilities connected by multilane highways, leading to unplanned development and environmental challenges.

- **Origins of Urban Sprawl:** Historically, cities were designed for walking, bicycles, and public transport, with compact neighborhoods offering easy access to amenities. However, with technological advancements and increased private transportation, cities expanded outward.

- **The Dream of Suburbs:** The desire for personal space and the convenience of private transportation led to a housing boom in suburbs. This shift from city centers to suburban areas resulted in haphazard development and increased dependence on cars.

**Urban Blight**

- **Challenges Faced:** Industrialization brought challenges like poor housing, pollution, and congestion in cities, prompting people to seek housing outside urban centers.

- **Consequences**: The rapid growth of housing colonies without proper planning led to a maze of developments, straining local governments to provide essential services like schools, water, and roads.

- **Vicious Cycle of Development:** Efforts to reduce congestion with new highways only fueled further sprawl, creating a cycle of increased commuting distances and traffic congestion.

**Moving Towards Sustainable Communities**

- **Smart Growth Initiatives:** To address these challenges, smart growth initiatives promote compact, walkable communities with mixed land use, reducing environmental impacts and improving quality of life.

- **Environmental Impacts of Urban Sprawl:** Urban sprawl leads to energy depletion, air and water pollution, loss of agricultural land, and fragmentation of wildlife habitats, impacting ecosystems and human health.

**Towards the Common Good**

- **Quality of Life vs. Environmental Concerns:** While sprawl offers benefits like larger homes and better services, it also increases vehicle ownership, fatal accidents, and health issues like obesity and high blood pressure.

- **Common Good Perspective:** Environmental costs are often overlooked in favor of personal benefits, highlighting the need for collective responsibility and sustainable choices for common good.

- **Reining in Urban Sprawl: Smart Growth Solutions:** By promoting compact development, public transport, and green spaces, communities can mitigate the negative impacts of sprawl and create livable, eco-friendly environments.

In conclusion, understanding the complexities of urban sprawl is crucial for creating sustainable cities that balance personal preferences with environmental conservation for the common good.

**Lesson 45**
**Urban Blight**

In the developed world the other end of ex-urban migration is the city from which people are moving. A completely different set of factors is responsible for urban blight in developing countries. In this case, people are moving *to* the cities at a rate that far exceeds the capacity of cities to assimilate them. The result is urban slums that surround virtually every city in the poorer developing countries.
The economies of rural areas, often based only on subsistence farming simply do not provide the jobs needed by a growing population. So, people move to the cities, where at least they have the hope for employment.

**City Housing**
City housing is overwhelmed by the influx of migrants who could not afford the rents even if housing was available this leads to the formation of slums

**Slums Surrounding the Cities**
Vast slums surrounding the cities are a great challenge to the institutional structure of developing countries Because slums are unauthorized areas, cities rarely provide them with electricity, water, sanitation, and other social amenities Yet the slums often represent essential workforce to the city taking low-paying jobs that keep the city and its inhabitants functioning People live in the fear of bulldozers coming in any time and leveling down the shanty town A great need in such neighborhood is home security People in the slums also need more jobs Greatest need for people living in these areas is government representation. Their voices are seldom heard because they are among the poorest and most powerless in society.

**What Makes Cities Livable**
A sustainable future will depend upon both reining the urban sprawl and revitalizing the cities. The urban blight in the cities in developing countries requires deliberate policies to address the social needs of people flocking to the shanty towns. The only possible way to sustain the global population is by having viable, resource-efficient cities, leaving the countryside for agriculture and the natural ecosystem.

**Viable**
Viable means livable No one wants to and should be required to live in the conditions that have come to typify the urban blight Livability is a general concept based on people's response to the question "Do you like living here, or would you rather live somewhere else?"

**Livable Cities around the World**
The common denominator of livable cities in the world is: Maintaining high population density Preserving heterogeneity of residences, businesses, stores, and shops keeping layouts on human dimension so that people can meet, visit or conduct business incidentally over coffee at side walk café or stroll through an open area.

**People VS Automobiles**

In livable cities the space is designed for and devoted to people In contrast development of the last fifty years has concentrated on accommodating automobiles and traffic Two thirds of the land in the cities that have grown up in the ear of automobiles is devoted to moving, parking or servicing the cars.

**A Matter of Design**

The world's most livable cities are not those with perfect auto access between all points. Instead they are the cities that have taken measures to reduce outward sprawl, diminish automobile traffic, and improve access by foot and bicycle in conjunction with mass transit.

**Examples of Cities around the World**

Geneva, Switzerland prohibits automobile parking at workplaces in the city's center forcing customers to use excellent public transport system Copenhagen bans all on-street parking in the downtown core Paris has removed 200,000 parking places in the downtown area Curitiba, Brazil with a population of 1.6 million is cited to be one of the most livable city in the entire Latin America because they have guided development around the idea of mass transit system rather than cars.

**Livable = Sustainable**

Livability of a city leads to its sustainability Reduction of auto traffic and greater reliance on foot and public transportation reduce energy consumption and pollution Urban heterogeneity can facilitate the recycling of materials Housing can be retrofitted with passive solar space heating and heating for hot water Landscaping can provide cooling Vacant or cleared areas can be converted into garden plots Rooftop hydroponic gardens can be made popular. Such gardens may not make the cities agriculturally self-sufficient but they add to urban livability. By making the urban areas more appealing and economically viable we not only improve the lives of city residents but also spare the surrounding areas Parks, wilderness, and farms will not be replaced by exurbs and shopping malls but rather will be saved for future generations.

**Moving Towards Sustainable Communities**

A joint facility of the UN Environment Program and the UN Human Settlement Program, the Sustainable Cities Program (SCP) is designed to foster the planning and management of needed to move cities in developing countries towards sustainability.

**Sustainable City**

Sustainable Cities Program (SCP) defines the sustainable city as a city in which achievements in social, economic, and physical development are made to last. Cities are the focus of the program because they are absorbing two-thirds of population growth in developing countries and in the process, we are experiencing serious environmental degradation in and around the growing urban centers.

**Bottom-up Approach**

The bottom-up approach calls for the involvement of all people at all economic and social levels and reconciling their interests when conflicts are evident Common principles seen in sustainable development theory are principles such as Social equity, Economic efficiency, and Environmental planning and management.

**SCP Cities**

To date, there are twenty SCP cities some of them are: Madras (India), Dar es Salam (Tanzania), Accra (Ghana), and Shenyang (China), Concepcion (Chile).

**Towards the Common Good**

How must we live on our planet? There are three strategic themes to answer the question: Sustainability, Stewardship, and Sound Science.